National Aeronautics and Space Administration



Hydrogen Contamination and Its Correlation with Enhanced Low-Dose-Rate Sensitivity

NASA Electronic Parts and Packaging (NEPP) Program
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ABSTRACT

Enhanced low-dose-rate sensitivity (ELDRS) investigations carried out by the Jet Propulsion Laboratory (JPL) and Arizona State University (ASU) have shown significant differences in the degradation of bipolar microcircuits with total dose in the presence of molecular hydrogen (H_2) in packages. This has a significant impact on radiation hardness assurance and provides opportunities to improve device performance.

The general objectives of this program were to:

1. determine the extent to which hydrogen contamination affects the total dose and dose rate response of linear bipolar circuits;
2. develop a model that will enable the prediction of high dose rate (HDR) and low dose rate (LDR) response asymptotes and transition dose rates as a function of total dose, temperature, pressurized hydrogen, defect precursors, and other process-dependent variables;
3. explore the possibility of an accelerated hardness assurance method and possible hardening approaches; and
4. extend the work to other technologies that have total dose response affected by hydrogen contamination.

This document summarizes the findings obtained from experimental and modeling studies done on bipolar technologies from FY07–FY12.

SUMMARY OF THE HYDROGEN EFFORT

Molecular hydrogen (H_2) is ubiquitous in today's semiconductor integrated circuit (IC) fabrication and packaging processes [1]. During IC fabrication, H_2 is present in wafer cleaning procedures, film depositions, etches, high- and low-temperature anneals, and an assortment of other processes. During IC packaging, it is introduced during die attachment and by forming gases during packaging processes [2]. H_2 can outgas from grain boundaries or structural imperfections in iron-nickel alloy (kovar, Alloy42) lead frame material. Electroplated metal components such as plated gold or nickel films are major sources of dissolved hydrogen. Moisture is often present and results from the absorption or adsorption of H_2O on the internal surfaces of the package prior to sealing or from moisture within the sealing gas itself.

Solutions to hydrogen contamination have been reported and include thermal treatment, the use of package materials with low hydrogen absorption, a change of barrier materials in gates, and the use of hydrogen getters inside the packaging to absorb the hydrogen. However, there is no clear guideline or limit as to what level of hydrogen might be considered acceptable in sealed packages. The military standard test method for internal gas analysis, MIL-STD-883 Test Method 1018, was designed to look for moisture and not hydrogen or other gas impurities. There is no specification limit available for H_2. This lack of specification introduces another unknown when dealing with the radiation response of commercial linear bipolar devices. As shown in this document, the devices' total ionizing dose (TID) response and sensitivity to enhanced low-dose-rate sensitivity (ELDRS) are affected.

In FY07, we reported on the impact of hydrogen contamination on the total dose response of linear circuits. A general investigation was performed on a selection of key parts from different manufacturers that both exhibit ELDRS as well as differences in the total dose degradation with bias conditions and dose rates. Residual gas analyses (RGAs) and die passivation analyses were performed on these devices. ***The results of this study clearly indicated that there is a correlation between packaging characteristics and hydrogen content***. They suggested that by only looking at the package characteristics (ceramic packaging with or without gold plating and with or without kovar lids, can packaging, passivation layers, etc.), it is possible to evaluate which category of device is likely to have a non-negligible amount **(~0.5–4%)** of hydrogen in the package and consequently might be sensitive to total-dose and low-dose-rate (LDR) enhancement. We showed that 1) devices in can packages exhibit low amounts of hydrogen; 2) ceramic frit glass devices show negligible amounts of hydrogen; 3) parts that also have a nitride passivation layer do not show a significant quantity of hydrogen (though there is not necessarily a correlation here); and 4) nitride-coated devices exhibit both cases of ELDRS and non-ELDRS. While silicon nitride is a very good barrier to hydrogen diffusion, the deposition processes are known to introduce hydrogen into device passivation layers. Thus, we believe it is critical to investigate the mechanisms of hydrogen absorption/desorption in nitride passivations. In addition, two parts, the HSYE-117RH linear voltage regulator from Intersil and the AD590 temperature transducer from Analog Devices, were identified as showing a significant amount of hydrogen (~0.6–3%) in their packages. Further experiments were conducted to identify the relationship between ***hydrogen content and total dose response***. Twelve screened space-qualified AD590s were irradiated at both high dose rate (HDR) and LDR unbiased with all leads grounded. Three flat packs (with 0.4–1% H_2) and three cans (~0% H_2) were irradiated up to 30 krad(Si) with a LDR of 0.01 rad(Si)/s. Three flatpacks and three cans were irradiated up to 100 krad(Si) with an HDR of 25 rad(Si)/s. In addition, two parts of the HYSE-117RH (~3% H_2) from the same wafer lot were irradiated unbiased at a dose rate of 0.05 rad(Si)/s. One part was opened for more than a week to release the hydrogen content. The results led to the following

conclusions: 1) flat pack devices degrade much more at both HDR and LDR compared to the cans due to hydrogen contamination; 2) devices in the HDR and LDR case degrade more as the amount of hydrogen content increases; 3) can devices can be made to degrade similarly to the flat pack when the die is exposed to H_2; 4) the devices in the HDR and LDR case degrade more as the amount of hydrogen content increases; and 5) parts that have an oxide passivation are more affected by H_2 in packages. ***The results clearly confirmed the correlation between total dose response, packaging, and hydrogen contamination***. For the HSYE-117 case, the same trends were observed but more experiments were needed with more devices to confirm. During FY09, an evaluation of 12 additional devices in three different packages (i.e., with different concentrations of H_2) was performed to compare the HDR and LDR behavior. Results, presented in this document, show the same impact of hydrogen on the total dose response. In order to explain the underlying mechanisms that relate to the role of ***hydrogen contamination in the total dose response of linear bipolar microcircuits,*** additional work was performed at Arizona State University. A combination of modeling and experiments were conducted on gated-lateral PNP (GLPNP) devices fabricated at National Semiconductor. These devices were specifically designed to study ELDRS. Experimental results showed a monotonic increase in radiation-induced interface traps as well as oxide-trapped charge with increasing molecular hydrogen concentration in the ambient during irradiations. Using chemical kinetics and previously developed models for interface trap formation, a first-order model was proposed to describe the relationship between interface trap formation and excess molecular hydrogen concentration in gas ambient during radiation exposure. This model provided an excellent fit to the data obtained from the experiments.

In FY08, we focused our efforts on providing a better understanding of how hydrogen impacts the ***total dose and dose rate response of linear bipolar circuits and its correlation with ELDRS***. Because hydrogen is a dominant factor in determining both the total dose and dose rate responses of linear bipolar circuits, we conducted experiments on both transistor structures and linear circuits to measure their response as a function of the externally introduced hydrogen concentration. The results of these experiments showed that the amount of hydrogen does two things: 1) it increases the degradation at LDR, and 2) it increases the dose rates region where the transition from HDR to LDR enhancement occurs. The mechanisms for these trends were explored with a code that incorporates the basic drift-diffusion as well as kinetic processes for hydrogen cracking and free electron-hole recombination. The results from this model also indicate saturation at LDR. However, further experiments at a lower dose rate were suggested to completely validate the results. The following are the main conclusions drawn from the FY08 study:

1. Bipolar linear circuits should be processed and packaged with a minimum amount of hydrogen to achieve reasonable total dose hardness and minimize ELDRS. If the amount of hydrogen introduced during processing through metallization ensures an acceptable response, then the post-metal processes should be designed to minimize any further introduction of hydrogen.
2. If the amount of hydrogen—both initially in the base oxide and in the introduction after metallization—is low, then the transition to ELDRS may occur at a very low dose rate. Hence, some parts that have only been tested at dose rates as low as 10 mrad(Si)/s may show enhanced degradation when taken to even lower dose rates. If this is the case, it would have severe implications for MIL-STD-883, Test Method 1019. Experiments conducted by NASA Goddard Space Flight Center are underway to explore this possibility and results are available upon request.

3. An accelerated hardness assurance test method was suggested by testing parts at HDR (100 rad(Si)/s) in a 100% H_2 atmosphere to set an upper bound to the LDR response in space [3]. The technique is to irradiate parts with package lids removed in a glass tube pressurized with high concentration of H_2 (10–100%). Parts with nitride layers will prevent any penetration of externally applied hydrogen. To use this approach for such parts, the nitride layer has to be removed. This approach has only been demonstrated on a GLPNP test transistor and one circuit type. As a result, in FY10, the proposed method of accelerated testing using HDR irradiations in environments with elevated concentrations of H_2 was tested on six different part types representing a wide variation in manufacturer, process technology, and circuit design (GLPNP, LM193, AD590, LT1019, OP-42, and HSYE-117). In all cases, results were very promising. Such a technique may have a large beneficial impact on radiation hardness assurance for bipolar linear technologies. Compared to LDR testing, hydrogen-enhanced testing at HDRs can be a very cost-effective approach for part selection during the design phase of space systems. It also may be considered for missions that require higher dose levels for qualification where LDR testing is not practical. For use as a qualification or lot acceptance test method, a characterization test would need to be performed to establish the optimum dose rate and H_2 concentration to bound the LDR response.

In FY11, further experiments were required to consolidate the method proposed particularly for non-ELDRS parts. This report evaluates if the method is too conservative for radiation-hardened, ELDRS-free parts and discusses radiation hardness implications. Experiments performed on a number of bipolar linear circuits delidded the parts and exposed the dies to molecular hydrogen at various concentrations from 1% to 100% H_2, prior to irradiation. The total dose response of these parts was compared to the HDR and LDR response of the packaged samples not exposed to H_2. The results, in all cases, were that the degradation was significantly higher for the parts subjected to H_2, even at 1%. The increased degradation ranged from a few percent to over two orders of magnitude. While the use of H_2 exposure followed by medium- to high-dose-rate irradiation, as an accelerated test for ELDRS-free parts, is shown to be overly conservative, the results demonstrate that the radiation hardness of the parts will be compromised if the parts are purchased in die form and placed in a package that contains even small amounts of H_2. This would be the case for hybrids, dies assembled by third-party packaging companies, or other non-manufacturer packaging options.

Finally, to support the experimental observations, we developed a theoretical model that describes the physical mechanisms contributing to the buildup of radiation-induced defects in a metal-oxide-semiconductor (MOS) system. The presented model helps in understanding the underlying mechanisms that describe the impact of dose rate effects. The model incorporates hole-trapping mechanisms as well as the formation of interface traps due to the release of hydrogen as described by the two-stage hydrogen radiolysis model [7]. Therefore, the effects of on the dose-rate response are captured by the model. Calculations suggest that dose-rate response and ELDRS can depend on the combined effect of space charge from trapped holes near the interface and bimolecular recombination. Transient calculations also show that the reaction of cracking at positively charged defects is important in modeling the effects of hydrogen on dose-rate sensitivity. Results from calculations using this model for dose rate and hydrogen effects have shown to be consistent with experimental data.

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1.0 INTRODUCTION

Organizations or space agencies involved in building spacecraft have to guarantee the survivability of electronics in order to fly missions successfully. With the various types of missions that exist for space exploration, such as deep space, Earth orbit, and outer planets, the risk assessment of electronics is an involved process. Each spacecraft uses a very large number of components, from basic transistor structures to hybrid implementations, which need to be evaluated and selected according to the design functions criteria. This assessment depends on the requirements of the mission (i.e., the characteristics of the orbit, the duration of the mission, the amount of shielding, and the radiation environment).

This document focuses on providing a set of recommendations to engineers and scientists on how to address the issue of hydrogen contamination through the involved process of total dose evaluation within a radiation hardness assurance flow. These recommendations are supported by many experimental data obtained from irradiation performed on various device types and thorough modeling work that captures the key reactions, including hydrogen reactions, that participate in the physical mechanisms of total dose degradation.

The document first covers the basic understanding about the physics of total dose degradation mechanisms in silicon dioxide (SiO_2)-based electronics. Section 2 introduces the mechanisms of creation of defects in SiO_2 induced by total ionizing dose (TID) (i.e., interface traps [N_{it}] and oxide traps [N_{ot}]). Section 3 then demonstrates how these defects affect the electrical properties of the most common SiO_2-based devices (i.e., complementary metal-oxide semiconductor [CMOS] and bipolar) that are used in integrated circuits (ICs) to design space systems. Many details are provided, particularly, about the influence of time-dependent biases and annealing effects, and how these challenge current total dose evaluation practices for old and emerging technologies.

Section 4 presents results obtained from in-situ hydrogen-soaking irradiations performed on transistor test structures and various circuit types from different manufacturers. These results help quantify the impact of hydrogen contamination of the total dose and dose rate response of microcircuits. This experimental work is supported by a theoretical model, described in Section 5, that has been developed to describe the physical mechanisms contributing to the buildup of radiation-induced defects in a metal-oxide-semiconductor (MOS) system. This model shows the key reactions that capture the role of H_2 and its correlation with ELDRS. A good correlation with experimental data is shown.

Section 6 summarizes the results of additional experimental studies that investigated the extent to which molecular H_2 can be used to accelerate the degradation induced by higher-rate laboratory sources in order to bound or perhaps predict the low dose rate (LDR) responses of linear bipolar circuits. Results from experiments performed on ELDRS-sensitive and ELDRS-free parts are summarized. The radiation hardness assurance implications as well as the limitations of the testing method are discussed.

The end of each section provides a summary of key findings and a set of recommendations on how to account for the issue of hydrogen contamination within a radiation hardness assurance flow.

2.0 OVERVIEW OF TOTAL IONIZING DOSE AND DAMAGE PROCESSES IN SiO_2

Ionization of a target material is caused by the interaction of high-energy photons or charged particles (mainly protons and electrons) with the atoms of that material [1]. Photon interactions are not a primary concern for satellites in the natural space environment. However, they are for TID evaluations since most laboratory sources used to simulate total-dose space environments emit either low-energy x-rays or high-energy gamma rays.

Photons interact with material through three different processes, namely the photoelectric effect, the Compton Effect, and pair production [2] as illustrated in Figure 1. For each of these processes, the primary result of the interaction is the creation of energetic secondary electrons.

The relative importance of the three processes as a function of photon energy and atomic mass of the target material is illustrated in Figure 2 [3]. Figure 2 shows regions where each process dominates. The solid lines correspond to equal probabilities for the different interactions. The dashed line corresponds to the atomic mass of silicon (Z=14). Thus, for silicon, x-rays emitted from a low-energy (typically 10 keV) x-ray irradiator will interact predominantly through the photoelectric effect, while high-energy gamma rays (typically 1.25 MeV) from a ^{60}Co source will interact predominantly through Compton scattering.

Based on these interactions (i.e., photon or charge particles), the ehps density generated is proportional to the energy transferred to the target material [4]. Stopping power or linear energy transfer (LET) is the particle energy loss per unit length (*dE/dx*). It is a function of the particle mass and energy as well as the target material density [5]. The units of LET are commonly expressed as $MeV.cm^2/mg$.

Figure 3 shows a plot of LET vs. particle energy for electrons and protons [5]. The LET for protons is higher than the one for electrons at lower energies, but decreases rapidly with increasing energy. Electrons show a non-monotonic response, decreasing as a function of particle energy before increasing for energies above 1 MeV. The total amount of energy deposited by a particle that results in ehps production is commonly referred to as TID. The SI unit for TID is Gy (1 Gy = 1 J/kg); however, rad (radiation-absorbed dose) is the conventional unit used in space industry (1 rad = 100 erg/g = 6.24×10^{13} eV/g = 1×10^{-2} Gy). One rad (material) is equivalent to 100 ergs absorbed by one gram of the target material.

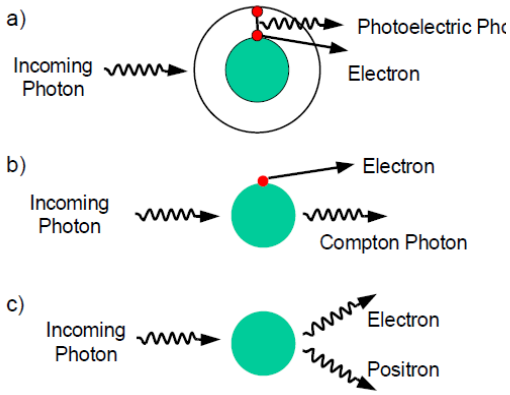


Fig. 1. Schematic drawing of three processes through which photons interact with material: a) photoelectric effect, b) Compton effect, and c) pair production. [1]

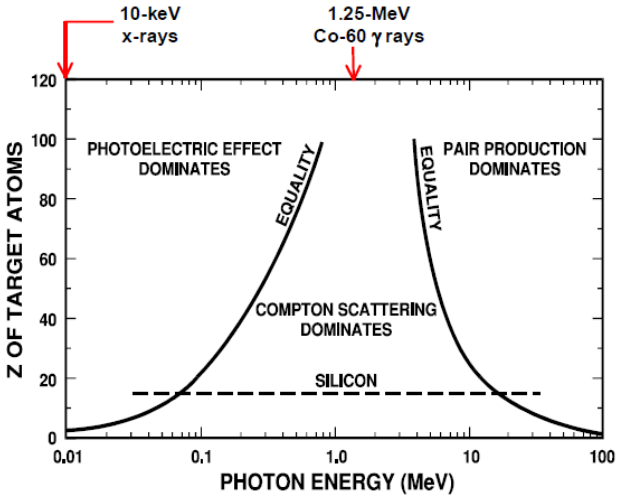


Fig. 2. Relative importance of the photoelectric effect, Compton scattering, and pair production as a function of photon energy (after [1]).

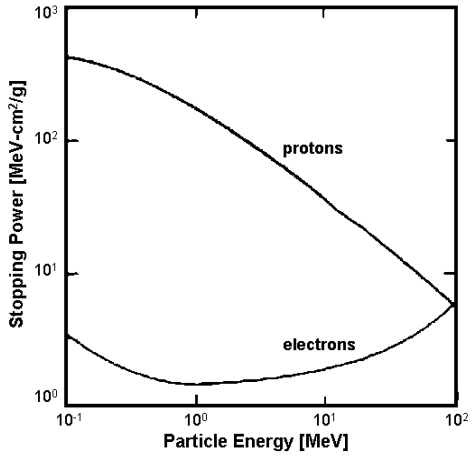


Fig. 3. Stopping power for electrons and protons as a function of particle energy [6].

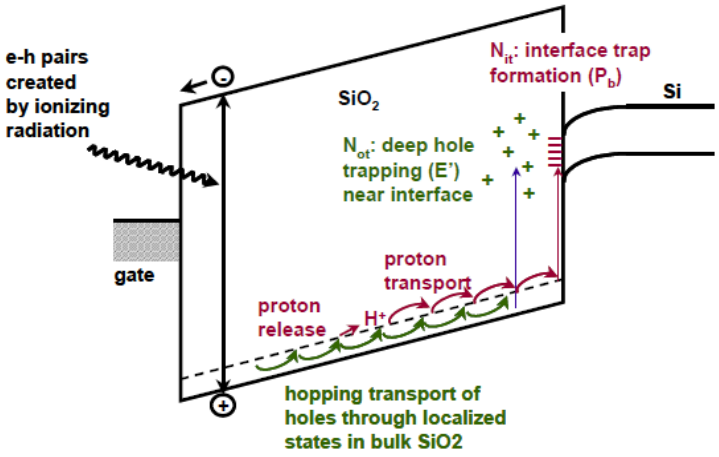


Fig. 4. Schematic energy band diagram for a MOS structure with a positive bias at the gate, indicating the basic radiation-induced processes [8].

The physical processes that lead from the initial deposition of energy by ionizing radiation to the creation of ionization defects in SiO_2 are the generation of ehps (**process 1**), the prompt recombination of a fraction of the generated ehps (**process 2**), the transport of free carriers remaining in the oxide (**process 3**), and either the formation of a trapped charge via hole trapping in defect precursor sites (**process 4a**) or the formation of interface traps via reactions involving hydrogen (**process 4b**) [6]. These processes are summarized graphically in Figure 4 [7].

In ehps generation (**process 1**), a fraction of the kinetic energy of the incident particle is lost to ehps creation. The mean energy, E_p, needed to ionize a material is dependent on the band-gap of the target material. The number of ehps generated for a given dose is thus strongly dependent on E_p as well as the material density. The relationships between ionization energy, material density, and generated carriers for three materials (GaAs, Si, and SiO_2) are listed in Table 1 [1]. The ehps density per rad (column 3 in Table 1) is obtained using the following conversion formula:

$$\kappa_g\left[\frac{\#\text{ehp}}{\text{cm}^3\text{rad}}\right]=100\left[\frac{\text{erg}}{\text{g}}\right]\left[\frac{1}{\text{rad}}\right]\bullet\frac{1}{1.6x10^{-12}}\left[\frac{\text{eV}}{\text{erg}}\right]\bullet\frac{1}{E_P}\left[\frac{\#\text{ehp}}{\text{eV}}\right]\bullet\rho\left[\frac{\text{g}}{\text{cm}^3}\right]. \quad (1)$$

Once generated, a fraction of the ehps is annihilated through either columnar or geminate recombination (**process 2**) [4]. Figure 5 plots the fractional charge yield (f_y) as a function of the radiation type [7, 8]. The figure also indicates that electron-hole pair recombination is electric field–dependent within the material [9]. The fractional yield of ehps increases monotonically as the local electric field increases (and is not zero at the zero electric field [4]). Electrons, having a much higher mobility than holes in oxides, are rapidly swept out of the dielectric [10]. Holes surviving this initial recombination undergo polaron hopping transport via shallow traps in the SiO_2 (**process 3**) [10]. A fraction of these transporting holes may fall into deep traps in the oxide bulk or near the Si/SiO_2 interface, thereby forming a trapped positive charge (N_{ot}) (**process 4a**) [10]. The hole trapping efficiency (f_{ot}) is also a function of the electric field in the oxide [10]. The trapped hole defect may,

Table 1. Ionization energy and carrier generation for a given material.

Material	E_p (eV)	Density (g/cm³)	Pair Density, g_0 (ehps/cm³ • rad)
GaAs	~4.8	5.32	7.6×10^{12}
Silicon	3.6	2.328	4×10^{13}
Silicon Dioxide	17	2.2	8.1×10^{12}

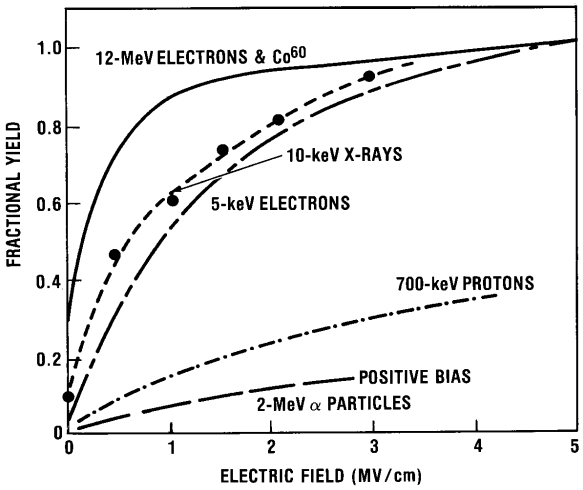


Fig. 5. Fractional yield of holes generated in SiO_2 as a function of the electric field in the material [7, 8].

depending on its proximity to the interface, exchange charges with the underlying Si via electron tunneling [11, 12]. Reactions between holes and hydrogen-containing defects or dopant complexes can also lead to the formation of a second type of ionization defect: the interface trap *(N_{it})* (**process 4b**) [13, 14].

2.1.1 Total Dose Induced Oxide-Trapped Charge and Interface Trap Formation in SiO_2

2.1.1.1 Oxide-Trapped Charge (N_{ot})

An oxide-trapped charge (N_{ot}) is typically net positive due to the capture of a hole in neutral oxygen vacancies and the subsequent formation of oxygen vacancy defects, or *E'* centers [15-20]. There are primarily two types of *E'* defects: E_δ' and E_γ'. The E_δ' center is a "dimmer" vacancy, which forms a relatively shallow trap for holes in the oxide. Most of the E_δ' centers have energies located in the SiO_2 band-gap within 1.0 eV of the oxide valence band [16]. The shallow trap level of the E_δ' makes it a good candidate for the defect type responsible for hole transport through SiO_2 (**process 3**) [16]. The E_γ' center is a significantly deeper trap than the E_δ' defect, residing at energy levels greater than 3 eV above the oxide valence band [15, 16]. While E_γ' centers may be located throughout the oxide, most are found near the Si/SiO_2 interface [10-12].

Both types of *E'* centers can exchange charges with an adjacent Si layer [15]. The ability of the *E'* defect to "communicate" with the Si is a strong function of its proximity to the interface [12]. E' centers that readily capture or emit carriers to the adjacent Si are often called *border traps* or *switching states*. They are generally located within a few nanometers of the Si/SiO_2 interface and can exchange charges via electron tunneling on time scales of microseconds to seconds [11-12]. E_γ' centers located at distances greater than a few nanometers from the interface (i.e., in the oxide bulk) may capture and emit carriers, but the probability of this process occurring is low. Thus, E_γ' defects in the oxide bulk are generally treated as a fixed (i.e., bias-independent) positive oxide charge (N_{ot}). Removal or compensation of N_{ot} may require elevated temperature and/or biased anneals over relatively long periods of time. A schematic illustration of the location of border traps (switching states) and oxide-trapped charge (fixed states) in the MOS system is illustrated in Figure 6. N_{ot} buildup in an oxide can be expressed as [21, 22]

$$\Delta N_{ot} = D\kappa_g f_y f_{ot} t_{ox}, \tag{2}$$

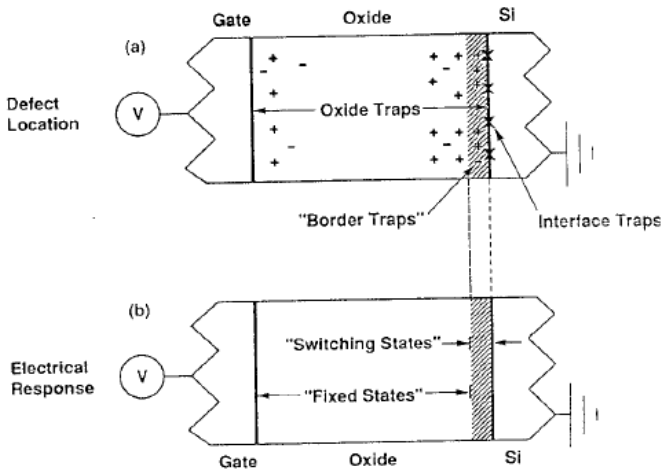


Fig. 6. Location and stability of the trapped charge in SiO_2 [11].

where D is the total ionizing dose deposited. As Eq. 2 shows, ΔN_{ot} is proportional to the oxide thickness.

As noted above, depending on their proximity to the Si/SiO_2 interface, E' centers can "communicate" with an adjacent Si layer [11, 12, 15]. The mechanism of charge exchange is dependent on the nature of the trap. For instance, the E_δ' defect can simply capture and re-emit trapped holes [11]. This process has been shown to more likely occur in p-type MOS transistors [11]. For E_γ' defects, the mechanisms are somewhat more complex. Most theories contend that the acting carriers for charge transfer in and out of the E_γ' defect are tunneling electrons. It is now widely held that the positively charged E_γ' centers can trap an electron, forming a highly stable dipole structure [10-12, 15, 26]. While it is possible for these trapped electrons to be re-emitted, the dipole will also act as a shallow electron trap that is relatively easy to fill or empty depending on the surface potential of the underlying Si [11, 26]. For a more detailed examination of the processes related to switching oxide traps, readers may want to consult [10-12, 15, and 26].

The impact of switching oxide (border) traps on CMOS dc parameters is different than the impact of fixed oxide charge discussed above. Unlike N_{ot}, the charge state of the switching trap can vary with bias. As such, the signature effect of border traps on metal-oxide-semiconductor field-effect transistor (MOSFET) dc parameters is similar to the effect of N_{it}.

2.1.2 Interface Traps

Like border traps, interface traps (N_{it}) will also exchange charges with an adjacent Si layer. However, unlike border traps, interface traps (N_{it}) are located exactly at the interface. There is essentially no barrier to trapping and de-trapping of carriers in the near-surface Si. N_{it} can therefore have a significant effect on carrier mobility and recombination rates of carriers at the semiconductor surface. N_{it} are primarily dangling bond defects called P_b centers [27]. The most important and abundant of these centers is called the P_{b0} center. A secondary contribution is provided by a closely related defect called P_{b1}. Figure 7 shows the schematic illustrations of P_{b0} and P_{b1} defects on (111), (110), and (100) silicon [28].

The three initial processes that lead to N_{it} formation are similar to processes that lead to the formation of oxide-trapped charges (i.e., ehps generation, recombination, and transport). However, the final formation of dangling bonds relies on several other reactions. The first

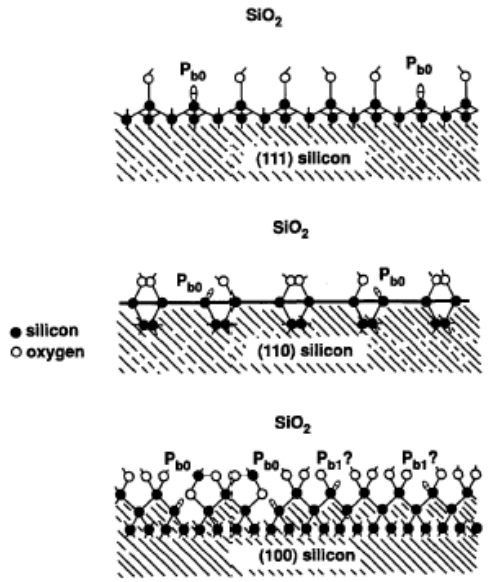


Fig. 7. Schematic illustration of P_{b0} and P_{b1} interface trap defects on (111), (110), and (100) silicon [35].

reaction is between transporting holes and hydrogen-containing oxide defects (D'H), which releases protons (H^+) [13]. Although it has been contended that direct reactions with holes can create N_{it}, it has been shown experimentally that (near and above room temperature) most N_{it} are created by protons [29]. Moreover, density-functional theory calculations confirm that the formation of N_{it} by direct hole interaction is not energetically favorable under most conditions [30]. The creation of a P_b center primarily relies on the presence of H^+ near the interface. Protons diffusing or driven by the electric field to the SiO_2 interface can remove hydrogen atoms from H-passivated dangling bonds (D) via the simple reaction [13, 30]

$$SiH + H^+ \Rightarrow D^+ + H_2 . \tag{3}$$

The resulting defect (D^+) is the interface trap. Further details regarding the formation of radiation-induced interface traps can be found in [13, 14, 27, and 30].

2.2 Annealing of Oxide Traps and Interface Traps

The annealing (i.e., compensation) of radiation-induced trapped holes in SiO_2 is a long-term process that is strongly dependent on temperature and applied electric field. The basic mechanisms for electron compensation are tunneling of an electron from the Si substrate and compensation by thermal excitation of an electron from the valence band [27]. Both tunneling and thermal emission have been combined into a single model by McWhorter et al. that describes a tunneling front and a thermal emission front, where the position of both varies logarithmically with time [33]. In other words, the distance into the oxide bulk from where trapped holes can be removed, measured from either the Si-SiO_2 interface for the case of tunneling or from the valence band edge for thermal emission, varies as ln(t). Figure 8 illustrates hole trapping along with the annealing/compensation processes. The trapped positive charge can be neutralized by adding an electron to the relaxed E'_γ center. The added electrons eliminate the unpaired spin and therefore compensate for the positive charge by creating a dipole structure by the transition from (b) → (c). The electron compensation process is reversible as illustrated by the transition from (c) → (b), where the electron has tunneled back into the substrate. True annealing can occur when the electrostatic force between the two ends of the dipole structure (c) is sufficient to reform the Si-Si bond. This situation is indicated by the transition from (c) → (a) [33].

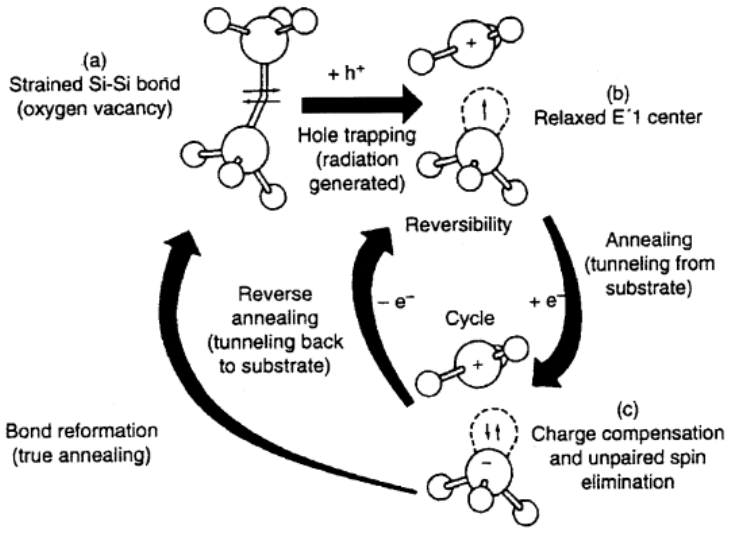


Fig. 8. Model for hole trapping and de-trapping (annealing) and for intermediate electron compensation and reverse annealing phenomenon [33].

The spatial and energy distributions of N_{ot} will strongly affect the rate at which charge neutralization occurs. For tunneling, the spatial distribution of N_{ot} must be close to the Si/SiO_2 interface. For thermal emission, the energy distribution needs to be close to the valence band energy. Unlike oxide-trapped charges, N_{it} buildup occurs on timeframes much slower than oxide-trapped charge build-up and can take thousands of seconds to saturate after a pulse of ionizing radiation. Interface-trapped charges do not anneal at room temperature, only at higher temperature [1]. These properties make N_{it} charge effects very important for LDR applications and total dose evaluations. Annealing behaviors have critical implication with total dose assessment of electronic devices.

3.0 TOTAL IONIZING DOSE (TID) EFFECTS IN MOS AND BIPOLAR TECHNOLOGIES

3.1 Total Ionizing Dose (TID) Effects in Metal-Oxide-Semiconductor (MOS) Devices

The basic structure of an n-channel MOSFET is illustrated in Figure 9. It is a four-terminal device with connection to the source, drain, gate, and silicon substrate (or bulk). It consists of a metal-oxide-silicon structure with a surface inversion layer or channel extending between two diffused junctions (i.e., source and drain). These diffused junctions are electrically disconnected unless there is an n-type inversion layer at the surface to provide a conducting channel between them. When the surface is inverted and a voltage is applied between the junctions, electrons can enter the channel at one junction, the source, and leave at the other, the drain.

The electrons flow primarily in regions of the silicon where they are majority carriers, a fundamental difference from bipolar transistors, in which current through the base is carried by injected minority carriers. Complementary-metal-oxide-semiconductor (CMOS) cells use MOS devices with smaller oxide thickness and are built by connecting in series p- and n-channel MOSFET devices. They are widely used in digital integrated circuits (ICs). Figure 10 shows a cross-sectional diagram for a generic deep-submicron bulk CMOS technology featuring shallow trench isolation (STI) oxides (that replaced LOCOS in old technologies).

The main degradation mechanism induced by total dose in MOS devices is caused by radiation-induced charge buildup (N_{ot} and N_{it}) in its gate oxide. N_{ot} in thicker gate oxide can invert the channel interface causing leakage current to flow in the OFF state condition (V_{GS} = 0 V). This typically results in an increase in the static power supply current of an IC and may also cause IC failure. Large concentrations of N_{it} charge can decrease the carriers' mobility and increase the threshold voltage of n-channel transistors. For advanced ICs with thinner gate oxides such as CMOS technology, radiation-induced charge buildup in field oxides (LOCOS or STI) normally dominates the radiation-induced degradation of ICs and induces large leakage currents.

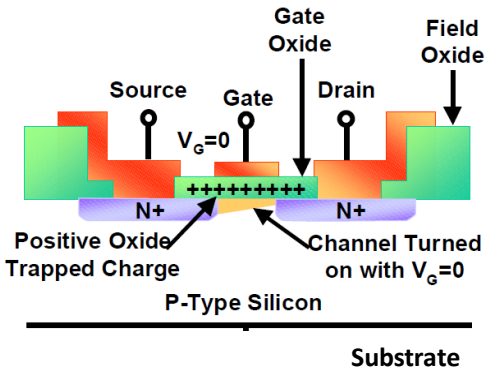


Fig. 9. Cross-section of an n-MOS device illustrating charge buildup in its gate oxide.

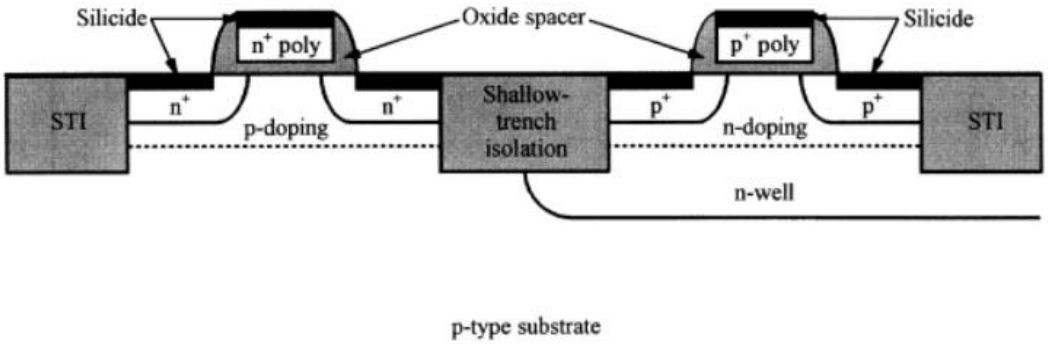


Fig. 10. Cross-sectional diagram for a generic deep-submicron bulk CMOS technology.

Effect of N_{ot} on I-V Characteristics

Assuming that gate oxides are thick enough, the primary impact of N_{ot} on the dc parameters of CMOS devices and ICs is the negative shift in the dc drain current vs. gate-to-source voltage (V_{gs}) for both n- and p-channel MOSFETs as illustrated in Figure 11. Figure 11 shows that for a fixed I_d, the radiation-induced buildup of N_{ot} shifts the V_{gs} bias point more negative (i.e., by ΔV_{ot}). In n-channel MOSFETs, this shift leads to a reduction in threshold voltage and an increase in off-state and drive currents. In p-channel MOSFETs, V_t increases while off-state and drive currents are reduced.

Radiation-induced dc voltage shifts can be calculated using the following equation:

$$\Delta V_{ot} = -\frac{t_{ox}}{k_{ox}\varepsilon_0} q\Delta N_{ot}, \tag{4}$$

where k_{ox} is the dielectric constant of SiO_2 and ε_0 is the permittivity of free space [34]. Given Eq. 1 and 3, the theory predicts that negative threshold voltage shifts caused by fixed oxide-trapped charge buildup are proportional to the square of oxide thickness, i.e.,

$$-\Delta V_t(N_{ot}) = -\Delta V_{ot} \propto t_{ox}^{\,2}. \tag{5}$$

This theoretical relationship has been verified through numerous experiments [21]. The relationship in Eq. 4 indicates that as the gate oxides of advanced CMOS technologies scale to thinner dimensions, the threat of shifts in dc parameters due to N_{ot} buildup in the gate oxide is reduced [23, 24]. Instead, hole trapping in the thicker STI dielectrics is now a greater radiation threat in modern CMOS technologies. Typical STI trenches are much thicker than gate oxides. For advanced CMOS technologies, the thicknesses range from approximately 300 nm to 450 nm [31].

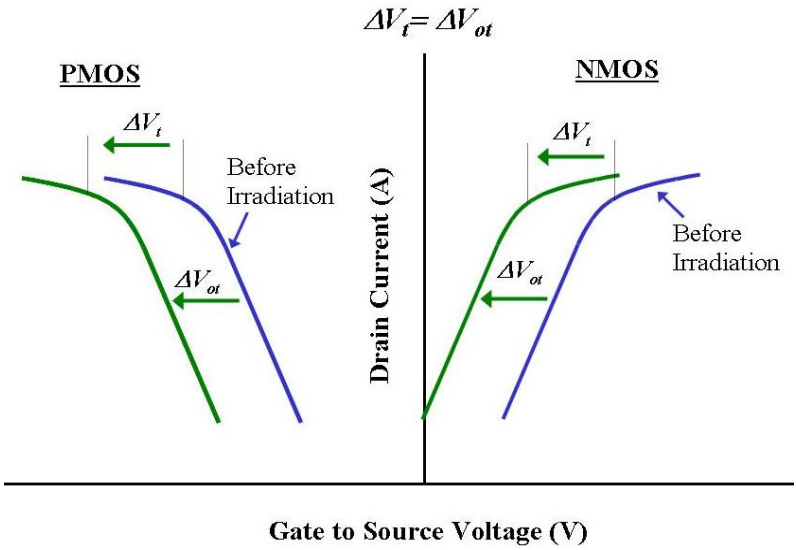


Fig. 11. Illustration of the effect of fixed oxide-trapped charge on n- and p-MOS devices.

Effect of N_{it} on I-V Characteristics

On the other hand, the primary effect of N_{it} is an increase in the sub-threshold swing of a CMOS device. This effect is illustrated in Figure 12, which shows a characteristic stretch out ΔV_{it} in the I_d vs. V_{gs} response for both n- and p-channel devices. The mechanism for this effect is the bias-dependent trapping or de-trapping of charge at the interface as the device surface is swept from accumulation to inversion by the gate voltage. Figure 12 shows that the threshold voltage is also impacted by N_{it} buildup. Also, depending on the pre-irradiation characteristics of the MOSFET, N_{it} buildup may also increase off-state drain current.

As stated above, switching oxide (border) traps can also change charge state as the dc gate-to-source bias is varied. As a result, in dc measurements, it is difficult to distinguish the effects of interface traps (P_b centers) and switching oxide traps (near interface *E'* centers). The key difference between the two defect types is that the charge exchange frequency at switching *E'* centers is low (<100 Hz) compared to P_b centers (>1 kHz) [31, 32].

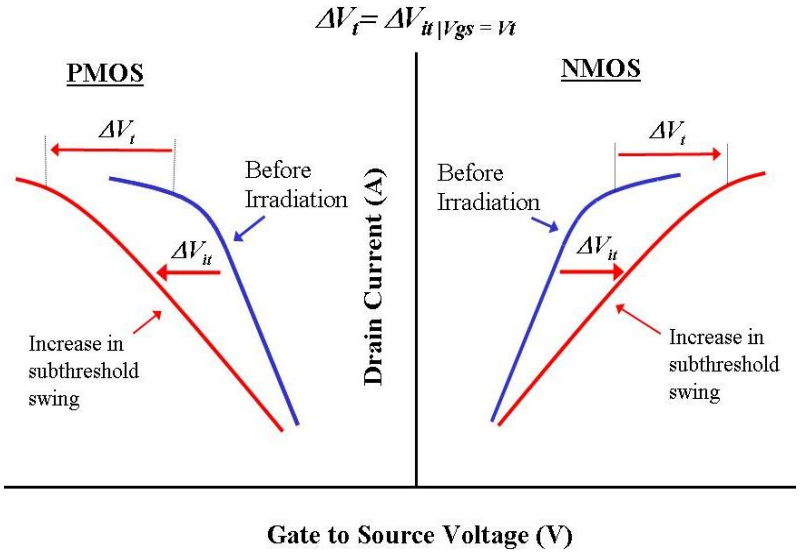


Fig. 12. Illustration of the effect of interface traps on n- and p-MOS devices.

3.1.1 Radiation Damage in Isolation Oxides

Defect buildup in gate oxides scales with t_{ox} (Eq. 4). Radiation tolerance in thin gate oxides is further enhanced by the increased likelihood of positive charge annihilation or compensation by tunneling electrons from the adjacent materials [35]. Inherent gate oxide hardness was demonstrated at the 0.25 μm technology node [45]. At this node, the oxide thickness is typically less than 6 nm, which is less than twice the approximate distance for high probability electron tunneling (i.e., ~3 nm) [12, 35]. Gate oxide hardness trends have continued to be observed in subsequent smaller technologies, e.g., 0.18 μm (t_{ox} = 3.2 nm) and 0.13 μm (t_{ox} ~2 nm) [37]. Defect buildup in thicker isolation oxides is typically the dominant cause of radiation-induced degradation in modern CMOS ICs. Figure 13 shows a representative cross-section of a modern n-channel MOSFET (n-FET). As the figure illustrates, the STI structures enclosing the active device are much thicker (by more than two orders of magnitude) than the gate oxide. In advanced CMOS technologies, the STI oxide thickness is typically greater than 300 nm.

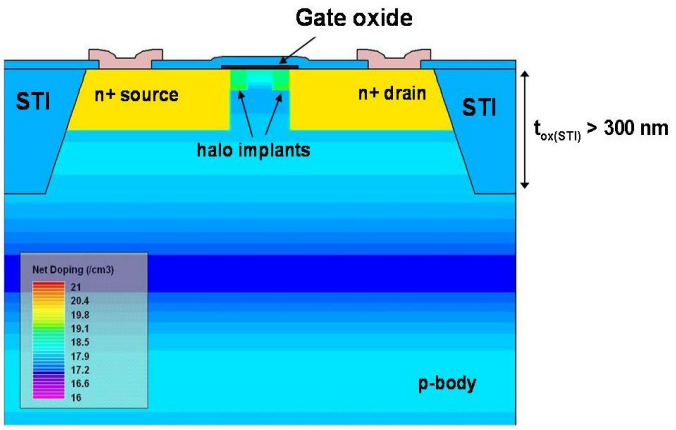


Fig. 13. Representative cross-section of an n-channel MOSFET in a modern CMOS technology.

Radiation Induces Leakage Path in CMOS

Radiation-induced degradation in STI oxides can cause a significant increase in the standby current in modern CMOS ICs. The causes of increased standby current are leakage paths created as a result of TID buildup. They include 1) drain-to-source leakage in a single n-channel MOSFET, 2) drain-to-source leakage between two different devices, and 3) source-to-well leakage between two different devices. The basic mechanisms for all field leakage phenomena are the same. Positively charged oxide defects invert an adjacent p-type Si layer, which enables the flow of current from one isolated region to another. Figure 14 plots the normalized increase in standby current in CMOS shift registers manufactured in a commercial 130-nm process [41].

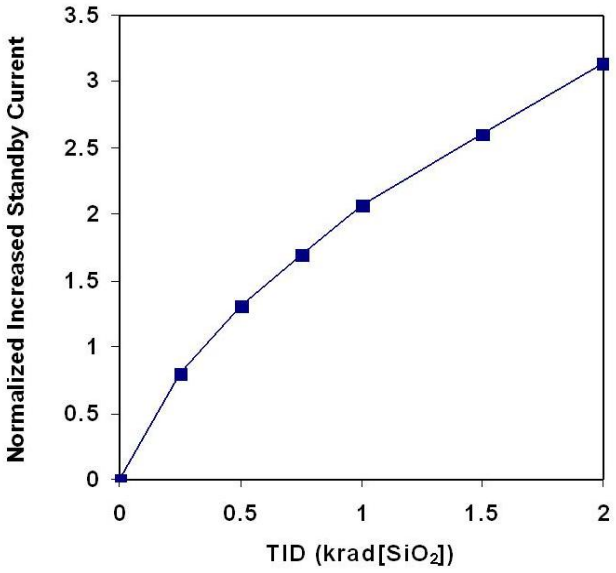


Fig. 14. Normalized increased standby current vs. total dose in shift registers fabricated in 130-nm process [42].

Charges trapped in the isolation dielectric, particularly at the Si/SiO_2 interface along the sidewalls of the trench oxide, create a leakage path, which becomes the dominant contributor to off-state drain-to-source leakage current in n-FETs [42]. This effect is illustrated schematically in Figure 15, which shows a) the edge leakage path from drain-to-source on the planer (top) view of the n-FET and b) the device cross-section with fixed oxide-trapped charge buildup in the STI [43].

The impact of STI radiation damage on the n-FET current-voltage characteristics of one deep sub-micron technology is shown in the experimental data in Figure 16 [37]. These data were obtained from TID measurements on 0.18-µm n-channel MOSFETs fabricated by the Taiwan Semiconductor Manufacturing Company (TSMC) [37]. The data show that off-state leakage current, $I_{d,off}$, exhibits a significant monotonic increase above 200 krad(SiO_2), reaching a level above 100 nA at 500 krad(SiO_2) of total dose [37].

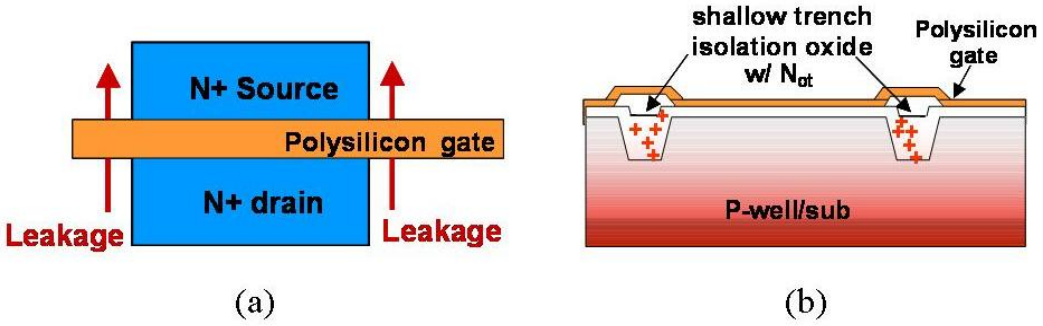


Fig. 15. a) Illustration of drain-source leakage path in n-FETs and **b)** its cause: oxide-trapped charge buildup in the isolation oxide [42].

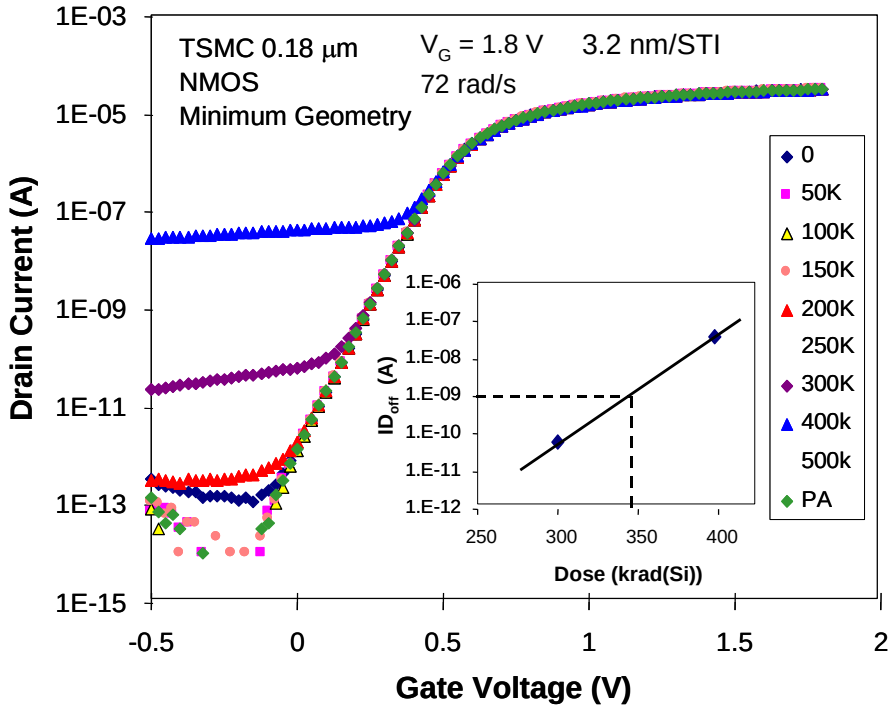


Fig. 16. Impact of STI radiation damage on the current-voltage characteristics of n-FET fabricated in TSMC 0.18-µm CMOS [37].

The primary cause of increased $I_{d,off}$ is the reduction in threshold voltage and increase in drive current in the parasitic n-FET formed along the two edges of the "as-drawn" device (Figure 17a). Prior to radiation exposure, the threshold voltage is high and drive current is low for the parasitic device relative to the as-drawn structure. Upon TID exposure, the parasitic device V_{th} is reduced significantly relative to the as-drawn transistor (Figure 17b). This is because the "gate" oxide of the parasitic structure is formed from the STI, which is much thicker than the as-drawn gate dielectric. In addition to negative voltage shifts, the drive current of the parasitic n-FET also increases significantly (Figure 17b). This is due to the fact that the effective width of the parasitic transistor ($W_{eff.}$), to which drive current is proportional, increases as surface along the STI sidewall inverts in response to N_{ot} buildup [43]. The degree to which ΔN_{ot} can invert this surface is also inversely proportional to the doping concentration along the sidewall [43]. Thus, a higher doping concentration in the p-type body will typically mitigate the effects of fixed oxide-trapped charge in the STI.

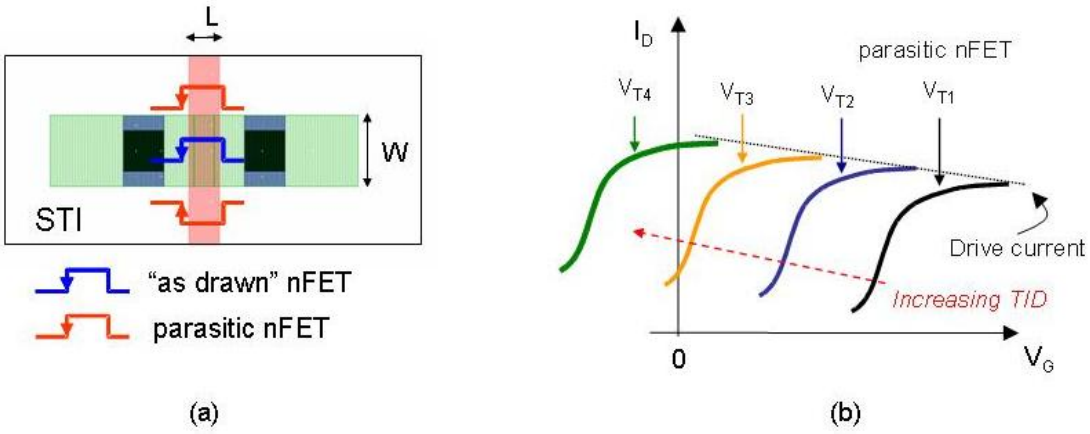


Fig. 17. a) Illustration of the circuit-level models associated with the n-channel MOSFET with parasitic n-FETs, and **b)** the effects of increasing TID radiation exposure on the threshold voltage and drive current of the parasitic n-FET [43].

Radiation and Scaling in CMOS Isolation Oxides

In order to examine the scaling trends of drain-to-source leakage, the pre- and post-irradiation I_d *vs.* V_{gs} characteristics of a commercial 130-nm CMOS were studied; the results are shown in Figure 18 [41]. These data indicate that negative voltage shifts in the parasitic edge devices are significantly smaller than the 180-nm TSMC technology (Figure 16). The increased radiation tolerance of the 130-nm technology may be due to the aggressive use of halo implants. In advanced CMOS technologies, highly doped halo implants (Figure 13) are used to suppress space charge effects (SCEs). Like the thinning of gate oxides, this deep-sub-micron processing technique seems to be fortuitously increasing the radiation tolerance of modern CMOS devices. At the 130-nm node, where the physical gate length is below 100 nm, the halo doping may extend across the entire channel for minimum gate length devices. This significantly increases the p-type doping concentration along the entire STI sidewall, between the drain and source of the n-FETs. The increased doping will inhibit the impact-fixed oxide charge, thereby increasing the inherent radiation hardness of ultra-small bulk CMOS devices.

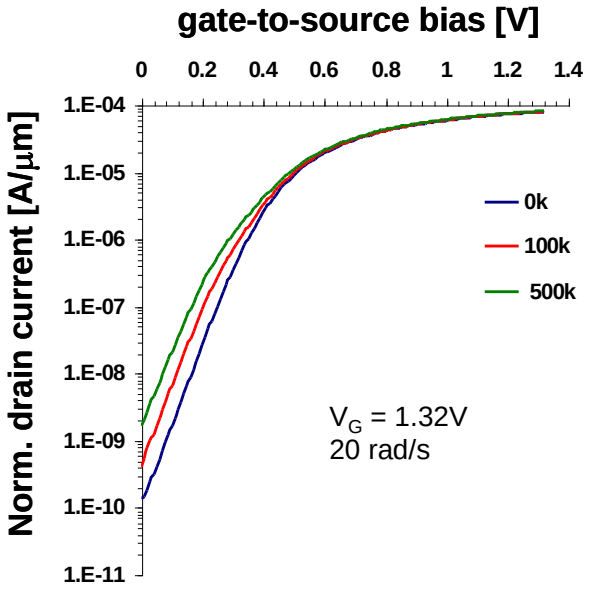


Fig. 18. Impact of STI radiation damage on the current-voltage characteristics of n-FET fabricated in 0.13 μm CMOS [41].

Unlike drain-to-source leakage, where the leakage path is associated with one n-channel device, inter-device leakage pertains to paths that are created between the n+ drains and sources of two adjacent n-FETs or between the n+ drain/source of one n-FET and the n-type well of an adjacent p-channel MOSFET. A schematic illustration of the n-FET device-to-device leakage mechanism is illustrated in Figure 19. Figure 20 shows a schematic illustration of the n-FET drain/source-to-n-well leakage mechanism.

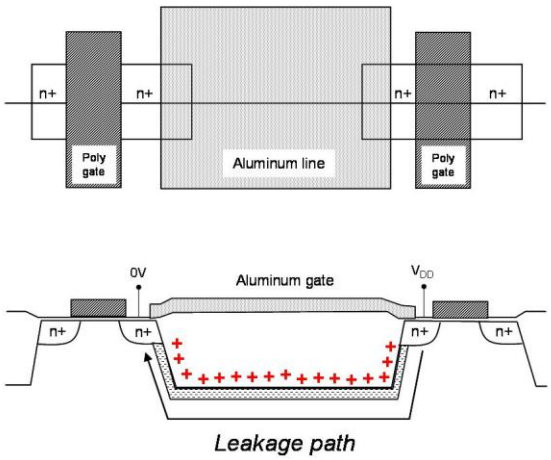


Fig. 19. Schematic illustration of n-FET device-to-device leakage [44].

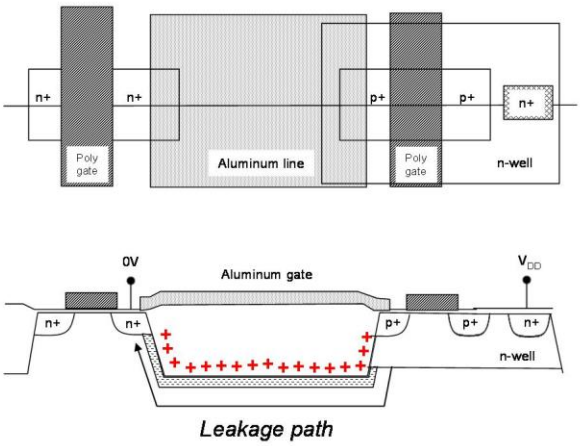


Fig. 20. Schematic illustration of n-FET drain/source-to-n-well leakage.

3.1.2 Time-Dependent Effects and Annealing

Time-dependent effects (TDEs) in MOS technologies are typically related to four mechanisms:

- Hole trapping process
- Interface traps formation
- Time dependencies of oxide trap annealing
- Time dependencies of interface trap annealing

These times vary with oxide thickness, oxide processes, oxide fields, and temperature [44-48]. Hole trapping is the fastest, followed by interface state growth, hole trap annealing, and lastly, interface trap annealing. The decrease in the magnitude of oxide-trapped charges generally follows a logarithmic time dependence and the rate at which N_{ot} (i.e., ΔV_{ot}) neutralized is dose-rate independent. As shown in Section 1, the neutralization of trapped charges mainly follows two mechanisms: 1) the tunneling of electrons from the silicon into oxide traps, and 2) the thermal emission of electrons from the oxide valence band into oxide traps. The rate and number of oxide traps neutralized by electron tunneling is highly dependent on the spatial distribution of the traps in the oxide, which is dependent on fabrication process. Traps with energy closer to the valence band will be neutralized more easily whereas traps farther from the valence band or in distance from the silicon interface will be less accessible. Unlike oxide-trapped charges, interface traps do not anneal at room temperature. Some interface-trap annealing at 100°C has been reported in several cases [44, 45]. However, higher temperatures are normally required to observe significant interface-trap annealing [44, 48]. To illustrate this latter point, Figure 21 shows an illustrative plot of the threshold voltage degradation for an n-channel MOS transistor during irradiation (anneal start at 1 Mrad) and anneal with V_{th} separated into shifts due to interface-trapped charge and oxide-trapped charge.

A more recent study led by Johnston [49], indicated that the annealing processes (i.e., tunneling and thermal) are still valid for most advanced CMOS technologies for which the TID degradation in isolation oxides is dominant. Indeed, the annealing behavior of the degraded leakage current in STI CMOS devices was converted in terms of threshold voltage recovery and compared to the gate and field oxides annealing behavior previously reported, similar to the data shown in Figure 22. Results clearly indicate that the annealing trends are very similar and that current annealing models seem to apply for advanced CMOS technologies as well. These results will have some good implication for total dose evaluations practices.

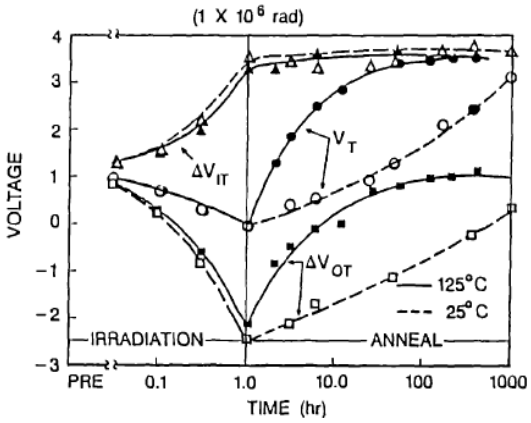


Fig. 21. Threshold voltage shift of an n-channel MOS transistor during irradiation and anneal with V_T separated into shifts due to interface-trapped charge and oxide-trapped charge.

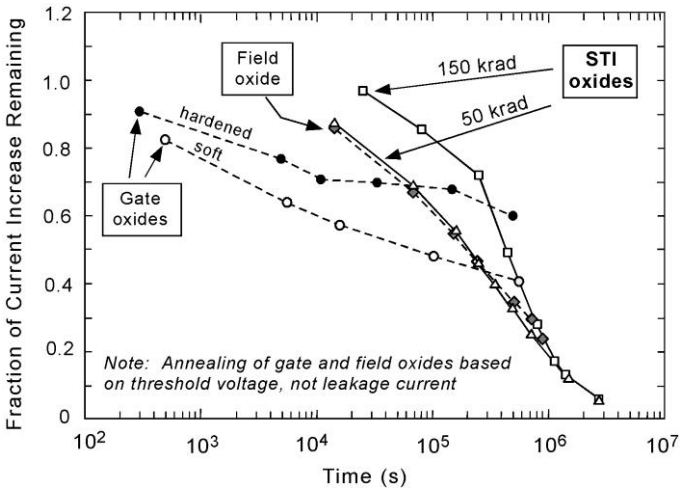


Fig. 22. Comparison between the annealing behavior of gate and field oxides with STI oxides based on threshold voltage, not leakage current [49].

KEY POINTS ABOUT TOTAL DOSE EFFECTS IN MOS DEVICES

* The main degradation mechanism induced by total dose in MOS devices is caused by radiation-induced charge buildup (N_{ot} and N_{it}) in its gate oxide. N_{ot} in thicker gate oxides can invert the channel interface causing leakage current to flow in the OFF state condition (V_{GS} = 0 V). N_{it} decreases the carriers' mobility and increases the threshold voltage of n-channel transistors.
* Threshold voltage shifts caused by fixed oxide-trapped charge buildup are proportional to the square of oxide thickness. For advanced ICs with thinner gate oxides (<6 nm), radiation-induced charge buildup in field oxides (LOCOS or STI) normally dominates the radiation-induced degradation of ICs and induces large leakage currents.
* Radiation-induced degradation in STI oxides can cause an increase in the standby current in modern CMOS ICs. The causes of this increase are leakage paths created as a result of total ionizing defect buildup. They include 1) drain-to-source leakage in a single n-channel MOSFET, 2) drain-to-source leakage between two different devices, and 3) source-to-well leakage between two different devices.
* TDEs in MOS technologies are critical to be considered for any total dose evaluation, particularly, for the dose rate selection and annealing consideration. TDEs are related to four mechanisms: 1) hole trapping process, 2) interface trap formation, 3) time dependencies of oxide trap annealing and 4) interface trap annealing. Hole trapping is the fastest, followed by interface state growth, hole traps annealing, and lastly, interface traps annealing. These times vary with oxide thickness, oxide processes, oxide fields and temperature.
* The annealing (i.e., compensation) of radiation-induced trapped holes (N_{ot}) in SiO_2 is a long-term process that is strongly dependent on temperature, applied electric field, and spatial distribution of the traps. It follows two mechanisms: 1) the tunneling of electrons from the silicon into oxide traps, and 2) the thermal emission of electrons from the oxide valence band into oxide traps. Interface-trapped charges do not anneal at room temperature, only at higher temperature. Compensation model applies for advanced CMOS STI oxides.

3.2 Total Ionizing Dose (TID) Effects in Bipolar Devices

Like MOS technologies, bipolar junction transistors (BJTs) are widely used within spacecraft systems. They are particularly important for analog-function, radio-frequency (RF), and mixed-

signal ICs. Much of the recent work on BJTs has focused on development of very high-speed transistors (f_T > 50 GHz) for RF-integrated circuits, particularly, BJTs based on silicon-germanium (SiGe) technology. SiGe BJTs have shown some very good total dose performance as well as very good electrical response at very low temperature [50].

The mechanisms of total dose degradation in bipolar technologies involve N_{ot} and N_{it} formation in SiO_2 as well [51-57]. Gain degradation is the primary effect with bipolar technologies. An illustrative plot of current gain vs. emitter-base voltage is shown in Figure 23. The top curve corresponds to the un-irradiated device, and the curves decrease monotonically with total dose. The decrease in current gain is particularly large at low bias levels, but the peak gain also decreases significantly.

In the space radiation environment, high-energy protons constitute a large part of the radiation environment. Protons cause displacement damage in addition to total-dose defects. The device-level effects due to these two types of damage are not simply additive; they interact in a complicated fashion [58]. It has been shown that proton irradiation can cause linear ICs to fail at much lower total-dose levels than can be tolerated for γ irradiation [59]. A comparison of heavy ions, electrons, and cobalt-60 γ rays showed that the heavy ions produce the most degradation at a given total-dose level because of the higher level of displacement damage compared to the other radiation sources [60]. When establishing the rule of a TID evaluation for radiation hardness assurance, the combined test of total dose and displacement damage effects must be taken into account.

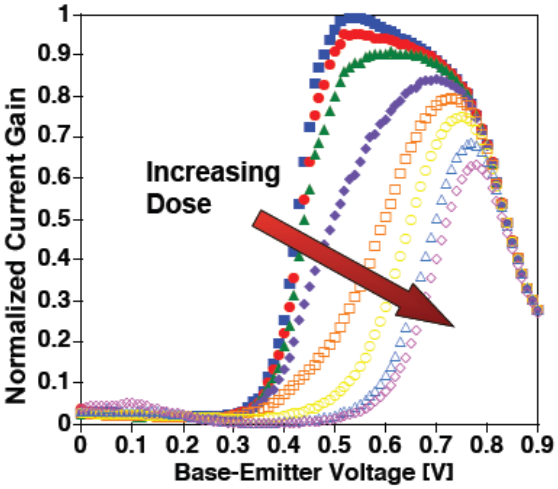


Fig. 23. Normalized current gain vs. base-emitter voltage for an *npn* BJT irradiated to various total doses.

3.2.1 Bipolar Junction Transistors Structures

The basic structure of a bipolar transistor consists of a three-layer sandwich of semiconductor material, with alternate layers having opposite doping types. The *npn* BJT is the most common bipolar transistor; it has two layers of *n*-type semiconductor separated by a thin layer of *p*-type material. In addition to having the doping types reversed in each layer, the *pnp* BJT typically has a different physical structure than that of the *npn* transistor because of the way in which it is fabricated. Figure 24 shows a cross-sectional view of a traditional *npn* BJT. The dashed line in this figure indicates the portion of the device that is most sensitive to ionizing radiation. The oxide layer of importance is typically a thick thermally grown oxide of poor quality (similar to field oxide in CMOS) compared to gate oxides in MOS devices.

Three basic *pnp* types of structures are used in ICs. Vertical *pnp* (VPNP) transistors are similar in structure to vertical *npn* (VNPN) devices. Lateral *pnp* (LPNP) transistors have the active region of the device at the silicon surface and the current flows laterally between emitter and collector regions that are both at the surface. Substrate *pnp* (SPNP) devices have a vertical current-flow pattern, but the substrate serves as the collector for the device. Figure 25 shows qualitative cross-sections of lateral and substrate *pnp* transistors. The physical structures of these two device types are very similar, but the current flow paths are different.

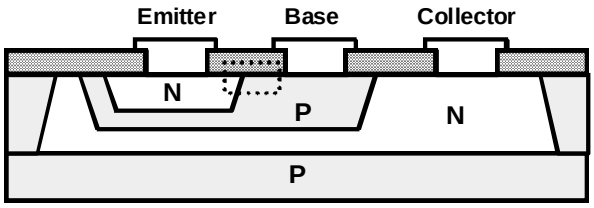


Fig. 24. Cross-sectional view of a conventional *npn* BJT. The dashed line indicates the portion of the device that is most sensitive to ionizing radiation.

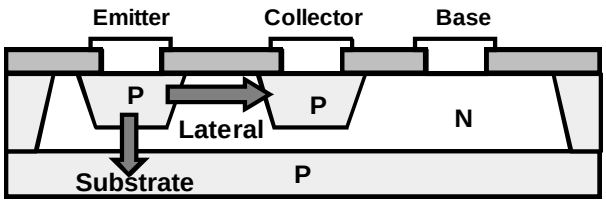


Fig. 25. Representative cross-sections of lateral and substrate *pnp* transistors. The arrows indicate the current-flow paths for each type of device.

The decreased current gain in irradiated BJTs is primarily a result of increased base current at a fixed emitter bias. Typically, the collector current remains roughly constant, although it may change somewhat [56, 61]. Figure 26 shows an example Gummel plot (log I_C and I_B vs. V_{BE}). For this device, the collector current remains unchanged except at very low bias levels, while the base current increases significantly. This causes a large reduction in the current gain, especially at low bias levels.

The mechanisms responsible for this increase in the base current result from the degradations of two out of three main components of the base current as illustrated in Figure 27: 1) I_{B1} is the current due to recombination in the emitter-base depletion region, 2) I_{B2} is the current due to back-injection of carriers from base to emitter, and 3) I_{B3} is the current due to recombination in the neutral base. For transistors that have not been irradiated, I_{B2} normally dominates. However, for devices that have been exposed to ionizing radiation, both I_{B1} and I_{B3} increase, with I_{B1} normally dominating. When the degradation is dominated by displacement damage, I_{B3} is often the current component that increases the most.

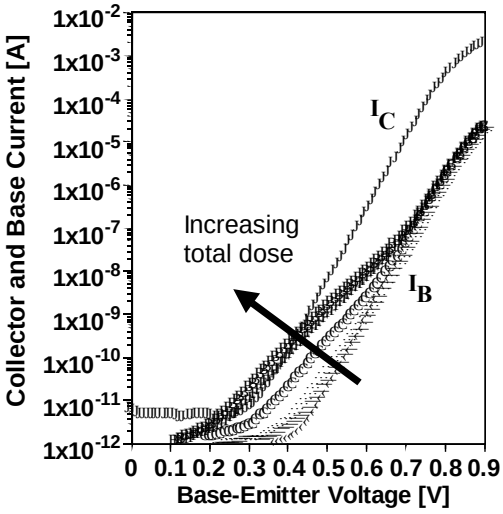


Fig. 26. Collector and base current vs. V_{BE} for an irradiated *npn* BJT.

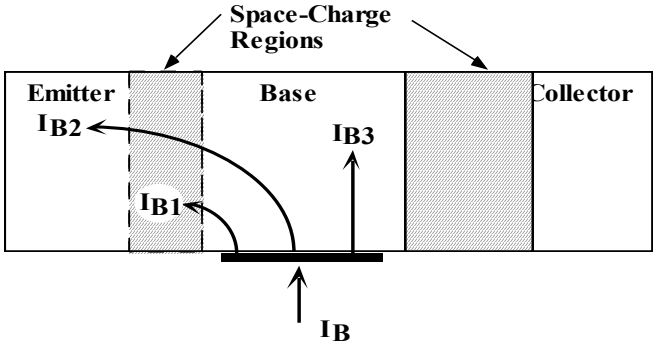


Fig. 27. Illustration of the different components of the base current in an *npn* BJT. The subscripts 1, 2, and 3 refer to recombination in the emitter-base depletion region, back-injection of minority carriers into the emitter, and recombination in the neutral base, respectively.

Equation 6 provides an expression of the excess-base current approximation induced by total dose that corresponds to an enhanced recombination in the emitter-base depletion region [53]. The recombination-rate increase occurs mainly where the depletion region intersects the Si/SiO_2 interface, due to N_{it} formation that serves as recombination centers. The increase of recombination centers is proportional to the surface recombination velocity at the Si/SiO_2 interface that covers the emitter-base junction.

$$J_{B1} \propto v_{surf} \exp\left(\frac{qV}{2kT}\right) \tag{6}$$

The recombination rate is a function of position along the surface and varies with total dose, exhibiting a peak where $n = p$, as illustrated in Figure 28. Four cases are represented: a) a recombination peak in a *pn* junction for an un-irradiated condition and a forward bias case, b) a recombination peak increase induced by N_{it} for a forward bias case, c) a recombination peak increase and extension of the depletion region induced by N_{ot} and N_{it} for a forward bias case, and d) an increase of generation term for a reverse bias junction. All of these conditions result in the Excess-base current and is proportional to the surface recombination velocity, which combines the effects of the different spatial locations at which the recombination takes place. The recombination rate increases more slowly with voltage than the back-injected (ideal) component of the base current. Its effect is greatest at low emitter-base voltages.

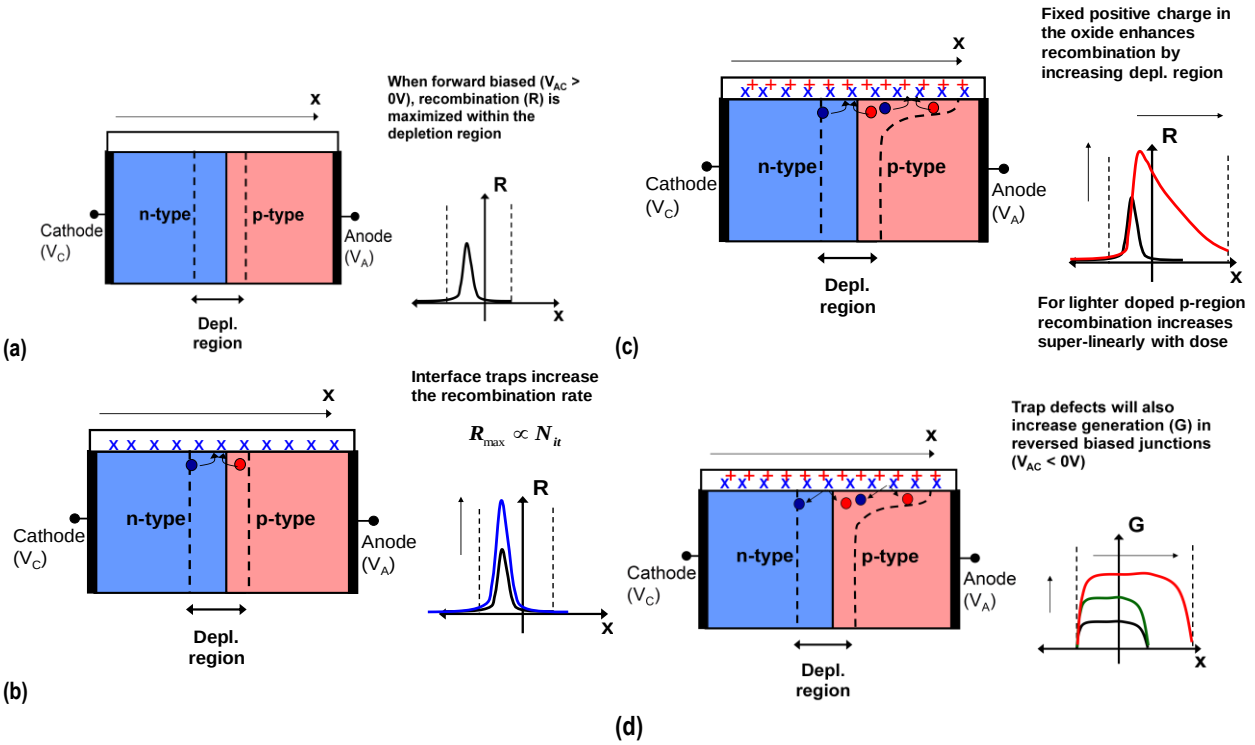


Fig. 28. Illustration of N_{it} and N_{ot} on recombination profile in the depletion region of a *pn* junction.

In the case of displacement damage, the recombination rate will increase throughout the depletion region, and not just where it intersects the surface. If the volume density of traps increases uniformly in the silicon, the peak recombination rate will always occur within the depletion region where $n = p$, as described above. The electrical effect of these traps can be described as a reduction in the minority-carrier lifetime. If recombination in the depletion region is dominated by bulk traps instead of interface traps, Eq. 6 becomes:

$$J_{B1} \propto \frac{1}{\tau} \exp\left(\frac{qV}{2kT}\right) \tag{7}$$

where μ is the minority-carrier lifetime in the depletion region.

Finally, TID can also affect the current due to recombination of minority carriers in the neutral base I_{B3}. Typically, pretty small for the un-irradiated condition, this current may become significant following irradiation (particularly for displacement damage). For instance, in a BJT operating in the forward-active region, the base current density required to supply the recombination process can be expressed as

$$J_{B3} = \frac{Q_B}{\tau_B} = \frac{qW_B n_{B0} \exp\left(\frac{qV_{BE}}{kT}\right)}{2\tau_B}, \tag{8}$$

where Q_B is the distribution of minority carriers in the neutral base, τ_B is the average lifetime of minority carriers in the neutral base, and W_B is the base width. When the device is irradiated, this component of the current increases as the lifetime in the base decreases. For ionizing radiation, the surface lifetime decreases due to N_{it} formation, but the bulk lifetime remains approximately constant. For displacement damage, the bulk lifetime decreases.

3.2.1.1 Radiation Dependence on the BJT Structure

VPNP transistors are usually relatively radiation-hard compared to VNPN devices [62, 63]. This is because the positive charge that accumulates in the oxide accumulates the *n*-type base region and tends to deplete the *p*-type emitter. However, the *p*-type emitter in VPNP transistors is heavily doped compared to the *p*-type base in VNPN transistors. In contrast, LPNP and SPNP transistors tend to be relatively soft [51, 64, 65]. The LPNP is commonly used as an active load or current source in microcircuits, such as operational amplifiers, voltage regulators, comparators, analog-to-digital (A/D) converters, and other analog IC types. LPNP transistors may degrade severely at low total dose levels [66, 67]. They are affected more significantly by ionizing radiation than SPNP transistors, since the current flow pattern in the LPNP devices is lateral and directly under the oxide where the recombination centers buildup after radiation, while the current flow path in the SPNP devices is vertical. The total dose required to degrade the lateral devices to half of their normalized current gain may be as much as 50 times less than that needed to observe current gain degradation of the same extent in vertical devices [67]. Figure 29 shows a plot comparing the base degradation for a fixed emitter-base voltage for the different BJT. It should be noted that this comparison varies with different bias levels and often BJTs that are biased at low levels are most sensitive to radiation-induced degradation.

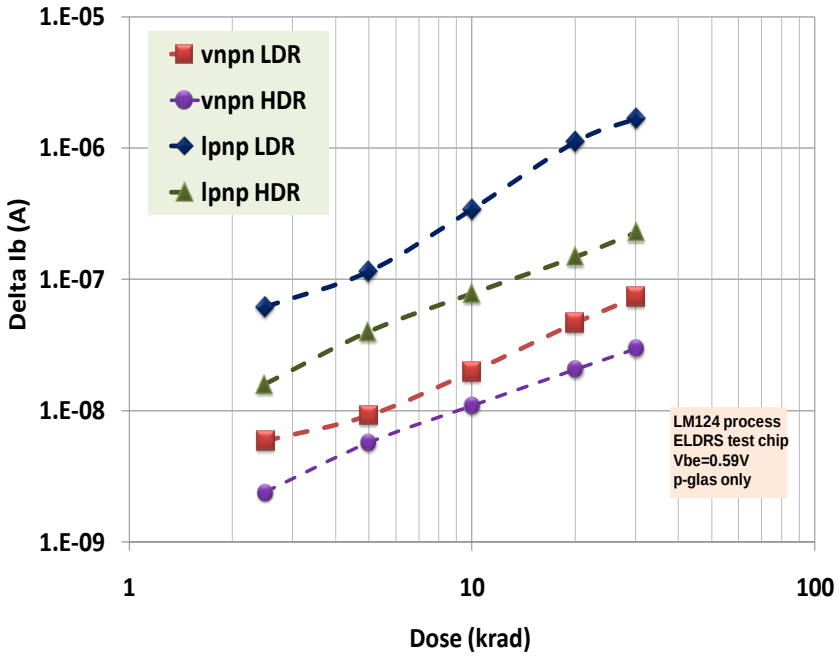


Fig. 29. Normalized current gain vs. base-emitter voltage for an *npn* BJT irradiated to various total doses.

3.2.2 Circuit Effects

While it is important to understand the mechanisms of degradation in the three types of bipolar structures, it should be noted that linear devices built with these structures exhibit total dose parametric degradation trends that are more complex and vary with bias, dose level, and dose rate [68-77]. Several studies have been performed to identify the circuit interactions that lead to the observed terminal response [68-77]. For almost all cases, the degradation of the circuit parameters is a result of the gain degradation of the various *npn* and *pnp* transistors in the circuit. For some circuit parameters, such as input bias current and output sink/source current, the degradation may be proportional to the gain degradation of a single transistor, in which case, the analysis is simple. The greatest total dose and dose rate effects occur when the input or output transistor is a simple LPNP or SPNP. Recent integrated linear circuits have improved their design and use combinations of *npn* and *pnp* transistors in the input stage circuitry for compensation, helping to extend their operation over a wide range of conditions while maintaining high input resistance, low offset voltage, and low offset current [76-77]. Since the *npn* and *pnp* transistors degrade differently as a function of both dose and dose rate, the degradation interactions can be complex. A first example is shown in Figure 30, where the change of input bias current as a function total dose of an OP-97 is plotted. The degradation shows a non-linearity that has been attributed to circuit effects [77]. Indeed, the OP-97 uses a compensated input stage where a current source (comprised of an LPNP current mirror) nearly compensates the base current of the *npn* transistors used at the input stage, reducing the input bias current while allowing the *npn* transistors to operate at relatively high current. Although this compensation technique reduces input bias current by several order of magnitude compared to old technologies, it shows a non-linear response with total dose due to the reversal of input current sign when the LPNP degrades sufficiently and reaches a threshold gain value. It has been shown that this threshold is power supply (V_{cc}) dependent and that the total dose level at which the non-linearity occurs varies as a function of V_{cc}. This is not too surprising since current sources are directly dependent on V_{cc} [77].

Figure 31 shows a second example of circuit effect, in which the output voltage degradation of an LM117 negative voltage regulator may start positive and then become negative at HDR, and start negative and then become positive at LDR [72]. The changes are related to circuit-level compensation of effects in one part of the circuit by effects in another part of the circuit, in this case, a current source.

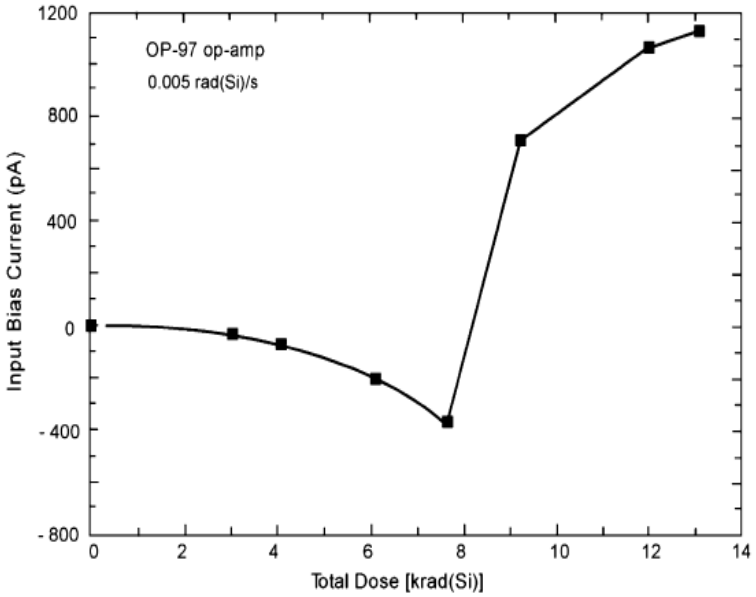


Fig. 30. Degradation of input bias current of an OP-97 op-amp showing the extreme non-linearity that occurs for compensated input stages [77].

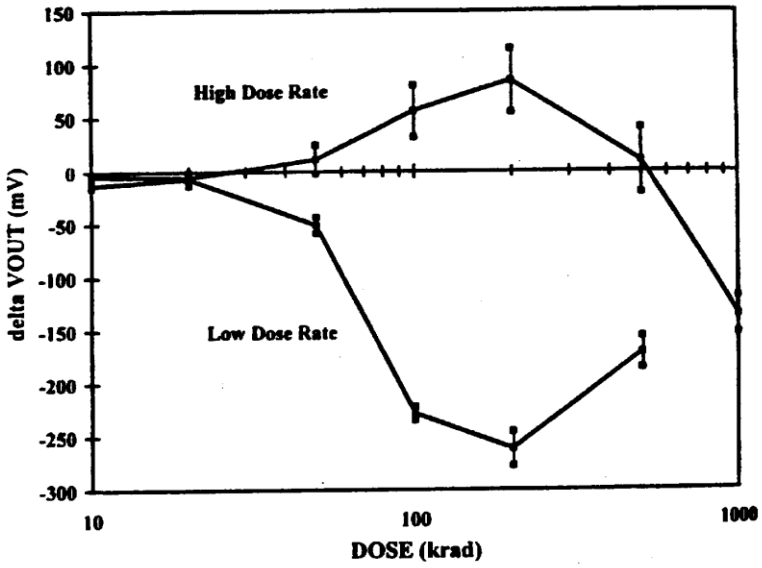


Fig. 31. Change in output voltage for an LM117 voltage regulator versus dose at HDR and LDR [72].

3.2.3 ELDRS in Bipolar Technology

Linear integrated circuits built in bipolar technologies are exposed to ionizing radiation at very low dose rates when used in space systems. With these conditions, their parametric degradation, more noticeable for dc parameters, exhibits a "true" dose rate effect known as enhanced low-dose-rate sensitivity (ELDRS) [78-81]. ELDRS is characterized by an LDR enhancement factor (EF) defined as the ratio of the parametric degradation at an LDR to the degradation at an HDR for a fixed dose level. EF is not a universal number and varies for different part types and for different parameters within the same part. This approach to characterize the dose rate response of bipolar devices and circuits as an EF versus dose rate was first used by Johnston et al. [80] and is illustrated in Figure 32. To demonstrate the ELDRS phenomena, it shows the input bias current degradation (or base current at a fixed collector current for the discrete transistors) relative to the degradation at 50 rad/s, as a function of dose rate for a fixed total dose level. Several linear circuits from different manufacturers were studied, the LM108 (op amp), LM101 (op amp), and LM111 (comparator) from National Semiconductor Corporation (NSC), and an LM324 (quad op amp) from Motorola. Several discrete bipolar transistors were also included. The number on the vertical scale is the LDR EF as a function of dose rate. The EF is greatest for the LM111 and LM324, both of which have LPNP/SPNP input transistors. The LM101 has VNPNP input transistors and the LM108 has super-beta *npn* input transistors.

To demonstrate that ELDRS is a real effect in space, an experiment was also run under the Microelectronics and Photonics TestBed (MPTB) program using an LM139 quad comparator. The experiment ran for approximately seven years, and the parts accumulated ~45 krad(Si), with most of the dose received in the 1-hour flight through the radiation belts during each 12-hour orbit [82-83]. This corresponds to an average dose rate of approximately 2.5 mrad/s. The results are shown in Figure 33, comparing various ground test data at constant dose rate to the space data. Although the space environment consists of protons and electrons of various energies and widely changing dose rates, the degradation falls between the degradation at a constant dose rate of 1 mrad(Si)/s and 10 mrad(Si)/s using Co-60. The results at 10 and 100 rad(Si)/s show much less degradation than in space; the elevated temperature irradiation at 100°C and 1 rad(Si)/s falls in between. Note that the input bias current that started at 25 nA reached 400 nA, a potential problem for some system applications. The pre-irradiation specification limit for this parameter is 100 nA.

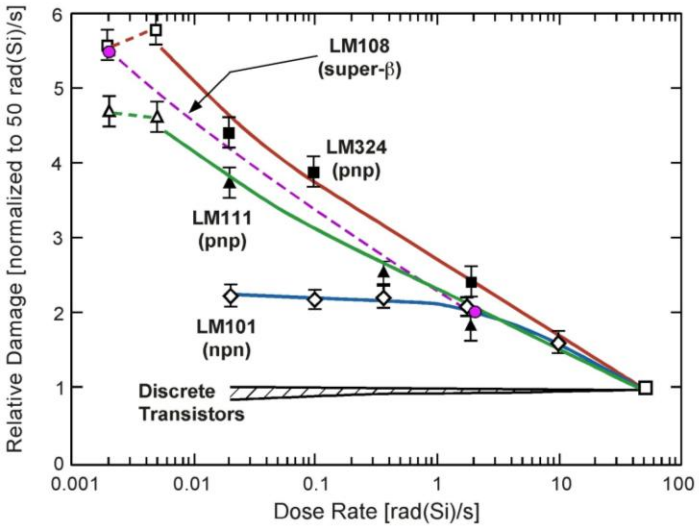


Fig. 32. Relative damage (EF) versus dose rate, normalized to 50 rad(Si)/s, for irradiation of several circuit types and discrete transistors to a fixed unspecified dose [80].

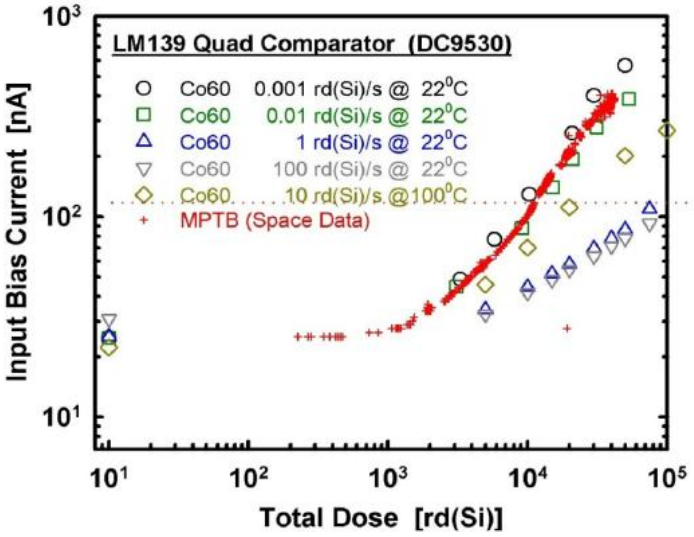


Fig. 33. MPTB experiments comparing space data with ground ^{60}Co data.

3.2.3.1 Characteristics of ELDRS

Research studies have indicated that the ELDRS response depends on a number of factors, such as final passivation (or processing tools) [82–84], pre-irradiation thermal stresses during burn-in or packaging [85]. However, the exact factors that determine the maximum EF and the dose rate where the transition between HDR and LDR response occurs has never been determined precisely. The largest LDR EFs that have been observed occur in bipolar linear circuits that incorporate LPNP and SPNP transistors [82]. Additional studies have shown that a small amount of H_2 can have a significant effect on the EF and LDR response and that 100% hydrogen can cause significant degradation at HDR and LDR [87]. The main characteristics of ELDRS listed below are described briefly in this subsection; hydrogen is described in detail in Section 4.

- True dose rate effect (TDRE) versus time-dependent effect (TDE)
- Bias conditions
- Passivation
- Pre-irradiation elevated temperature stress (PETS)
- Molecular hydrogen contamination

True Dose Rate Effect versus Time-Dependent Effect

When ELDRS was first discovered, it was observed as a "true" dose rate effect (TDRE) and not a time-dependent effect (TDE) often observed in CMOS. Presence of a TDRE means that the degradation at the end of an LDR irradiation is greater than the degradation measured after irradiation to the same dose at HDR followed by a room temperature anneal (RTA) for a time at least as long as the irradiation time at LDR and for the same bias conditions. Presence of a TDE means that the degradation at HDR and LDR is essentially the same when measured at the same time from the beginning of irradiation (including anneal time). The LDR EF shown in Figure 34 includes both the contribution from the TDRE and TDE. In this figure, an LM111 comparator has been subjected to a 175°C, 300-hour pre-irradiation elevated temperature stress (PETS), then irradiated to various total dose values at 50 rad(Si)/s and 10 mrad(Si)/s. Following each set of HDR irradiations, the samples were annealed at room temperature for the same amount of time as the LDR irradiation to the same dose to determine the TDE. The results show that above 10 krad(Si), there is a TDRE as well as a TDE component of the total EF.

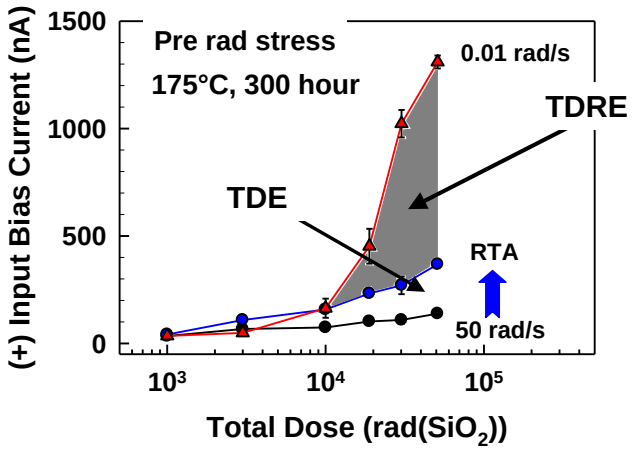


Fig. 34. Input bias current of LM111 versus dose for HDR and LDR irradiation plus RTA after HDR. Parts subjected to PETS of 175°C for 300 hours prior to irradiation [89].

Unfortunately, in the literature, almost all of the data on bipolar linear circuits as a function of dose rate are only shown as the result immediately after the irradiation and not as a function of time, so the EF cannot be separated into components of TDRE and TDE. This could have some implications when defining the rules for total dose evaluations.

Bias Conditions

Many of the experiments examining ELDRS are performed under a single irradiation bias condition, either with nominal or maximum supply voltage (V_{CC}) or with all leads grounded or shorted. The rationale for using the no bias condition is that the ELDRS effect occurs under very low oxide electric fields [92]. It has been shown that for many sensitive electrical parameters (e.g., input bias current, output source/sink current, and reference or output voltage, no bias is the worst case [93, 94]). However, there are other parameters, such as offset parameters, that are more sensitive to LDR effects with irradiation bias. In addition, because of circuit effects associated with degradation, it is recommended to always perform ELDRS testing both with and without bias. This testing practice with and without bias at HDR and LDR is incorporated in the ELDRS characterization test in MIL-STD-883/TM1019.8. Figure 35 illustrates a case (low dropout regulator) where the worst-case bias at LDR is with no bias.

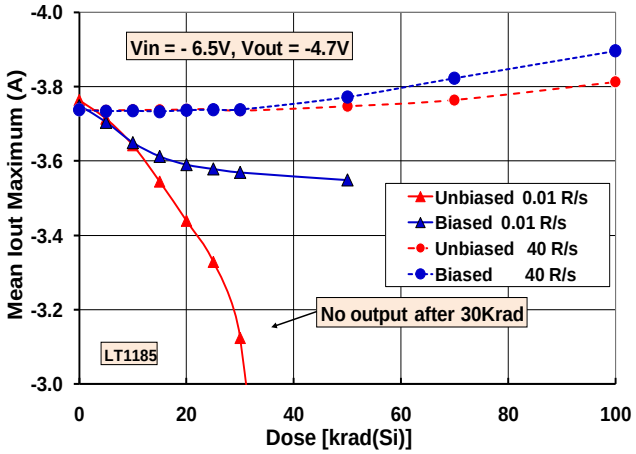


Fig. 35. Maximum output current of Linear Tech LT1185 versus dose for two dose rates and two irradiation bias conditions [93-94].

ELDRS and Passivation

A study led by Shaneyfelt in 2002 [102] showed that the final passivation had a significant impact on ELDRS. The study was performed on NSC LM111 comparators, which had the nitride passivation removed by chemical etching. Figure 36 shows the results for the LM111 for input bias current versus dose for parts with and without the nitride passivation at HDR and LDR.

The results clearly demonstrate that removing the nitride eliminates ELDRS. An additional study was done with LM139, in which a wafer lot was fabricated with various final passivation splits, including the standard nitride over phosphorous glass (p-glass), no passivation, p-glass only, and a new nitride over p-glass process to reduce the hydrogen content in the oxide. The results were that both nitride-passivated parts degraded severely at both HDR and LDR, the p-glass part showed ELDRS, and the parts with no passivation were hard at HDR and LDR [102].

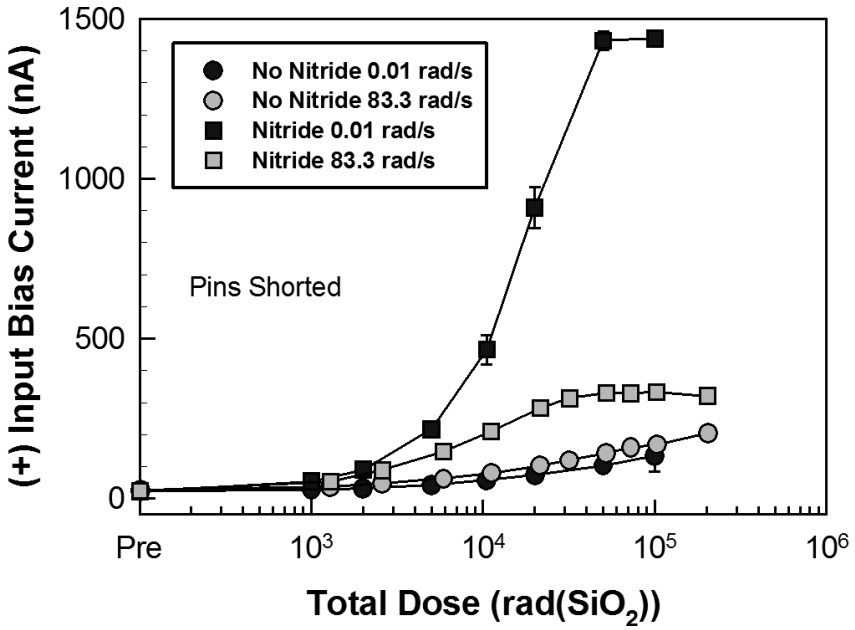


Fig. 36. NSC LM111 input bias current vs. dose for two dose rates for parts with nitride passivation and with the nitride removed [102].

PETS and ELDRS

PETS can also have a major impact on the total dose and dose rate response of bipolar linear circuits. The term PETS has been used to include any exposure to elevated temperature that occurs after processing, and before irradiation. This can occur during packaging, where parts are subject to short temperature cycles with temperatures as high as 450°C to simulate die attach or lid seal processes or power burn-in, which is often done at 125–150°C for 168 hours. The effect of PETS on the total dose response can result in either more or less degradation depending on the temperature and time. The increased degradation usually peaks at a time for each pre-irradiation temperature that decreases as the temperature is increased. For pre-irradiation temperatures as high as 450°C, this time may be as short as a few minutes, and for temperatures of 150°C, it may be many tens of hours. For a temperature of 175°C, it is shown in [91] that pre-irradiation bakes can eliminate the true dose rate effect in the NSC LM111 for PETS times longer than 10 hours, as illustrated in Figure 37.

In Figure 37, on the left (Virgin Devices), degradation is shown following irradiation to 100 krad(SiO_2) at 50 rad(SiO_2)/s and 0.1 rad(SiO_2)/s as well as the result for an RTA of 10^6 seconds following HDR irradiation. These results illustrate that the EF consists of both TDRE and TDE. The horizontal axis shows the pre-irradiation stress time at 175°C. While the TDE remains relatively constant, the TDRE is seen to go away after approximately 10 hours. After 4×10^6 seconds, the degradation at LDR is actually less than at HDR plus RTA.

The effects of PETS for higher temperatures and shorter times, as might be seen in some packaging steps, are shown in Figure 38 [102]. In Figure 38, the increase in input bias current at HDR (83.3 rad(SiO_2)/s) for the NSC LM111 is shown for parts with the nitride passivation and for parts with the nitride removed by wet etch. PETS for 200 seconds at 250°C and 450°C causes a significant increase in the degradation of the parts with nitride, whereas it has little effect on the parts with nitride removed. The peaking of the degradation and subsequent decrease at high doses is explained as a competition between degradation in the input LPNP and SPNP transistors and the *npn* transistors in the input stage [103]. These results would seem to imply that, since the parts with nitride have a much higher concentration of hydrogen, the PETS causes a redistribution of the hydrogen, a change in the way the hydrogen is incorporated in the oxide, or a change in the defects that act to release the hydrogen in the form of protons. In addition, these temperatures are high enough to break Si-H and N-H bonds to form molecular hydrogen.

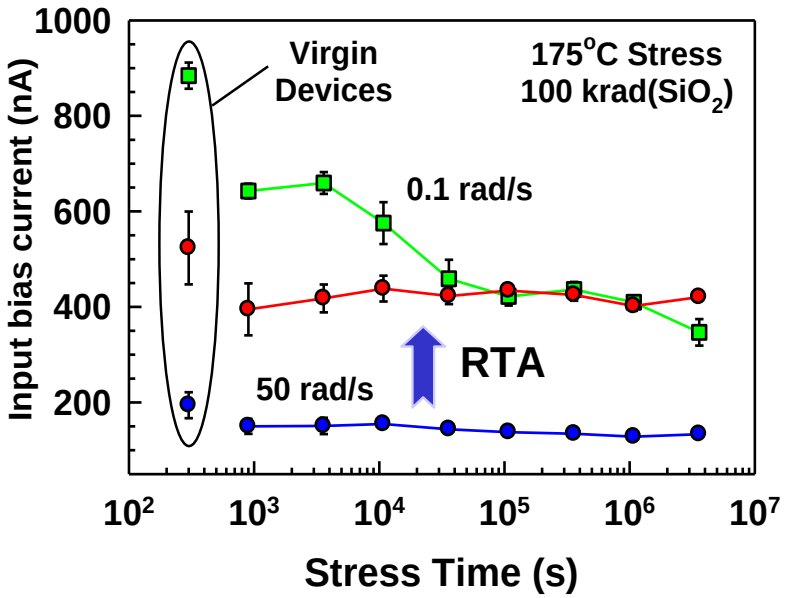


Fig. 37. Input bias current of LM111 after 100 krad(SiO_2) at HDR and LDR versus pre-irradiation stress time at 175°C [91].

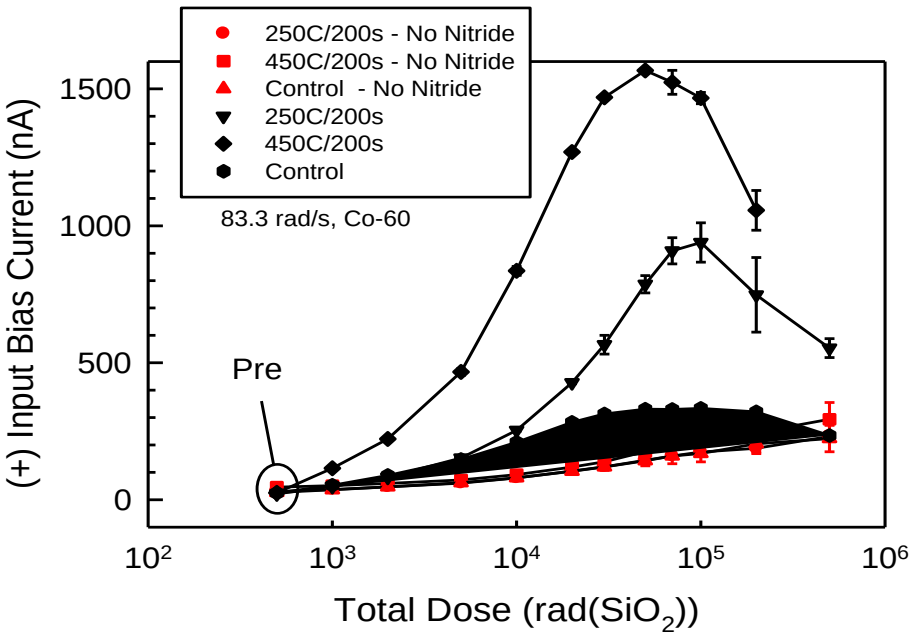


Fig. 38. Input bias current of LM111 vs. dose for parts with and without nitride subject to PETS of 250°C and 450°C for 200 seconds [102].

KEY POINTS ABOUT TOTAL DOSE EFFECTS IN BIPOLAR LINEAR CIRCUITS

* The mechanisms of total dose degradation in bipolar technologies involve N_{ot} and N_{it} formation in SiO_2. Gain degradation is the primary effect with bipolar technologies and is particularly large at low bias levels.
* The gain decrease is due to an increase in the base current resulting from the degradations of the current due to recombination in the emitter-base depletion region and the current due to recombination in the neutral base. Both are affected because: 1) base surface current increases due to oxide-trapped charges, $N_{ot,}$ and interface traps, N_{it}; and 2) N_{ot} changes surface potential at base surface and N_{it} increases surface recombination velocity.
* LPNP and SPNP transistors degrade much more than *npn* transistors because of surface current flow (directly under the oxide where the recombination centers buildup after irradiation) and lower doped base region.
* Bipolar technologies are sensitive to ELDRS and their electrical degradation varies with final passivation (or processing tools), pre-irradiation thermal stresses during burn-in or packaging.
* Bipolar ICs see some circuit effects associated with their total dose and dose rate responses. Non-linearity occurs with most recent designs that involve new compensation circuit scheme at their input stage. Because of the dose rate dependence, it is recommended to always perform ELDRS testing with and without bias at HDR and LDR.

4.0 IMPACT OF HYDROGEN CONTAMINATION ON TID RESPONSE OF MOS STRUCTURES

Now that we have explained the basic mechanisms and electrical degradation for SiO_2-based devices, the next sections focus on how residual H_2 present in hermetic packages influences the total dose and dose rate response of linear bipolar circuits (i.e., the transition dose rate curve and the EF [95-100]). Hydrogen contamination is an involved problem since device type, passivation, packaging processes, gate structure/metallurgy, and operating ambient temperature, all can impact the sensitivity to the hydrogen concentration level.

These studies indicate that there are several likely sources of H_2:

1. In many cases, piece parts are baked or fired and then sealed in a reducing environment of forming gas containing >4% H_2 in N_2.
2. H_2 can also outgas from grain boundaries or structural imperfections in iron-nickel alloy (kovar, Alloy42) lead-frame materials.
3. Electro-plated metal components such as plated gold or nickel films are major sources of dissolved hydrogen; H_2 can be trapped during this process.
4. Moisture is often present and results from the absorption or adsorption of H_2O on the internal surfaces of the package prior to sealing or from the sealing gas itself that is moist.

Solutions to hydrogen contamination have been proposed and include thermal treatment, the use of package materials with low hydrogen absorption, a change of barrier materials in gates, and the use of hydrogen getters inside the packaging to absorb the hydrogen. However, the problem still exists and there is no clear guideline or limit as to what level of hydrogen might be considered acceptable in hermetically sealed packages. H_2 concentration levels as high as 4% have been detected in hermetic packages through residual gas analysis (RGA) [97].

The military standard test method for internal gas analysis, MIL-STD-883 Test Method 1018, was designed to look for moisture and not hydrogen or other gas impurities. There is no specification limit on H_2. In addition, recent research findings about modeling ELDRS have shown that hydrogen transport and reactions play a key role in the radiation response and long-term reliability of bipolar microelectronics technologies [96, 98]. Due to its high mobility, hydrogen may diffuse into SiO_2 over-layers and participate in the creation of interface traps upon ionizing radiation exposure, thereby enhancing the total dose and dose rate response of bipolar devices. Modeling of ELDRS is still under investigation; however, the main mechanisms that involve hydrogen reaction and cracking have been identified [96, 98].

4.1 Experiments Showing the Correlation between Hydrogen, Packaging, and Dose Rates

The first set of experiments that relate to hydrogen contamination and its effects on the radiation response of bipolar devices started with radiation experiments conducted on specially designed gated-lateral *pnp* (GLPNP) bipolar fabricated in NSC's standard linear bipolar process and packaged in hermetically sealed packages. These transistors, shown in Figure 39, were part of an ELDRS test chip taken from a single wafer lot, designed to study the buildup of oxide-trapped charge and interface traps with NSC linear circuit technology. This special wafer lot was fabricated with several different combinations of the final passivation [99].

In the first set of experiments, measurements were conducted on samples with five different package conditions: 1) hermetically sealed with 1% in package H_2, 2) lid/seal removed but with metal lid taped back on, 3) lid/seal removed but with ceramic lid taped back on, 4) never-sealed package with taped-on lid. Figure 40 shows the radiation response of these devices after

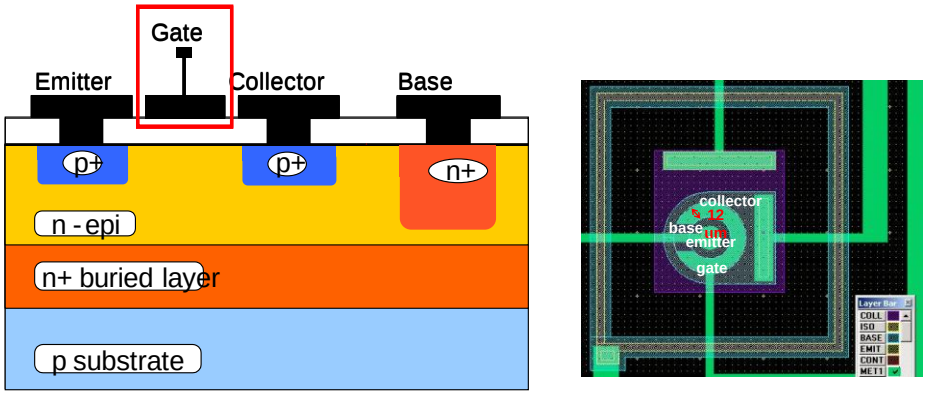


Fig. 39. GLPNP 2-D cross-section and layout.

30 krad[Si] of total dose exposure. Time-dependent experiments were also conducted to study the diffusion of hydrogen over time from the packages as the package lids are removed. Data on Figure 41 show the decrease in radiation damage as H_2 is diffused out of the package.

Further experiments were conducted on GLPNP devices with different ambient concentrations of H_2; the relationship between excess ambient H_2 concentration and radiation-induced interface trap density (N_{it}) was established using analytical modeling [3]. The experimental results in Figure 42 shows monotonic increase of both radiation-induced interface trap and oxide-trapped charge formation as ambient H_2 concentrations are increased.

A first study conducted by Crane, Raytheon, Boeing, and TRW (now Northrop Grumman) shows that the Analog Devices AD590 temperature transducer exhibits ELDRS when irradiated without bias and when packaged in a flatpack (FP). The part is available in both a FP and a can. Several lots of both cans and FPs were evaluated and the FPs all exhibited ELDRS whereas the cans were hard at both HDR and LDR. After seeing the results on the NSC GLPNP ELDRS test chip, it was suggested that an RGA be run on the AD590 FPs. The RGA showed that the FP parts had on the order of 0.6% of H_2. This amount of H_2 was apparently enough to cause the FP parts to degrade significantly at LDR under zero bias. Tests run on the AD590 with 100% H_2 did not show any greater degradation than that of parts with 0.6% H_2, indicating a saturating effect with the smaller amount of hydrogen. However, these results were obtained with parts from different diffusion lots and, therefore, the claim of a causal relationship between H_2 and total dose

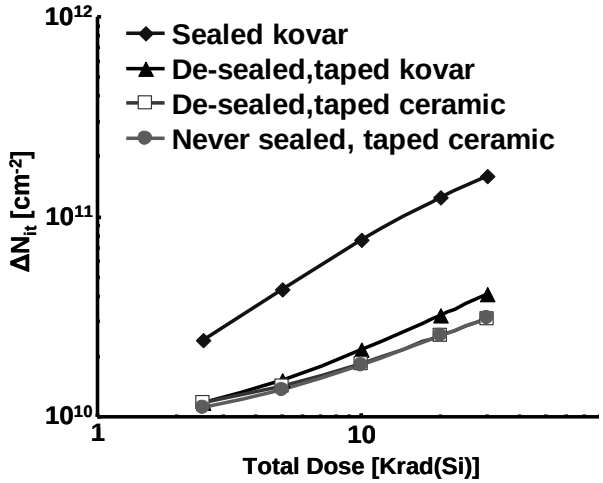


Fig. 40. Radiation induced interface trap build-up in devices with unsealed and hermetically sealed packages.

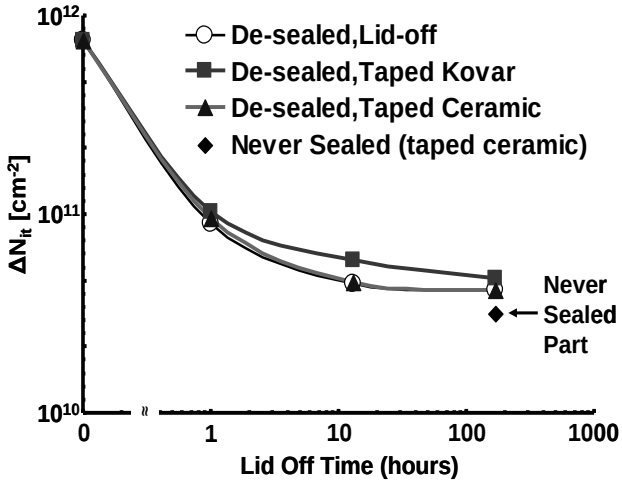


Fig. 41. Decrease of radiation induced interface traps as H_2 diffuses out of the sealed package over time.

sensitivity needed to be confirmed. As a result, additional experiments were performed on another twelve space-qualified AD590s from the same diffusion lot provided by Analog Devices. Parts were irradiated at HDR and LDR for comparison: 1) three FPs and three cans were irradiated up to 30 krad(Si) at a LDR of 0.01 rad(Si)/s; 2) three FPs and three cans were irradiated up to 100 krad(Si) at a HDR of 25 rad(Si)/s. Varying levels of H_2 were detected through RGA, respectively, 0.2%, 0.27%, and 0.88% of H_2 for the FP ceramic packaged devices, whereas 0% was detected within the cans devices. These results are not surprising because the FPs had several potential sources of hydrogen such as gold/nickel/tungsten metallization and kovar lid with gold-nickel plating. The cans did not show any hydrogen content because of their seal process (i.e., a welded process in which parts are sealed in dry air preventing the introduction of hydrogen). All irradiated parts were unbiased during irradiation with all leads grounded or in conductive foam.

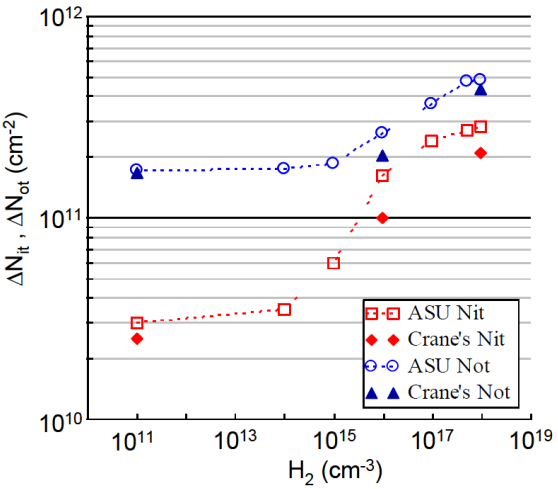


Fig. 42. Radiation-induced oxide-trapped charge (N_{ot}) and interface-trap (N_{it}) buildup.

The output current (or error in temperature) was recorded for each dose step with a V_{cc} of 20 V. The results are shown in Figure 43 and reveal that FP devices degrade significantly more than the cans at LDR and HDR when irradiated unbiased. At HDR, the degradation in the output current (or error in temperature) tracks the increase in hydrogen content (0.2%, 0.27%, and 0.88%) and, at LDR, the impact of hydrogen is even worse on the degradation (see 0.33% and 1.04% of H_2 for the LDR case). An additional data set was obtained by irradiating another device packaged in a can with 100% hydrogen at HDR to make it degrade similar to the FP devices. This latter data set suggests that an LDR-accelerated testing methodology could be developed by using hydrogen soaking at HDR [97].

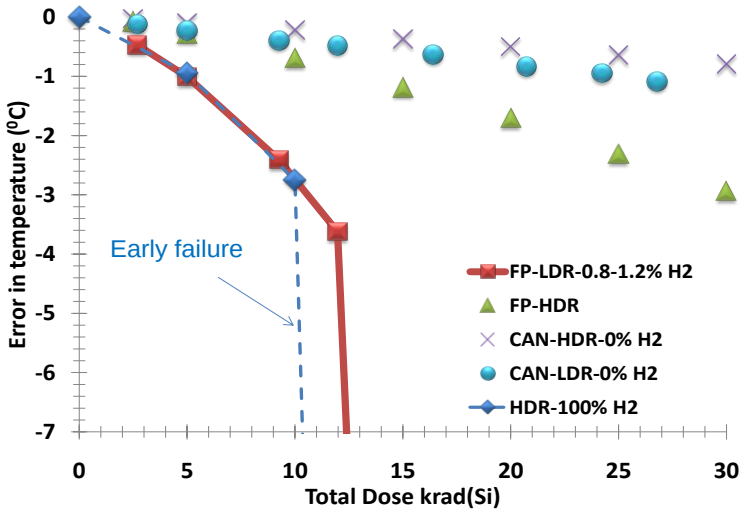


Fig. 43. Degradation of the AD590 error temperature (or output current) as a function of total dose: **1)** FPs degrade more at LDR and HDR compared to the cans due to H_2 contamination. **2)** At HDR and LDR, there is more degradation when H_2 content increases. **3)** Cans can be made to degrade similar to the FPs when the die is exposed to H_2 and HDR [97-101].

To confirm the correlation between molecular hydrogen and ELDRS, additional experiments were conducted to study the effects of externally applied hydrogen on both the total dose and dose rate response of bipolar test structures and ICs [100]. The hydrogen was applied to the parts in a glass tube that was evacuated and then back-filled with H_2 to a controlled pressure that determines the concentration of H_2 (%) to which the part is exposed. The parts were irradiated in the glass tube at different dose rates and to different total dose levels. The vacuum level in each tube before being soaked by H_2 was 10^{-5} torr. A picture of the glass tubes used by JPL for the H_2 exposures is shown in Figure 44.



Fig. 44. Picture of glass tubes used by JPL for exposure in H_2.

Several experiments were performed with different percentages of H_2 as well as for different amounts of H_2 exposure time before being irradiated. It has been shown that H_2 diffuses quite rapidly through oxides [100] but does not penetrate nitride [97]. Hence, it is reasonable to assume that when the parts with nitride have the nitride removed, the H_2 can outgas from the layers near the active device regions and increase the total dose hardness.

4.1.1 LM193

The circuit characterized in this additional set of experiments is an NSC LM193 dual voltage comparator. The LM193 is a dual version of the well-characterized LM139 quad comparator and has the same circuit design and layout, utilizing substrate and lateral *pnp* transistors in the input stage circuit. They are packaged in 8-pin dual in-line ceramic packages with sealed ceramic lids and have a standard final passivation of nitride over p-glass. The LM193 devices were from a single date code lot (date code 0551) and were purchased as commercial parts. An RGA showed the presence of <0.01% H_2 in the package cavity. Thirty of the parts had the lid removed and were subjected to a CF_4/O_2 plasma etch (150 mTorr CF_4, 50 mTorr O_2, 125 Watt RF for three minutes) to remove the top nitride layer, leaving the p-glass passivation. The nitride was removed because it is a barrier to hydrogen. The irradiations were carried out at JPL in a Shepherd 81 Co-60 irradiator. All irradiations were conducted with all leads shorted and grounded. All of the parts in the glass tube were soaked in the H_2 gas ambient for a minimum of 48 hours before irradiation to allow the H_2 to penetrate the p-glass. The range of dose rates used varied from 2 mrad/s to 100 rad/s, while the maximum total dose varied from ~8 krad to 100 krad. The LM193 was characterized for all dc specification parameters using an LTS2020 linear circuit tester.

Figure 45 displays the results of the increase in input bias current at 10 krad(Si) for an NSC LM193 dual comparator irradiated at various dose rates and various H_2 levels. LM193 data includes include the as-packaged parts from date code 0551 (virgin), and the parts with the nitride plasma etched off, leaving only the p-glass on the top of the SiO_2 layer, and irradiated in air (PG-0%), in a glass tube with 1% H_2 (PG-1%), and in a glass tube with 100% H_2 (PG-100%). The individual data points illustrate the spread in the data (the lines simply provide a guide). Unlike the other NSC parts with nitride, these LM193s are relatively hard down to 10 mrad(Si)/s before exhibiting ELDRS. The parts with p-glass and no externally applied H_2 exhibit ELDRS at a much higher dose rate. As the amount of H_2 is increased in the p-glass parts, the saturated degradation at LDR increases and the transition dose rate also increases.

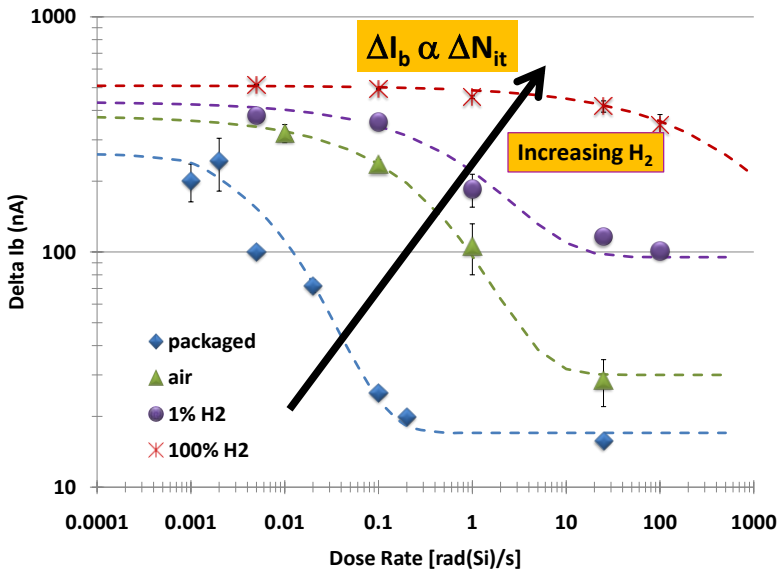


Figure 45. Increase in input bias current at 10 krad for NSC LM193 as packaged and with the nitride removed and exposed to 0%, 1%, and 100% H_2 versus dose rate. Individual data points are shown [100-101].

4.1.2 HSYE-117RH Linear Regulator

The HSYE-117RH linear regulator from Intersil was also evaluated and showed a significant amount of hydrogen (~3%) in its package. It presents some interesting packaging characteristics that are potential sources of hydrogen, such as brazed kovar lids with gold plating and dry nitrogen gas. It is likely that a large amount of hydrogen was introduced during the sealing process in a reducing environment of forming gas containing some H_2 in N_2. The HYSE-117RH has no passivation layer other than SiO_2. Two parts of the HYSE-117RH were available for this test. Parts were from the same wafer lot and were irradiated unbiased at a dose rate of 0.05 rad(Si)/s. One part was opened for more than a week to release the hydrogen content. Parts were electrically tested with an LTS2020 mixed signal automated test system located adjacent to the Co-60 range source. Pre- and post- irradiation tests were performed at ambient temperature according to the DC test parameters listed in the vendor or military specifications. Special precautions were taken to allow the temperature of the HYSE-117RH to stabilize before electrical testing was conducted. Three critical parameters were considered to assess the HYSE-117RH total dose response: changes in V_{ref}, I_{out} and V_{out}. Degradation of I_{out} is a good indicator of the loss of gain in the output transistor, which, in this case is a vertical *pnp*. V_{out} and V_{ref} are also good parameters to be considered so that the DC regulation of the device can be estimated. The results shown in Figures 46, 47, and 48, clearly indicate significant differences in the response between the sealed part (~3% of hydrogen content) and the open one (~0%). First, as shown in Figure 47, the voltage reference starts degrading at 100 krad. This is not surprising since this process is a "hardened" process; therefore, large variations were not expected below 100 krad. Small variations in the voltage reference have a non-negligible effect on the DC regulation of the linear regulator.

The most significant difference is shown for the output current change and dropout voltage. From the results show in Figure 47 and Figure 48, a very large difference is observed between the two parts: 1) an approximate 200 mA difference at 150 krad for the output current and 2) an approximate 0.9 V difference at 150 krad in the dropout voltage. These results clearly indicate a correlation between hydrogen content, packaging and total dose response, with a significant increase in degradation when hydrogen is present.

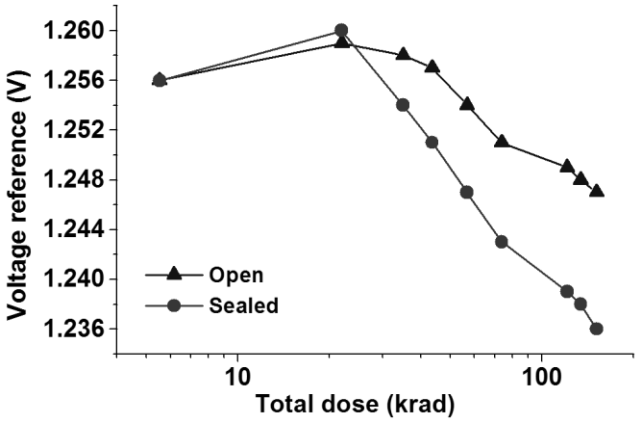


Fig. 46. Reference voltage degradation as a function of total dose for the HYSE-117RH regulator. Two devices were tested: sealed package with ~3% hydrogen and open package with ~0% hydrogen. The dose rate was 0.05 rad/s.

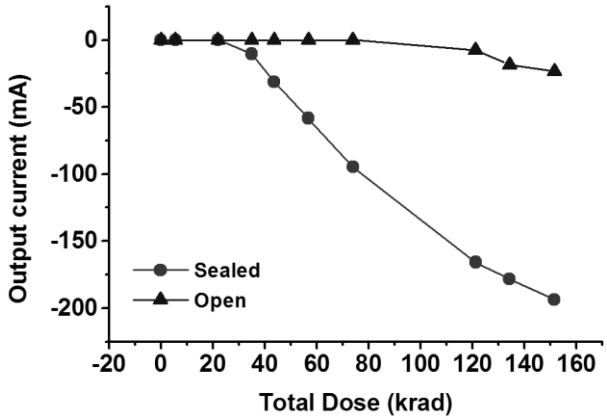


Fig. 47. Maximum output current variation as a function of total dose for the HYSE-117RH regulator. Two devices were tested: 1) sealed (~3% of hydrogen) and 2) open (0% of hydrogen). The dose rate was 0.05 rad/s.

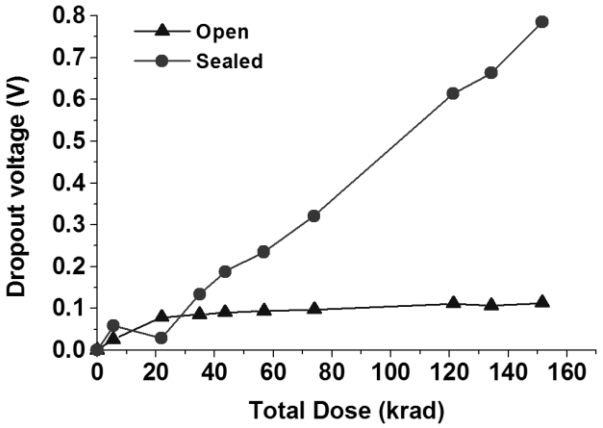


Fig. 48. Dropout voltage variation as a function of total dose for the HYSE-117RH regulator. Two devices were tested: 1) sealed (~3%) of hydrogen and 2) open (0%) of hydrogen. The dose rate was 0.05 rad/s.

4.1.3 Experimental Results—GLPNPs

To provide insight into the physical mechanisms of the total dose degradation in the presence of H_2, additional experiments were performed on the GLPNP devices. Experiments were performed both at ASU and JPL facilities. Data were taken on the transistors in the sealed kovar lid packages (i.e., ~1.2 to 1.4% H_2 in the cavity), in air (with the lid off), and in the glass tube at 100% H_2 over a range of dose rates from 0.02 to 100 rad/s, all to a total dose of 30 krad. The sample size was three for each set of conditions. For each data set, because of the GLPNP structures, the value of N_{it} was extracted using the method discussed in [99]. Figure 49 is a plot of the average increase in N_{it} versus the hydrogen concentration, including data taken in an earlier experiment by NAVSEA Crane at 20 mrad(Si)/s [99]. The error for each data point is within the symbol. It is clear that the value of N_{it} increases with the amount of externally applied H_2 and with decreasing dose rate. With 100% H_2 (atmospheric pressure), N_{it} is greater at HDR than at LDR for parts with no externally applied H_2.

Figure 50 shows the data for the increase in N_{it} (average of three samples) versus the dose rate for irradiation to 30 krad. The lines provide a guide for the eye and represent suggested hypothetical curves based on previous experimental data [100, 101] showing saturation at very low dose rates, current ELDRS models [14, 104, 105], and the hydrogen/ELDRS model presented in this document. There are two clear trends in the data. As the H_2 concentration is increased, the maximum N_{it} value at LDR increases and the transition between HDR and LDR response moves to higher dose rate. The highest dose rate that could be achieved with the available sources was 100 rad(Si)/s. This dose rate did not exhibit a significant decrease in the N_{it} value of the 100% H_2 sample, suggesting that N_{it} value will decrease at higher dose rates.

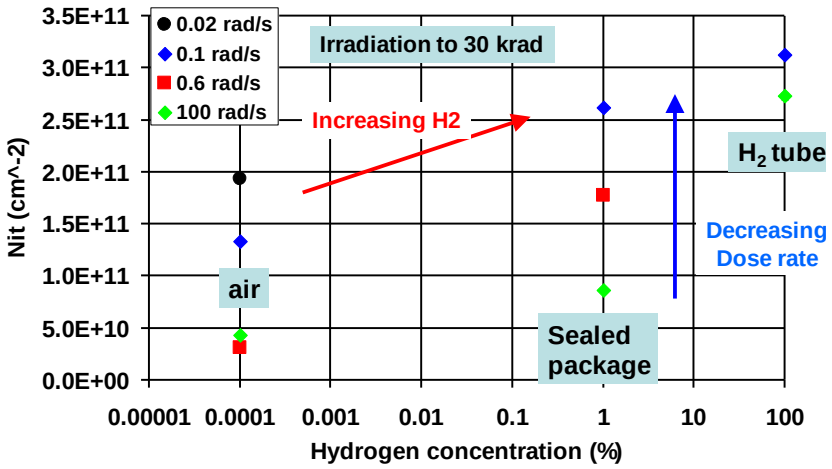


Fig. 49. Extracted value of increase in N_{it} versus hydrogen concentration for GLPNPs at 30 krad for irradiation at several dose rates.

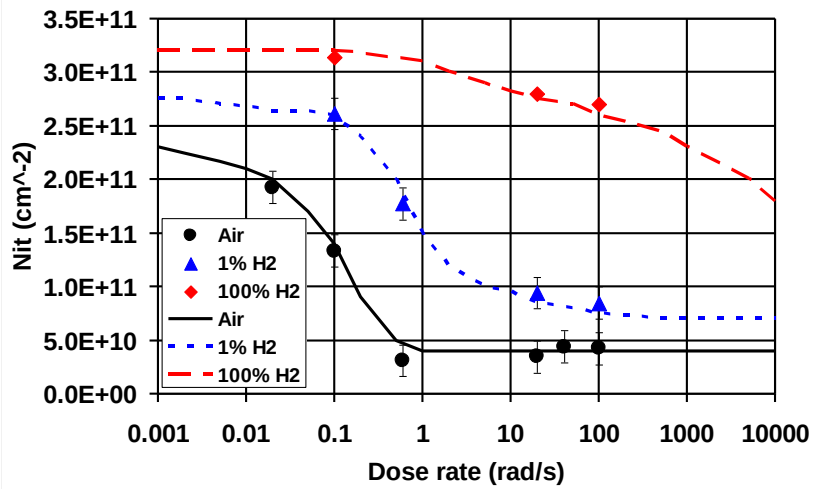


Fig 50. N_{it} versus dose rate for irradiation to 30 krad for the GLPNPs with p-glass. Lines are hypothetical curves to guide the eye.

A similar trend is seen for the average increase in base current for a base-emitter voltage of 0.5 V and a gate bias of 0 V, as shown in Figure 51. Again the error for each data point is within the symbol. The correlation between the increase in N_{it} and the increase in base current establishes the dependence of LPNP base current on the increased interface trap density. We note that the input bias current of the LM193 is primarily the base current of an LPNP transistor in the input circuitry and, therefore, would also be expected to depend on the increase in N_{it}.

In addition, post-irradiation annealing experiments were conducted in room temperature with another set of GLPNP devices irradiated both in air and in 100% H_2 environments. The annealing experiments also took place both in air and in 100% H_2 environments. Higher temperature anneals were conducted in air for samples irradiated in 100% H_2 as well. The annealing results indicate that changes in interface-trap and oxide-trapped charge densities are strong functions of H_2 content in both exposure and post-exposure ambient. For devices irradiated in air, Figure 52 shows the annealing behaviors of a 60-day air anneal followed by a 3-day 100% H_2 anneal. The decrease of both N_{it} and N_{ot} after a 60-day air anneal agrees with past results. After a 3-day 100% H_2 anneal, a decrease in N_{ot} corresponding to an increase in N_{it} was observed in these devices. This annealing behavior coincides with the results and mechanisms described by Stahbush and Mrstik [106, 107], indicating that H_2 reacts at trapped hole sites to anneal N_{ot} and produce additional H^+ to enhance N_{it} generation during annealing.

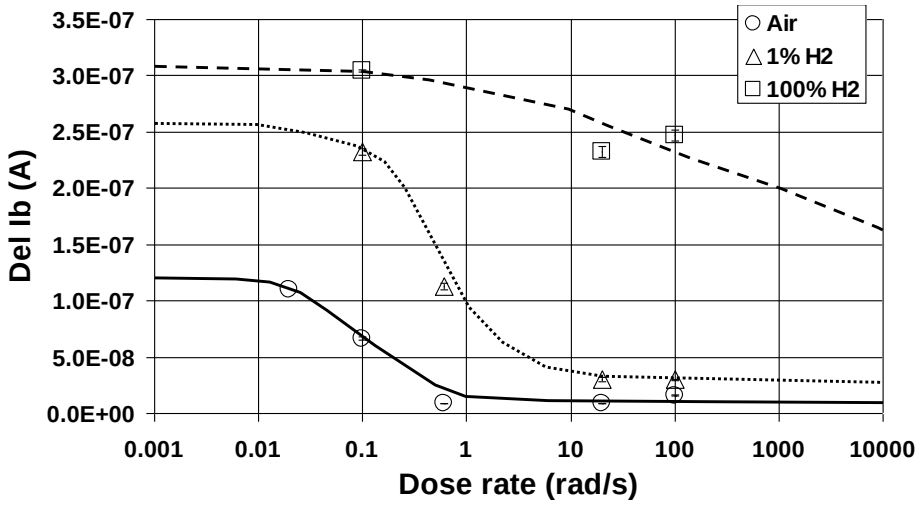


Fig. 51. Increase in base current at a base-emitter voltage of 0.5 V versus dose rate for irradiation to 30 krad for the GLPNPs with p-glass. Lines are hypothetical curves to guide the eye.

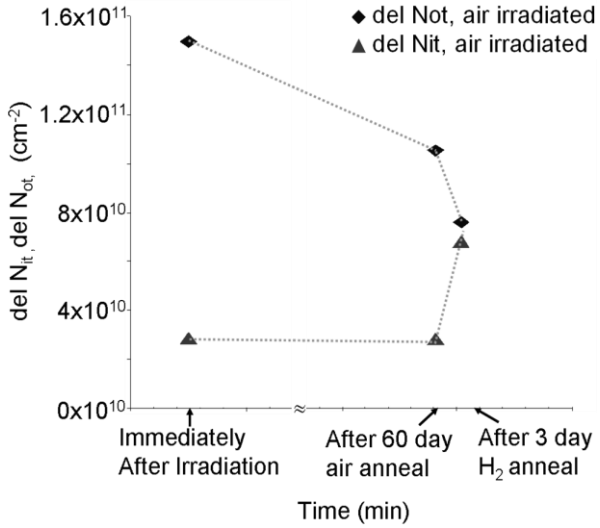


Fig. 52. Extracted ΔN_{it} and ΔN_{ot} values for GLPNP devices irradiated in air, then annealed first in air for 60 days and then in 100% H_2 for 3 days.

KEY POINTS ABOUT THE CORRELATION OF H_2 AND TOTAL DOSE EFFECTS IN BIPOLAR CIRCUITS

- Molecular hydrogen (H_2) influences the total dose and dose rate response of linear bipolar ICs. Residual H_2 as low as 0.1% present in hermetic packages can significantly affect the degradation. Hydrogen introduced during the passivation process influences ELDRS.
- ELDRS-free parts provided by manufacturers can be made ELDRS in the presence of hydrogen. This can have a significant impact on hardness assurance for parts purchased in die form and packaged later, or directly introduced in hybrid circuits.
- Increasing H_2 concentration increases device total dose degradation for both HDR and LDR.
- Exposure at HDR in 100% H_2 causes more degradation than exposure in air at LDR.
- The transition between HDR and LDR response moves to higher dose rate with increasing H_2.

5.0 MODELING

The previous sections detail the mechanisms of radiation effects in SiO_2-based devices that are relevant for bipolar and CMOS technologies and show experimental evidence that a correlation exists between residual H_2 present in package and ELDRS. This section focuses on the physical mechanisms responsible for "true" dose rate effects and hydrogen contamination. Up to now, two types of models have been proposed to explain ELDRS: the space charge model that focuses on the dose rate–dependent effect of positive charge (e.g., trapped holes) on the transport of protons toward the interface [92, 40] and the bimolecular reaction model that describes a reduction of proton density in the oxide during HDR exposures [109, 110]. The combined effect of the mechanisms described by these two models yields an enhancement in N_{it} buildup at LDR when compared to HDR irradiations. A good review of these models has been presented by Pease et al. [111]. Additional work has been presented by Rowsey et al [112] and provides a very good description of the true dose rate effect and hydrogen contamination as well. This latter model was based on first principles calculations using density functional theory.

Here, we present a generalized 1-D model that simulates the physical mechanisms that contribute to the buildup of defects in SiO_2 structures following exposure to ionizing radiation. This model was developed by ASU under JPL supervision and captures both dose rate effects and the impact of hydrogen contamination on the dose rate response. It incorporates hole trapping mechanisms as well as the formation of interface traps due to the release of hydrogen as described by the two-stage hydrogen model [108]. Dose-rate effects are simulated following the approach described by Hjalmarson et al. in [110], which adopts most of the formalisms presented in [49, 113, 114]. The influence of molecular hydrogen on dose rate response is incorporated by using the key reactions that include molecular hydrogen (H_2) cracking at positively charged defects and other bimolecular reactions. Modeling results show that space charge effects arise naturally from these bimolecular reactions. As a result, the model therefore captures the combined effect of the two dose rate mechanisms described above. The dose rate effects in SiO_2 technologies can be analyzed and modeled by processes that use kinetic equations of those mechanisms. Numerical calculations are obtained through a finite difference representation of the model.

The model is validated with the data presented in the previous sections from dose rate experiments on GLPNP bipolar transistors. We used their electrical characteristics to extract the densities of radiation-induced defects (i.e., oxide trap charge, N_{ot}, and interface trap, N_{it}). The 1-D model is used to compare the dose rate sensitivity of the GLPNP structures and investigate the mechanisms that contribute to the dose rate–dependent buildup of these defects. Finally, we extend the discussion based on additional calculations that provide insight into the key factors that determine total dose and dose rate effects in SiO_2-based devices, not only bipolar technologies. We show that degradation is a strong function of different parameters, such as applied bias, dose-rate, hydrogen contamination, and distribution of trapping precursors (i.e., processing defects), that all can affect the dose and dose rate responses.

5.1 1-D Model Description

The theoretical model presented below describes the physical mechanisms contributing to the buildup of radiation-induced defects in a MOS system. This model helps to understand the underlying mechanisms that describe the dose rate sensitivity and the effect of hydrogen on dose rate effects. The model incorporates hole trapping mechanisms as well as the formation of interface traps due to the release of hydrogen, as described by the two-stage hydrogen radiolysis model [1].

The model can be divided into two set of key reactions that captures *dose rate effects* and *hydrogen contamination effects*. A more complete set of equations is available upon request.

5.1.1 Dose Rate Effects

The mechanisms included in the model can be described by the following set of reactions between the mobile species and defects [109, 110]:

a) Hole trapping:

$$D_A + p \rightarrow D_A^+ \quad (1)$$

$$D_A^+ + n \rightarrow D_A \quad (2)$$

b) Proton Release:

$$D_B H + p \rightarrow D_B H^+ \quad (3)$$

$$D_B H^+ \rightarrow D_B + H^+ \quad (4)$$

$$D_B H^+ + n \rightarrow D_B H \quad (5)$$

c) De-passivation:

$$P_b H + H^+ \rightarrow P_b^+ + H_2 \quad (6)$$

Reactions (1)–(6) describe the mechanisms contributing to the radiation response following the generation of electron-hole pairs and initial "prompt" recombination, as described in the first part of this document. In Reactions (1) and (2), neutral hole trapping defects (~oxygen vacancies) are denoted by D_A, and positively charged hole trapping defects are denoted by D_A^+ . As described by Reaction (2), a positively charged defect can be neutralized by capturing an electron. The formation of interface traps occurs through the "depassivation" of P_b-centers at the Si-SiO_2 interface. As described in the two-stage model [111], protons (H^+) are first released within the oxide and then migrate toward the interface where they can react with the passivated dangling bond to form interface traps. It is commonly assumed that the proton is released following hole capture [114]. The first stage of the proton release model is described by Reactions (3)–(5). In these reactions, D_BH is a hydrogenated neutral hole trapping defect. In the model, the atomic nature of D_B is not determined. Reactions (3) and (4) describe proton release following hole capture. Reaction (5) describes electron capture at a positively charged hydrogenated defect (i.e., D_BH^+). *The competition between the mechanisms described by Reactions* (*4*) *and* (5) *result in dose rate effects* [109]. In the second stage, protons that have reached the Si-SiO_2 interface can react with passivated P_b centers (P_bH) as described by Reaction (6). The P_bH centers are dangling bonds that have been passivated by hydrogen during processing. This reaction will produce a dangling bond and release a neutral hydrogen molecule (i.e., H_2).

Reactions are formulated into continuity equations describing the reactive transport for each mobile species. Following the notation in [109-110], the continuity equations are given by

$$\frac{dn_i}{dt} + \nabla \cdot \mathbf{J}_{si} = \sum_j \nu_{ij} R_j \,. \quad (7)$$

In Reaction (7), $n_i \equiv n_i(\mathbf{r},t)$ is the density for each species i defined as a function of position $\mathbf{r}$ and time t, $\mathbf{J}_{si}$ is the species current density, R_j is the reaction rate, and ν_{ij} is the stoichiometric coefficient giving the contribution from reaction j to species i [109-110]. The flux of each mobile species is given by $f_i = |\mathbf{J}_{si}|/q$. The kinetic equations describing the rates of the radiation induced defects buildup are

$$\frac{dp_{t,A}}{dt} = \sigma_{pta} f_p N_{TA} - \sigma_{npta} f_n p_{t,A}, \quad (8)$$

$$\frac{dp_{t,B}}{dt} = \sigma_{ptb} f_p N_{TB} - \sigma_{nptb} f_n p_{t,B} - r_{pth} p_{t,B}, \quad (9)$$

$$\frac{dP_b}{dt} = \sigma_{it} f_{H^+} N_{P_bH}. \quad (10)$$

In Reactions (8)–(10), N_{TA} and N_{TB} are the density of hole traps (i.e., D_A) and the density hydrogenated defects (i.e., D_BH); $p_{t,A}$ and $p_{t,B}$ are the density of trapped holes at hole traps and hydrogenated defects, respectively; σ_{pta} and σ_{ptb} are the capture cross-sections for holes at hole traps and at hydrogenated defects; σ_{nptb} and σ_{nptb} are the capture cross-sections for electrons at positively charged hole traps and at positively charged hydrogenated defects; r_{pth} is the proton-release coefficients from positively charged hydrogenated defects; σ_{it} and N_{PbH} are the capture cross-section for protons at passivated P_b centers and the density of passivated P_b centers at the Si-SiO_2 interface, respectively. The electrostatic potential (ψ) is obtained by solving Poisson's equation given by

$$\begin{aligned} \frac{\partial^2 \psi}{\partial x^2} &= -\frac{\rho_{ox}}{\varepsilon_{ox}} \\ &= -\frac{q}{\varepsilon_{ox}} \left(p_{t,A} + p_{t,B} + n_{H^+} + p - n \right), \end{aligned} \tag{11}$$

where all charged particles are included in the charge density term (ρ_{ox}).

A finite-difference methodology is used to compute solutions for the densities of the mobile species as well as for the electrostatic potential (ψ) at nodes contained within a mesh superimposed on the solution domain . With this set of reactions, calculations for the buildup of N_{it} can be plotted for any SiO_2 system as a function of dose rate for a fixed total dose level.

5.1.2 Hydrogen Effects

To capture the effects of H_2 on dose rate response, the model follows the approach in [108] which adopts most of the formalisms presented by Stahlbush et al. and Mrstik et al. in [106-107]. In the model, hydrogen cracking occurs at positively charged defects. Therefore, H_2 disassociates to form hydrogenated defects (i.e., *DH* centers) by releasing a proton following hole capture (i.e., the positive charging of a defect). The H_2 cracking mechanisms are described by the following set of reactions [109-110]:

H_2 cracking:

$$D_C + p \rightleftarrows D_C^+ \tag{12}$$

$$D_C^+ + H_2 \rightleftarrows D_C H + H^+ \tag{13}$$

$$D_C H + p \rightarrow D_C H^+ \tag{14}$$

$$D_C H^+ \rightleftarrows D_C + H^+ \tag{15}$$

$$D_C^+ + n \rightarrow D_C \tag{16}$$

In Reactions (12)–(15), a third kind of hole trapping defect is introduced, i.e., D_C. In the hydrogen cracking process, (12) describes hole capture resulting in positively charging of the D_C defect. Reaction (13) describes the cracking of H_2 at the positively charged defect creating a DH center (D_CH) and releasing a proton. The resulting DH center can release additional protons as described by Reactions (14) and (15). Reaction (16) describes electron compensation at positively charged D_C defects. In this case, dose rate dependence results from the competition of Reactions (13) and (16). By introducing the hydrogen cracking mechanisms into the calculations, it is possible to describe the effect of H_2 on the buildup of interface traps and on the dose rate response. Section 5.1.3 describes the experimental results from irradiation performed on GLPNP devices and how the model can reproduce the experimental data. In addition, we employed the model to determine the key factors responsible for total dose and dose rate effects in SiO_2-based devices. Experimental data obtained on FOXFETs devices are also presented.

5.1.3 Calculations and Discussions

To explain and extrapolate the results from experiments obtained from irradiation of GLPNP devices, we use the 1-D model described above and employ a finite-difference methodology for the numerical calculations. This methodology computes solutions for the densities of the mobile species as well as for the electrostatic potential at nodes contained within a mesh superimposed on the solution domain [115]. These calculations allow for studying the key mechanisms responsible for differences in enhancement factor (EF) between devices as well as identifying the key mechanisms responsible for the build-up of N_{it} at LDR in the presence of H_2.

5.1.3.1 Calculation 1: Dose Rate Effects

The first set of calculations, shown in Figure 53, plots the buildup of N_{it} as a function of dose rate at a total dose of 30 krad(SiO_2) for different values of t_{ox}. All of the calculations are obtained from *transient simulations* performed up to a maximum total time of 3×10^7 s, which is the time required to reach a dose of 30 krad(SiO_2) at the lowest dose rate. These calculations are obtained using a uniform density of hydrogenated defects, $N_{TB} = 10^{15}$ cm^{-3}, distributed throughout the oxide and a uniform density of hole traps, $N_{TA} = 10^{19}$ cm^{-3}, located within 25 nm of the Si-SiO_2 interface. Other model parameters used in the calculations are given in Table 2. The defect densities and capture cross-sections used here are consistent with values reported in [13, 110, 114, 116, 117]. The results plotted in Figure 53 show that at low dose rates, the buildup of N_{it} is suppressed as t_{ox} is reduced from 1,000 nm to 400 nm (i.e., ΔN_{it} is lower for t_{ox} = 400 nm for dose rates below 0.1 rad(SiO_2)/s). This reduction is a result of fewer protons being released within the thinner oxides, as described by Reaction (4), and consequently a lower ΔN_{it}. For increasing dose rates, the competition between Reactions (4) and (5) becomes more significant as the density of radiation-induced generated ehps becomes comparable to N_{TB}. In this case, electron recombination at D_BH^+ reduces the amount of protons being released within the oxide, resulting in a reduction in ΔN_{it} at higher dose rates. This neutralization process is described by Reaction (5). However, the results in Figure 53 show that the reduction in ΔN_{it} as a function of dose rate (i.e., the dose rate sensitivity) is more significant for t_{ox} = 1,000 nm than for t_{ox} = 400 nm. The LDR to HDR EF is reduced from approximately 6.2 for t_{ox} = 1,000 nm to approximately 1.8 for t_{ox} = 400 nm. The EF is obtained from the ratio of ΔN_{it} for dose rates of 10^{-3} rad(SiO_2)/s and 10^2 rad(SiO_2)/s. The difference in dose rate sensitivity as a function of t_{ox} can be explained by space charge effects.

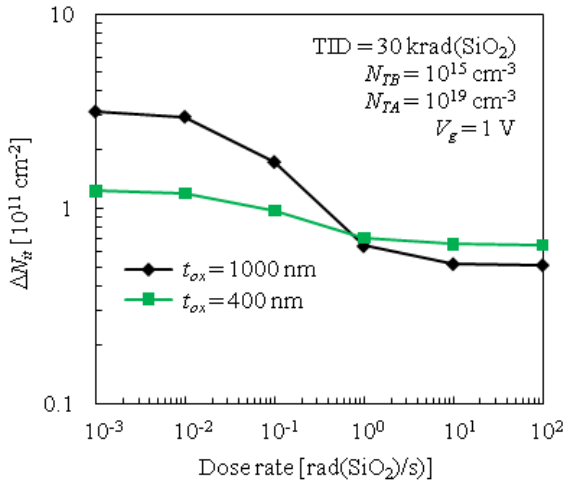


Fig. 53. Calculations for the buildup of N_{it} plotted as a function of dose rate for a total dose of 30 krad(Si) and for different values of t_{ox}.

Space charge effects that contribute to dose rate sensitivity are investigated by using different densities of hole trapping defects (i.e., N_{TA}). A higher N_{TA} results in more fixed positive

Table 2. Simulation input parameters for calculations in Figures 54 and 55.

Parameter	Value	Units
N_{PbH}	10^{13}	cm^{-2}
σ_{pta}	5.5×10^{-14}	cm^2
σ_{ptb}	5.5×10^{-14}	cm^2
σ_{npta}	5.0×10^{-13}	cm^2
σ_{nptb}	2.0×10^{-12}	cm^2
σ_{it}	10^{-11}	cm^2
r_{pth}	10^{-5}	s^{-1}

charge near the Si-SiO_2 interface. Localized electric fields caused by the fixed positive charge and other radiation-generated species can result in the confinement of electrons within the oxide bulk [13, 92]. The confinement of electrons enhances recombination at D_BH^+ sites, and contributes to dose rate sensitivity. Figure 54 is a plot of the EF obtained by the ratio of ΔN_{it} at a total dose of 30 krad(Si) for dose rates of 10^{-3} rad(Si)/s and 10^{3} rad(Si)/s. The EF is plotted as a function of t_{ox} for two different densities of hole traps, N_{TA} = 10^{19} cm^{-3} and 10^{18} cm^{-3}, located within 25 nm of the Si-SiO_2 interface. The results in Figure 54 show that *a higher density of hole trapping defects near the Si-SiO_2 interface enhances dose rate sensitivity as a function oxide thickness*. Space charge effects are due to localized electric fields caused by radiation-generated species and can alter the transport of charged particles during irradiation. However, as t_{ox} is reduced, space charge effects become less significant because the transport of charged particles is predominantly determined by the electric field set by the gate bias and the gate-to-semiconductor work-function difference. This effect is demonstrated by the calculation results shown in Figure 55. Here, for the case of N_{TA} = 10^{19} cm^{-3}, dose rate sensitivity is significantly impacted by oxide thickness as determined by the increase in EF as a function of t_{ox}. On the other hand, for N_{TA} = 10^{18} cm^{-3}, the impact of t_{ox} on dose rate sensitivity is less significant as determined by the nearly constant EF as a function of t_{ox}.

Additional data from numerical calculations are shown in Figure 55. Data show a plot of the LDR to HDR EF calculated by the ratio of N_{it} at a total dose of 30 krad(Si) for dose rates of 10^{-3} rad(Si)/s and 100 rad(Si)/s. The EF is plotted as a function of the electron capture cross-section at positively charged hydrogenated defects (i.e., σ_{nptb}) and for two different densities of hole traps, N_{TA} = 10^{19} cm^{-3} and 10^{18} cm^{-3}, located within 25 nm of the Si-SiO_2 interface. For the lower values of σ_{nptb}, electron compensation at positively charged hydrogenated defects, i.e., Reaction (5), is not significant. Consequently, the EF, and therefore dose rate sensitivity, is independent of electron compensation at positively charged hydrogenated defects for values below 10^{-13} cm^2. For these values of σ_{nptb}, the simulated EF is simply the result of space charge effects that arise from buildup of fixed positive charges at hole traps near the Si-SiO_2 interface, and thus increases with N_{TA}. For σ_{nptb} = 10^{-14} cm^2, the EF is ~1 for N_{TA} = 10^{18} cm^{-3} and ~3.5 for N_{TA} = 10^{19} cm^{-3}. As

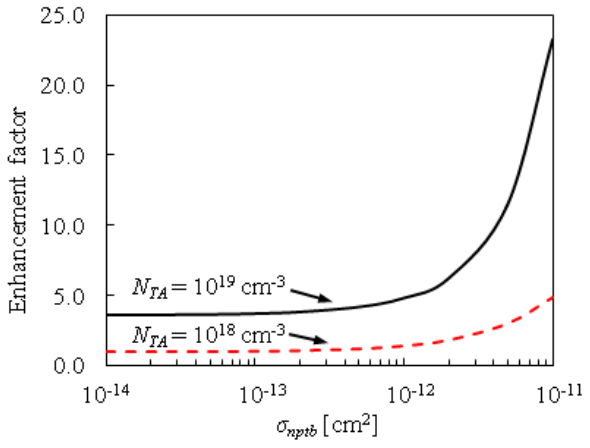


Fig. 54. LDR to HDR enhancement factor obtained by the ratio of N_{it} for dose rates of 10^{-3} rad(Si)/s and 100 rad(Si)/s for two different densities of hole traps, N_{TA} = 10^{19} cm^{-3} and 10^{18} cm^{-3}, located within 25 nm of the Si-SiO_2 interface. For these calculations N_{TB} = 10^{16} cm^{-3} and is uniformly distributed in the oxide.

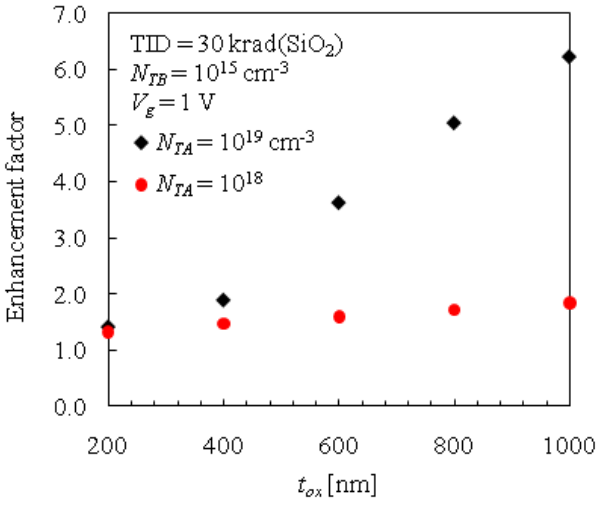


Fig. 55. LDR to HDR enhancement factor given by the ratio of N_{it} for dose rates of 10^{-3} rad(Si)/s and 10^{3} rad(Si)/s plotted as a function of t_{ox}, for two different densities of hole traps, N_{TA} = 10^{19} cm^{-3} and 10^{18} cm^{-3}, located within 25 nm of the Si-SiO_2 interface. For these calculations N_{TB} = 10^{15} cm^{-3} and is uniformly distributed in the oxide.

σ_{nptb} increases above 10^{-13} cm^2, the EF increases as the reaction between electrons and D_BH+ centers becomes more significant at higher dose rates. However, the enhancement is greater for the case of $N_{TA} = 10^{19}$ cm^{-3} because space charge confines electrons in the oxide bulk, allowing more recombination to occur. A higher N_{TA} results in more fixed positive charge near the Si-SiO$_2$ interface and therefore more confinement of electrons in the oxide bulk.

5.1.3.2 Calculation 2: Effect of Hydrogen

Effect of Hydrogen on Dose Rate Sensitivity

We performed additional calculations to study the impact of hydrogen on the total dose and dose rate response using the 1-D model on a MOS structure that simulates the gated base oxide region of the GLPNP bipolar transistors. The simulation parameters include a fixed gate work-function, uniform Si substrate doping concentration and a base oxide thickness of 1.22 µm. Other parameters are the same as listed in Table 1, except $N_{TC} = 10^{15}$ cm^{-3}, which is the density of defects D_C. For these calculations, no initial density of hydrogenated defects is specified (i.e., $N_{TB} = 0$). Figure 56 shows the model calculations of N_{it} plotted as a function of dose rate for three different concentrations of H_2. The results show a shift to the right in the dose rate response of N_{it} (i.e., the N_{it} vs. dose rate curve) as the concentration of H_2 is increased. This shift is consistent with the experimental data shown in Figure 46. As mentioned above, dose rate effects arise from the competing contributions of Reactions (13) and (16). Increasing the concentration of H_2 favors Reaction (13) resulting in more protons being released. Therefore, the dose rate response is shifted to right as a greater density of electrons is required for recombination mechanisms to become significant.

It should be noted that the saturation point can be controlled depending on the oxide parameters, indicating that by using a known oxide process, it would be possible to predict the expected EF for a specific device at a specific dose for LDR conditions. As a result, it is not surprising that we see saturation for the GLPNP devices and no saturation for the LM193 dual comparator from NSC reported by Pease [100].

With this model, additional calculations were also performed and results were plotted against the experimental data obtained in Figure 43. Those data indicate that with minimum amount of hydrogen, between 0.1%–0.5%, the degradation can be significantly enhanced.

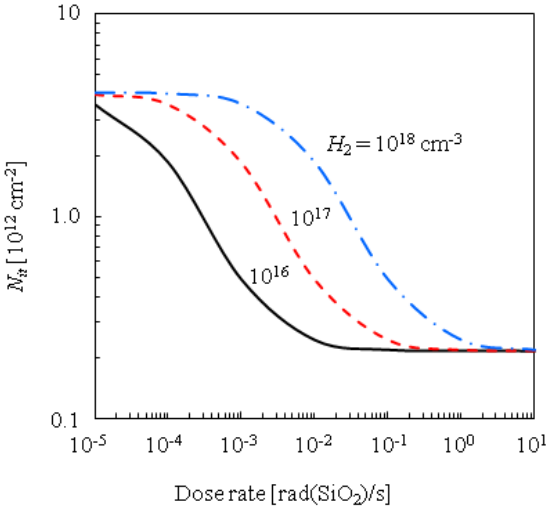


Fig. 56. Calculations of N_{it} plotted as a function of dose rate for three different concentrations of H_2.

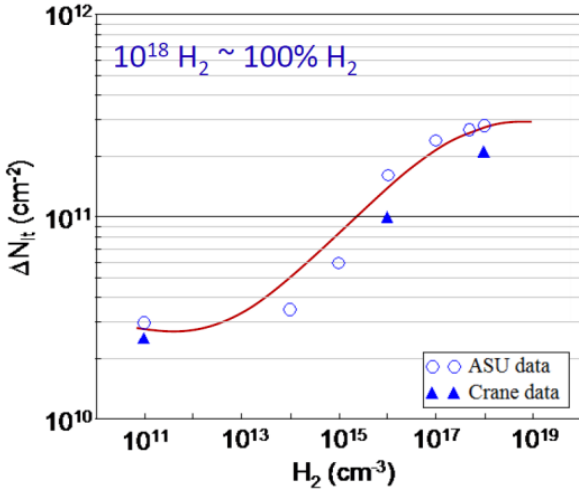


Fig. 57. Fit of analytical model describing interface trap formation due to excess H_2.

5.1.3.3 Calculation 3: Comparison between STI Deposited Oxides vs. Thermal Oxides

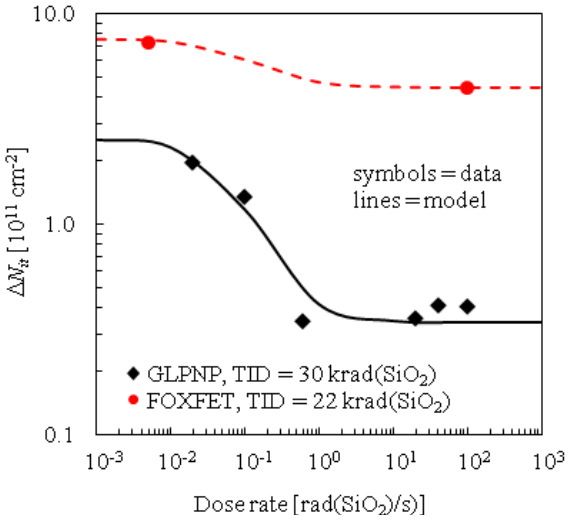


Fig. 58. ΔN_{it} plotted as a function of dose rate for the FOXFETs (from data in Fig. 4) and for GLPNP transistors fabricated in the NSC linear bipolar circuit technology. FOXFET data corresponds to a total dose of 21 krad(SiO_2) and the GLPNP transistor data corresponds to a total dose of 30 krad(SiO_2). Symbols indicate data and solid lines model calculations.

Figure 58 plots the buildup of interface trap density as a function of dose rate for data presented in [100] from irradiation performed FOXFET test structures and GLPNP transistors [24]. The FOXFET data corresponds to a total dose of 21 krad(SiO_2) and the GLPNP transistor data corresponds to a total dose of 30 krad(SiO_2). ΔN_{it} for the GLPNP transistors is extracted using the methods discussed in [99, 100]. Degradation in the GLPNP transistors is a result of interface trap buildup at the base oxide-silicon interface. The base oxide thickness for the GLPNP transistors is t_{ox} = 1.22 μm. As described in the previous section and shown in Figure 40, oxide thickness and the density of hole traps (i.e., N_{TA}), can affect dose rate sensitivity. However, trapping properties that result from different oxide processing may also affect dose rate sensitivity. For example, as reported in [118], most of the hole trapping in thermal oxides occurs near the Si-SiO_2 interface whereas deposited oxides can have a significant amount of hole trapping deeper within the bulk. Additionally, the capture cross-section for hole traps (i.e., σ_{pta}) appears to be smaller in deposited oxides than in thermal oxides [118]. The GLPNP base oxide consists of a thermally grown oxide covered by a deposited oxide that is used to assure adequate oxide thickness over the emitter [100]. For STI oxides, typical processing involves etching a trench pattern through a nitride layer, sidewall oxidation to grow a thin oxide liner, chemical vapor deposition to fill the trench, and chemical-mechanical planarization polishing [119]. Therefore, the bulk of the STI will consist of a deposited oxide.

Additional calculations have been completed to investigate how the differences in oxide processing between base oxides in GLPNP transistors and STI oxides in FOXFETs affect the dose rate sensitivity. The solid and dashed lines in Figure 59 correspond to these calculations. The calculations fitted to the GLPNP transistor data (i.e., the solid line) are obtained using a uniform density of hydrogenated defects, $N_{TB} = 8.5 \times 10^{14}$ cm^{-3}, distributed throughout the oxide and a uniform density of hole traps, $N_{TA} = 10^{19}$ cm^{-3}, located within 25 nm of the Si-SiO_2 interface. All other parameters are the same as listed in Table 1. For the case of the FOXFETs, the calculations are fitted using $N_{TB} = 6.5 \times 10^{15}$ cm^{-3} distributed uniformly throughout the oxide and $N_{TA} = 10^{19}$ cm^{-3} located within 75 nm of the Si-SiO_2 interface (dashed line in Figure 59). In this case, all other parameters are the same as listed in Table 1 except hole capture cross-section, which is reduced to $\sigma_{pta} = 5.5 \times 10^{-15}$ cm^2. Calculations show that increasing the depth within the oxide where hole traps are located (as measured from the Si-SiO_2 interface) slightly increases dose rate sensitivity. However, the reduction in hole capture cross-section reduces the buildup of fixed positive charge near the Si-SiO_2 interface, significantly reducing dose rate sensitivity. This results in a better agreement with the FOXFET data.

The impact of hole capture cross-section on dose rate sensitivity can be observed in the calculations shown in Figure 59 where EF is plotted as a function of σ_{pta}. EF is obtained from the ratio of ΔN_{it} for dose rates of 10^{-3} rad(Si)/s and 10^3 rad(Si)/s. The results in Figure 59 show that

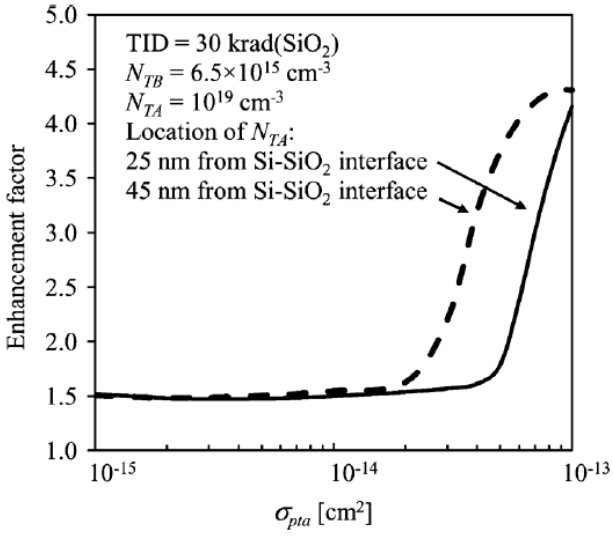


Fig. 59. Enhancement factor in the buildup of interface trap density obtained by the ratio of ΔN_{it} at a dose rate of 10^{-3} rad(Si)/s to ΔN_{it} at a dose rate and 10^{3} rad(Si)/s plotted as a function of the hole capture cross-section (σ_{pta}). The solid lines are calculations for NTA located within 25 nm of the Si-SiO_2 interface and dashed lined for NTA located within 45 nm of the interface.

for a fixed location and density of hole traps (e.g., NTA = 10^{19} cm^{-3} located within 25 nm of the Si-SiO_2 interface) the dose rate sensitivity is independent of σ_{pta} for values below ~4×10^{-14} cm^2. However, as σ_{pta} increases, significant hole trapping in regions near the Si-SiO_2 interface increases dose rate sensitivity as determined by the increased EF obtained for values above ~4×10^{-14} cm^2. The calculations also show that the value for σ_{pta} where the transition in dose rate sensitivity occurs is reduced by extending the location of the hole traps from within 25 nm of the Si-SiO_2 interface to 45 nm.

5.1.4 Discussion and Conclusions

The impact of different model parameters on the dose rate–dependent buildup of interface traps can be investigated using the numerical calculations given in Section 2. This analysis makes it possible to understand how differences in technological and processing characteristics affect the dose rate sensitivity of SiO_2-based devices. From the model calculations, it has been determined that oxide thickness, density and location of hole traps and hole capture cross-section affect dose rate sensitivity. The density and location of hole traps, as well as the hole capture cross-section, affect the buildup of fixed charge near the Si-SiO_2 interface, which can result in space charge effects that enhance recombination mechanisms and impact the dose rate sensitivity. The numerical calculations presented in this section indicate that dose rate sensitivity can result from the combined effects of space charge and recombination mechanisms. However, the contribution from space charge effects becomes less significant as t_{ox} is reduced because the transport of charged particles during irradiation is determined by the electric field set by the gate bias and the gate-to-semiconductor work-function difference.

Comparison of model calculations with experimental data results in excellent agreement for the description of dose rate sensitivity. The difference in dose rate sensitivity between base oxides of bipolar devices and advanced CMOS STI oxides is captured through differences in oxide thickness, in the location of hole traps (i.e., the location of N_{TA}) and in the value of the hole capture cross-section. The differences in the location of N_{TA}, and in the values of σ_{pta} are consistent with observations made by Mrstik et al. in [119] based on experimental characterization of deposited oxides using photo-assisted injection techniques.

Other model parameters such as the electron capture cross-section at positively charged hydrogenated defects (i.e., σ_{nptb}) and proton release coefficient from positively charged hydrogenated defects (i.e., r_{pth}) can also impact dose rate sensitivity. As described in [117], important factors that impact dose rate sensitivity in bipolar technologies are final passivation, packaging and post-packaging thermal treatments, and hydrogen contamination in the package. However, the work presented here provides strong evidence that specific technology and processing characteristics of isolation oxides can explain the differences observed in dose rate sensitivity between bipolar and CMOS technologies.

In addition, with respect to the hydrogen contamination issue, results in Figures 56 and 57 demonstrate good qualitative agreement between the analytical model and the experimental data shown in Figure 46, particularly in the transition region between HDR and LDR responses. At present, the model does not capture the impact of hydrogen on the HDR and LDR asymptotes. One explanation for the increase in interface trap buildup at the HDR and LDR extremes is that hydrogen alters the density of precursor species and other reactants. However, as observed in Figure 56, saturation in the enhancement of ΔN_{it} at LDR in the presence of H_2 occurs in some cases. A contributing factor to the differences in saturation of ΔN_{it} at measured at LDR may be the different concentrations of molecular hydrogen and hole trapping defects (i.e., N_{TC}) in oxide regions for a given hydrogen ambient condition. Experimental results and model calculations indicate that saturation of ΔN_{it} in the presence of hydrogen for devices exposed at LDR may vary based on the trapping characteristics and concentration of H_2 in the oxide.

KEY POINTS ABOUT THE HYDROGEN-ELDRS MODEL

* Oxide thickness, density, and location of hole traps and hole capture cross-sections affect the buildup of fixed charges near the Si-SiO_2 interface, which can result in space charge effects that enhance recombination mechanisms and impact the dose rate sensitivity.
* The dose rate sensitivity can result from the combined effects of space charge and recombination mechanisms. However, the contribution from space charge effects becomes less significant as t_{ox} is reduced.
* The difference in dose rate sensitivity between base oxides of bipolar devices and advanced CMOS STI oxides is captured through differences in oxide thickness, the location of hole traps (i.e., the location of N_{TA}) and the values of hole capture cross-sections.
* Experimental results and model calculations indicate that saturation of ΔN_{it} in the presence of hydrogen for devices exposed at LDR may vary based on the trapping characteristics and concentration of H_2 in the oxide.
* With minimum amount of hydrogen, between 0.1%–0.5%, the degradation can be significantly enhanced.

6.0 IRRADIATION IN AN HYDROGENATED ENVIRONMENT AS AN ACCELERATED TESTING TECHNIQUE

6.1 Introduction

The previous sections show that the ELDRS phenomena is dependent on the processing of the base oxide, oxide thickness, final passivation, packaging and post-packaging thermal treatments, and hydrogen contamination. Through the use of modeling and experiments, we showed that hydrogen is a dominant factor determining both the total dose and dose rate characteristics of linear bipolar circuits. Experiments conducted on both transistor structures and linear circuits indicated that the percent of hydrogen 1) increases the degradation at LDR and 2) increases the dose rate region where the transition from a HDR response to an enhanced LDR response occurs.

These results suggested that a new accelerated hardness assurance test method might be possible, whereby parts are tested at a higher dose rate while exposed in a rich H_2 environment. Data obtained from such a method could rapidly establish an upper bound to the LDR response in space. If these results are reproducible in other part types, then a general method of accelerated testing using hydrogen stress may be developed and help in the parts selection for systems designed for space. Up to now, several accelerated testing methods have been proposed: 1) elevated temperature irradiation (ETI) initially proposed by Fleetwood et al. [92] and investigated by others [104, 110, 121]; 2) alternate HDR irradiation and elevated temperature anneals initially proposed by Freitag and Brown [122] and further investigated by Pershenkov et al. [123]; and 3) switched dose rate experiments proposed by Boch et al. [124, 125]. In the ETI technique, irradiation is usually performed at a temperature of ~100°C at a dose rate of 1 rad(Si)/s or less. In the alternate HDR irradiation and elevated temperature anneal approach, Freitag and Brown found that for two types of op amps the following procedure worked: irradiation at HDR to half the specification dose followed by an elevated temperature anneal at 100°C for 3 hours, followed by an additional irradiation at HDR to half the specification dose, followed by another elevated temperature anneal at 100°C for 4.4 hours [122]. The switched dose rate technique consists of irradiation at HDR to increasing values of total dose and then switching to LDR and continuing the irradiation [124, 125]. The results at LDR are then transposed along the dose axis to construct the LDR response. Although it takes many more test samples to use this approach, the total irradiation time is reduced by the number of steps used.

All of these techniques are useful; however, they all have their limitations. The number of part types investigated for each of the techniques is limited and, at least for the first two techniques, there is no set of variables that is universal. Hence, a characterization would be required to establish the parameters and procedures for each process technology and part type to bound the LDR response. Also, in the case of the ETI technique, the total dose is limited because irradiation at elevated temperatures for extended times results in annealing that competes with the additional degradation. Hence, combining this approach with over-test will usually not work. Nevertheless, accelerated hardness assurance method development is highly desired because of the cost and time constraints associated with LDR testing.

The data shown in Section 4.1.1 is again plotted in Figure 60. The data represent the average increase in input bias current, Ib, versus dose rate (three or four parts for each data point) after a total dose exposure of 10 krad(Si). The four curves represent the dose rate response characteristics of 1) unmodified commercial-off-the-shelf (COTS) parts with a nitride over p-glass final passivation (nitride), 2) parts with nitride stripped off (p-glass left intact) and irradiated in a natural air environment with ~0 % H_2 (air), 3) parts soaked in 1.0 % H_2 (1%), and 4) parts soaked in 100% H_2 (100%). One data point at 1 mrad (Si)/s for the parts with nitride over p-glass has been added and shows the predicted saturation at LDR. The data provide an excellent illustration of how H_2 can

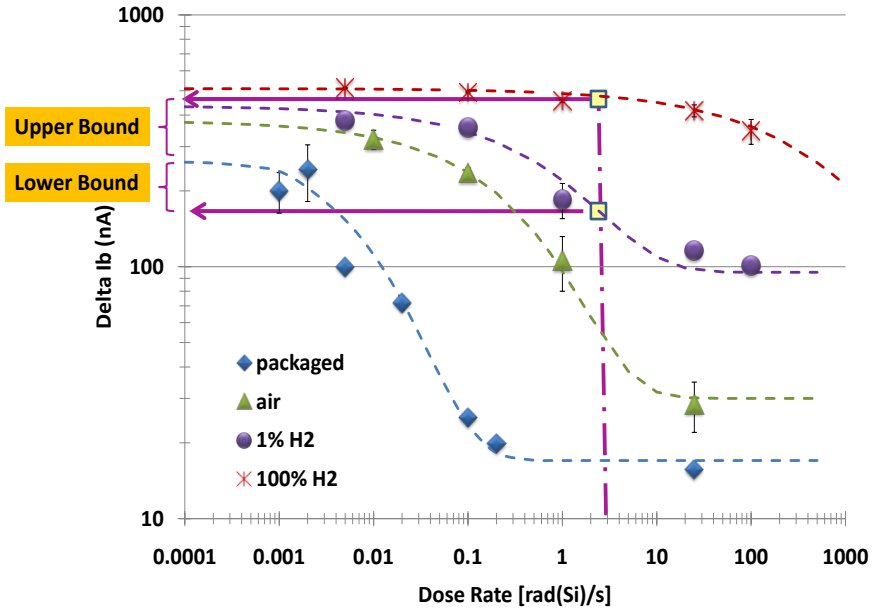


Fig. 60. Increase in input bias current vs. dose rate for 1) irradiation to 10 krad for the LM193s with p-glass and different percentage of H_2, and 2) nitride with no H_2. At 1 rad/s the 1% H_2 underestimates the LDR response whereas the 100% H_2 bounds it.

be used to accelerate the dose rate response characteristics. The unmodified part's bias current represents a typical dose rate profile. At the higher rate exposures, ΔIb for the COTS part is fairly low (well within the specification limit of the part). At a dose rate near 100 mrad(Si)/s, called the transition dose rate, excess current increases dramatically. Near 2 mrad(Si)/s, the degradation seems to be saturating. This type of leveling off in the dose rate behavior is consistent with most ELDRS signatures.

As shown in the figure, these data suggest that with the right combination of dose rate and hydrogen concentration, we should be able to bound or perhaps predict LDR responses of linear bipolar circuits.

6.2 Accelerated Testing Validation

The proposed accelerated hardness assurance technique using H_2 has the following characteristics: It is best used on parts that are packaged in hermetic packages. The package lids must be removed so that the microcircuit die can be exposed directly to H_2. If plastic packages are used the plastic must be etched away over the surface of the die. Parts with a silicon nitride passivation must have the nitride removed, for example, by plasma etching. The exposure to H_2 is done in evacuated glass tubes (10^{-5} torr), that are backfilled to various controlled partial pressures of H_2. While the test method appears to be true for the data shown in Figure 60 from various experiments performed on the LM193 ELDRS sensitive part, it needed to be tested against various ELDRS devices covering several circuit types and manufacturers and some claimed "ELDRS-free" parts to see if the method is too conservative or not.

6.2.1 ELDRS Parts

Experimental data presented in this section were taken at JPL facilities. LDR and HDR irradiations were performed using a Shepherd 81 Co-60 range irradiator. The irradiations with concentrations of H_2 (1–100%) were all conducted with the parts inside a sealed glass tube that was evacuated and then filled with H_2 to the appropriate partial pressure. The vacuum level in

each tube before being soaked by H_2 was 10^{-3} torr. All the parts in the glass tube were soaked in the H_2 gas ambient for a minimum of 48 hours before irradiation. All DC specification parameters were measured at each dose step using an LTS2020 or ETS300 linear circuit tester. Parts under consideration are all ELDRS sensitive candidates commonly used in space: the LT1019 voltage reference from Linear Technology, the HSYE-117 linear voltage regulator from Intersil, the AD590 temperature transducer and OP42 JFET-input operational amplifier from Analog Devices. These five part types represent a wide range of manufacturer process technologies and circuit designs.

6.2.1.1 Experimental Validation on Four Candidates (LT1019, HSYE-117RH, AD590 and OP42)

Figure 61 shows experimental data obtained from irradiations performed on the LT1019 voltage reference from Linear Technology. The change in reference voltage as a function of total dose is plotted for five different conditions of irradiation: HDR (50 rad/s), LDR (10 mrad/s) and HDR (50 rad/s) with 1%, 10%, and 100% H_2. Irradiations were performed unbiased with all leads grounded. Every data point is an average of 2–3 samples. At 30 krad, the ratio between the HDR-100% data point and the LDR data point is approximately 1.9. The ratio between the HDR-10% data point and the LDR data point is approximately 1, indicating that 10% could be a more appropriate testing choice because it is much less conservative.

Figure 62 is a plot of experimental data obtained from irradiation performed on the HSYE-117 linear voltage regulator from Intersil. The dropout voltage at 100 mA load current is plotted as a function of total dose for two conditions of irradiation: 1) twelve parts in three different packages (can TO-39 with 0% H_2, TO-220 with ~1.66% H_2, and flatpack-SCC with ~1.9% H_2) were irradiated unbiased at a dose rate of 0.05 rad/s, and 2) two additional TO-220 parts (with the lid off and nitride removed) were irradiated with 100% H_2 at a dose rate of 50 rad/s. Once again, the HDR-100% H_2 data bound the LDR responses for all dose steps. This also applies to other susceptible parameters that exhibit enhanced degradation at LDR, such as the reference voltage, V_{ref}, and adjust current, I_{adj}. The interesting result is that unlike other parts, the response between the ~2% and 100% hydrogen data is saturating. The fact that Intersil uses a hardened (RSG) process with better oxide properties (possibly lower numbers of passivated interfacial

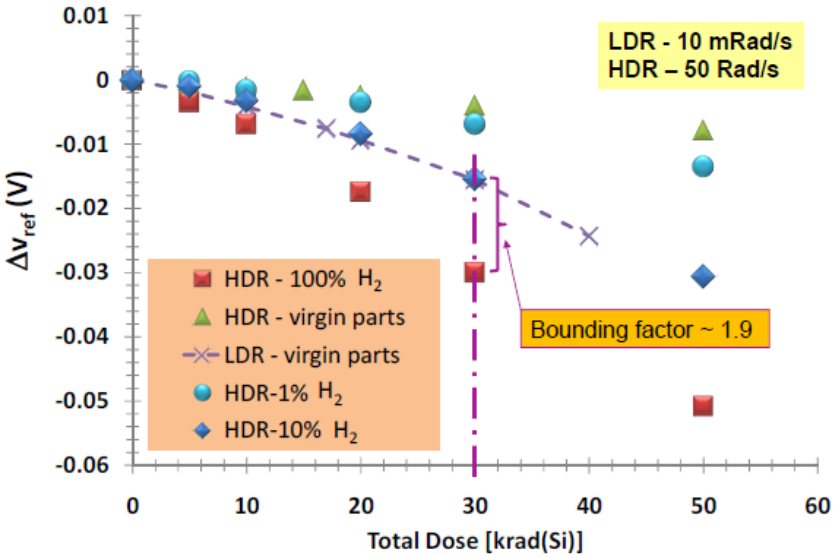


Fig. 61. Change in reference voltage for the LT1019 as a function of total dose for four irradiation conditions: HDR (50 rad/s), LDR (10 mrad/s) and (50 rad/s with 1%, 10%, and 100% H_2).

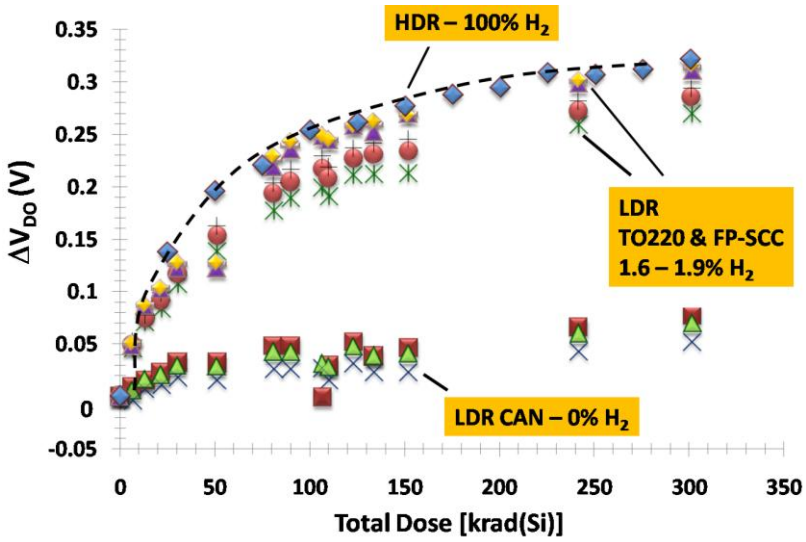


Fig. 62. Change in dropout voltage versus total dose for the HSYE-117 linear voltage regulator in three different packages (TO39, TO-220 and FP-SCC). Twelve parts were irradiated at LDR (0.05 rad/s). Two additional parts (TO-220) with the lid off were irradiated at HDR (50 rad/s) with 100% H_2.

bonds limiting N_{it} creation or fewer defect centers in the oxide to release protons in the presence of H_2) might explain the results. In addition, these results could indicate that the dose rate used for these experiments was somewhat high, and going to a lower dose rate (i.e., 1 rad/s) could have shown a bounding factor > 1.

The AD590 temperature transducer has been shown to exhibit ELDRS when irradiated in a FP containing H_2 and with all leads shorted. Additional tests have been performed on AD590s in both cans (no H_2) and FPs. Initially, twelve screened space-qualified AD590s were irradiated at both high (25 rad/s) and low (10 mrad/s) dose rates with all leads grounded. Three FPs (with 0.4–1% H_2) and three cans (~0% H_2) were irradiated up to 30 krad. The results show that FPs degrade significantly more than cans at LDR and HDR when irradiated unbiased due to the amount of hydrogen introduced into the package. The results are compared to an accelerated test and show similar trends as those observed for the other devices. Indeed, the AD590 in cans (~0% H_2) were made to degrade similarly to the FPs when the die was exposed to 100% H_2 with a dose rate of 1 rad/s. Unlike with the LT1019 and HSYE-117, the HDR exposure in H_2 had to be lowered to 1 rad/s to bound the LDR response. Once again, the HDR-100% H_2 data bounds the LDR response for all dose steps. Note that the parts irradiated after being soaked with 100% H_2 failed earlier (dashed line on Figure 63) than the FPs.

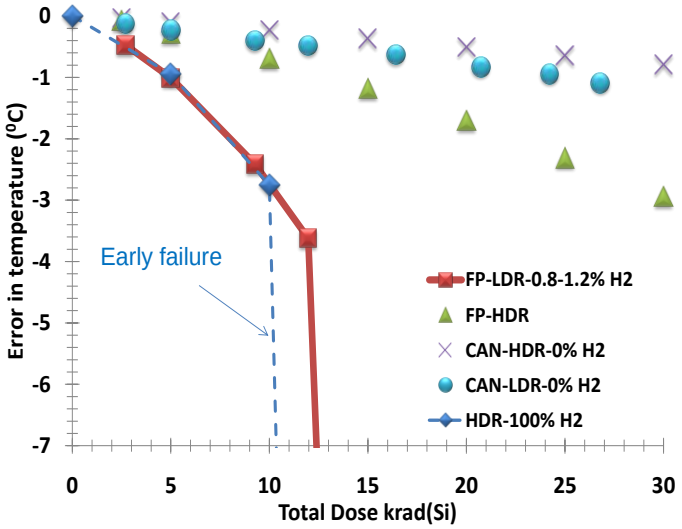


Fig. 63. Degradation of the AD590 temperature error as a function of total dose. Two groups of three flatpacks and three cans were irradiated unbiased at both HDR and LDR (respectively 25 rad/s and 0.01 rad/s). RGA was performed after irradiation on the flatpack devices to evaluate the amount of hydrogen. Three additional parts (Cans) with the lid off were irradiated at HDR (1 rad/s) with 100% H_2.

Finally, a set of irradiations were performed on the OP42 JFET input op-amps from Analog Devices. The irradiation condition was similar to the ones used in [65], which was a voltage-follower configuration with ±15 Volt on the supply pins. The offset voltage under biased irradiations is the critical parameter to look at since this is the only parameter that shows an ELDRS effect and only under bias [65]. Four samples were soaked in 100% H_2 and irradiated with a dose rate of 1 rad/s. As shown in Figure 64, the hydrogenated devices show an ELDRS effect compared to the HDR response. The LDR curve at 0.04 rad/s also shows an ELDRS effect. In this case the bounding factor between the LDR curve and the HDR-H_2 curve is approximately 1. Once again, using hydrogen stress during irradiation at HDR can make the part degrade similarly to the one irradiated at LDR. We should indicate that additional testing is needed to find out if these OP42 devices behave the same way as compared to data presented in [65]. If for lower dose rate (<0.04 rad/s) the device shows a larger ELDRS effect, then additional data at a lower dose rate (<1 rad/s) will be needed to bound the LDR response. (Note that this op-amp has JFET input transistors and that irradiations were performed under bias; two different conditions compared to the other bipolar devices were evaluated for this study).

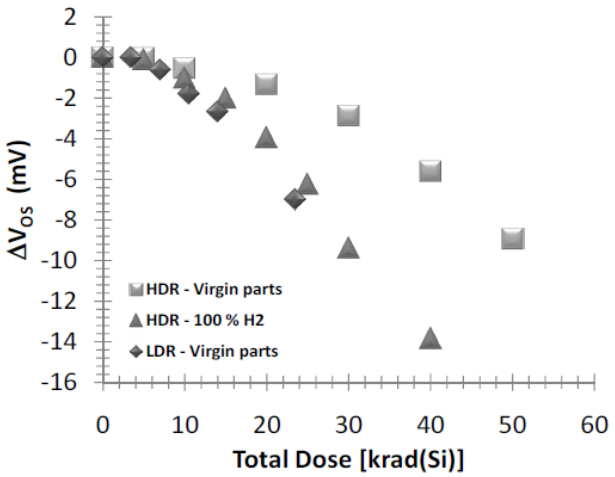


Fig. 64. Change in offset voltage versus total dose for the Analog Devices OP42. The HDR results are shown for irradiation at 50 rad/s (HDR—virgin parts) and 1 rad/s in 100% H_2 (HDR—100% H_2), as well as a LDR result at 0.04 rad/s (LDR—virgin parts).

6.2.2 ELDRS-Free Parts

Several space-qualified ELDRS-free part types were selected and obtained from NSC. These included the LM136 voltage reference, the LM2941 low-dropout voltage regulator, the LM124, quad operational amplifier, and the LM139, quad voltage comparator. LDR data on the LM2941 and LM136 were published in the Radiation Effects Data Workshop last 2009 by NSC [126, 127]. LDR data on the LM124 and LM139 were published in 2008 [127]. All four of these part types have been demonstrated to be hard to 100 krad at both HDR (50–300 rad/s) and LDR (10 mrad/s) for irradiation under both biased and unbiased conditions [126, 127, 128]. In addition, the LM2941 was tested at 1 mrad/s up to 20 krad and was shown to be ELDRS-free. To ensure that the results are not unique to one process technology, we added several ELDRS-free part types from other manufacturers: one ELDRS-free RH1009 voltage reference was obtained from Linear Technologies; and two radiation-hardened discrete bipolar transistors, a 2N2222A and a 2N2907A, were obtained from Semicoa.

Samples of 3–6 parts of each type were delidded and placed inside evacuated glass tubes. Note that all parts had no silicon nitride passivation, which is a barrier to hydrogen. Parts were soaked in the H_2 gas ambient for a minimum of 48 hours before irradiation. The electrical parameters that were monitored, before and after exposure to H_2 and after each irradiation step, were the same ones used by the manufacturers in their irradiation testing. All irradiations were performed with all leads shorted at dose rates of 1 rad/s or 10 rad/s using a Co-60 source at JPL. Post-irradiation electrical measurements were made at several intermediate dose levels up to 100 krad. Table 3 lists the part types, manufacturers, package types, dose rates, percent H_2, and step-stress dose levels used in the study.

Table 3. List of part types used in the study along with irradiation information.

Manufacturer	Part Type	Description	Package	Date Code	H_2 (%)	Dose Rate (rad/s)	Dose (krad)	Sample Size
NSC	RM124AJRQMLV	Quad op amp	14-pin CDIP	*0514*	100	10	10, 20, 30, 50	3
NSC	RM139AJRQMLV	Quad comparator	14-pin CDIP	*0527*	1	10	10, 20, 30, 50, 75, 100	6
NSC	RM139AJRQMLV	Quad comparator	14-pin CDIP	*0527*	100	10	10, 20, 30, 50	3
NSC	RM136AH2.5RQMLV	2.5 V reference	TO-46	N/A	100	1	10, 20, 30, 40, 50, 75, 100	3
NSC	RM2941JXQMLV	LDO regulator	16-pin CDIP	*0911*	100	1	10, 20, 30, 40, 50, 75, 100	3
Linear Tech	RM1009MW	2.5 V reference	10-pin CDIP	*0649*	1	10	10, 20, 30, 50, 75, 100	6
Semicoa	2N2222A	npn BJT	TO-18	*0739*	10	10	10, 20, 30, 50, 75, 100	6
Semicoa	2N2907A	pnp BJT	TO-18	*0804*	10	10	10, 20, 30, 50, 75, 100	6

6.2.2.1 NSC LM124

The LM124 quad op amp is a general purpose amplifier that can be operated either single-sided at 5 V or with ±15 V supplies. The NSC part has been extensively characterized for ionizing radiation response and was the basis for the ELDRS study using a specially designed test chip that included gated lateral *pnp* transistors [127]. NSC has since produced an ELDRS-free version of this part that is qualified for space application by modifying the final passivation and making some circuit design changes [128]. The HDR and LDR data taken by NSC on ELDRS-free parts for the unbiased case are used for comparison to the results for exposure of parts from the same wafer lot subjected to 100% H_2 and irradiated at 10 rad/s.

The two most sensitive parameters for the LM124 are the input bias current and the input offset voltage. Figure 65 shows the total dose response of the increase in positive input bias current, ΔIb+, for the HDR and LDR response measured by NSC for the space-qualified packaged parts compared to parts soaked in 100% H_2 before exposure. Note that the scale is log-log. In all of the plots to follow, the average ±1 standard deviation is shown for either 3 or 6 samples (sample size shown in Table 3). Exposure to H_2 results in over an order of magnitude increase in degradation.

Figure 66 shows a comparison of the NSC ELDRS-free data to parts soaked in 100% H_2 for the increase input offset voltage, ΔV_{io}. Again there is over an order of magnitude increase in the degradation for the parts soaked in H_2 and the difference increases with increasing dose.

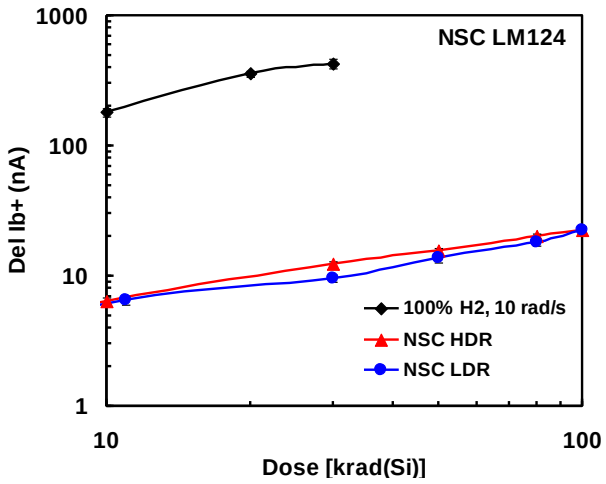


Fig. 65. Increase in input bias current for the NSC LM124, comparing NSC data to exposure to 100% H_2.

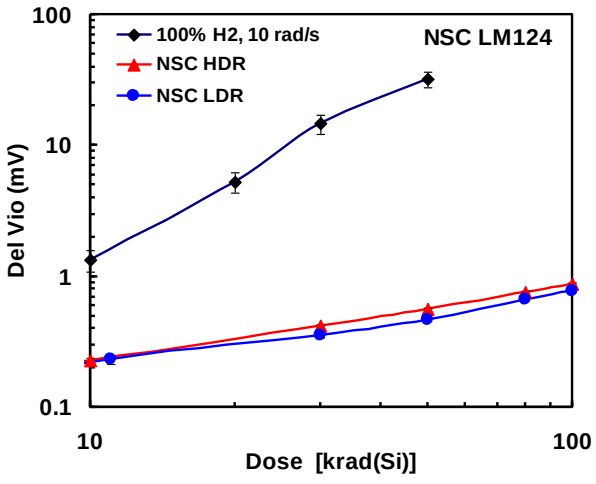


Fig. 66. Increase in input offset voltage for the NSC LM124, comparing NSC data to exposure to 100% H_2.

6.2.2.2 NSC LM139

The quad comparator, LM139, is another bipolar linear circuit that has been extensively characterized for total dose and dose rate response [89, 92]. This part was also modified by process and circuit design to be ELDRS-free [128]. For unbiased irradiation the two most sensitive parameters are the input bias current, Ib+, and the sink current, Isink. The LM139 was irradiated at 10 rad/s for two values of H_2 soaking, 100% and 1%. Figures 67 and 68 show the increase in ΔIb+ and ΔIsink, respectively, versus dose, compared to the HDR and LDR NSC ELDRS-free data.

As for the LM124, the exposure to 100% H_2 prior to irradiation resulted in well over an order of magnitude increase in the degradation. What is more significant is that with only a 1% H_2 exposure the increase in degradation over the ELDRS-free parts was still over an order of magnitude for ΔIb+ and about a factor of 8 for Isink.

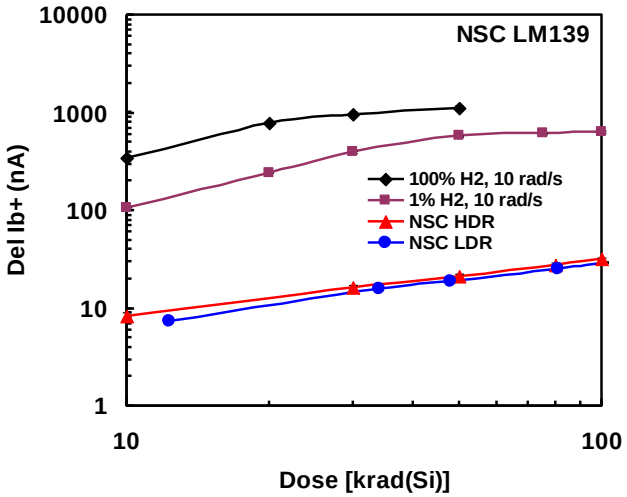


Fig. 67. Increase in input bias current for the NSC LM139, comparing NSC data to exposure to 100% and 1% H_2.

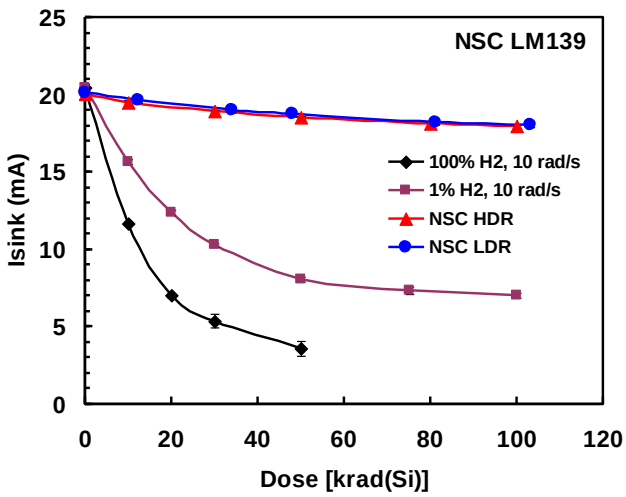


Fig. 68. Increase in sink current for the NSC LM139, comparing NSC data to exposure to 100% and 1% H_2.

6.2.2.3 NSC LM136-2.5 and LTC RH1009.

The NSC LM136 and LTC RH1009 are both 2.5 V references. NSC offers an ELDRS-free version of this reference as described in [127]. The RH1009 was radiation tested by ICS at high (50 rad/s) and low (8.2 mrad/s) dose rates both biased and unbiased and demonstrated to be ELDRS-free. We compare the results of the HDR and LDR tests on the ELDRS-free parts to irradiation at 1 rad/s after exposure to 100% H_2 for the LM136 and to irradiation at 10 rad/s after exposure to 1% H_2 for the RH1009. Figures 69 and 70 show the results for the average change in V_{ref} (at 1 mA) for the LM136 and RH1009, respectively.

Comparing the two references, we see that the V_{ref} decreases with irradiation for the LM136 and increases for the RH1009. With only 1% H_2 the RH1009 shows a significant increase in V_{ref} compared to the parts not exposed to H_2. The changes are much greater at 100% H_2 for the LM136.

Another sensitive parameter is the change in V_{ref} between two different current values. For these 2.5 V references, this parameter, BVR, is measured between currents of 400 µA and 10 mA. This parameter is also known as load regulation. Figures 71 and 72 show the results for BVR versus dose for the LM136 and RG1009, respectively.

Again, with BVR or load regulation, the degradation with H_2 is much greater than for the packaged ELDRS-free parts and the changes are in the opposite direction for the two part types.

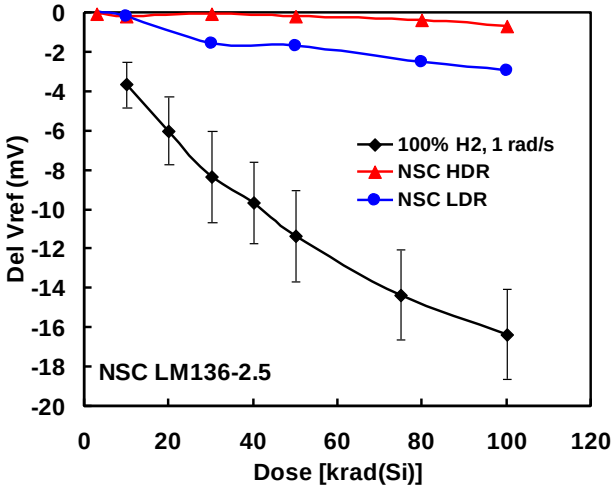


Fig. 69. Change in V_{ref} versus dose for LM136, comparing NSC data to exposure to 100% H_2.

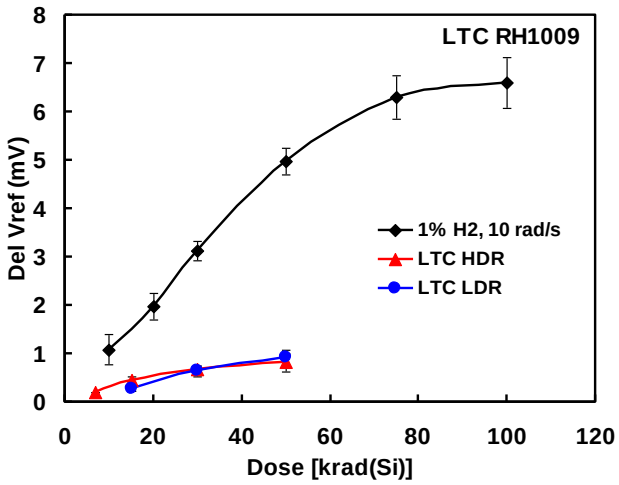


Fig. 70. Change in V_{ref} versus dose for RH1009, comparing LTC (ICS) data to exposure to 1% H_2.

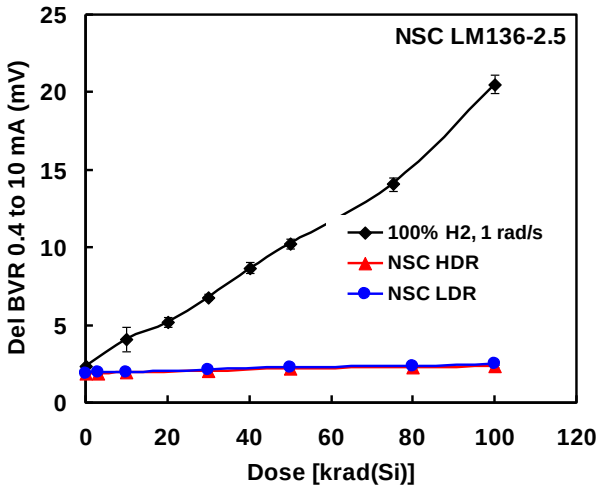


Fig. 71. Change in BVR (0.4 to 10 mA) versus dose for LM136, comparing NSC data to exposure to 100% H_2.

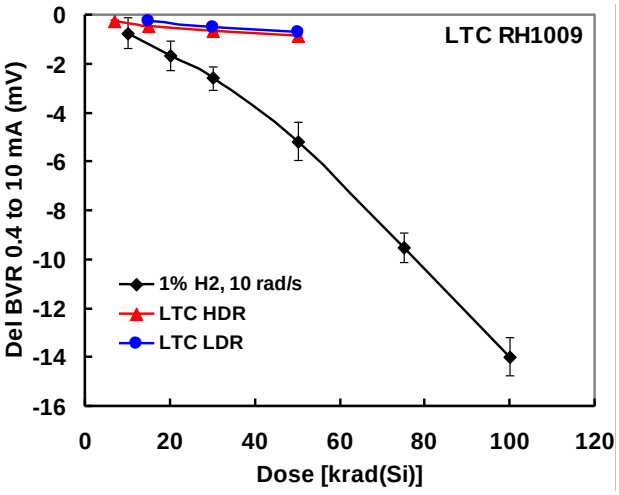


Fig. 72. Change in BVR (0.4 to 10 mA) versus dose for RH1009, comparing LTC (ICS) data to exposure to 1% H_2.

6.2.2.4 NSC LM2941

The LM2941 is a 1A adjustable low dropout regulator. The ELDRS-free version is described in [126]. This part was exposed to 100% H_2 prior to irradiation and irradiated at 10 rad/s. The output voltage was set at 5 V for the electrical tests. The reference voltage, measured at the adjust pin, is typically 1.275 V. The average change in V_{out} versus dose is shown in Figure 73 and the average change in V_{ref} is shown versus dose in Figure 74.

While there is only a few mV change at HDR and LDR for the ELDRS-free parts, there is several hundred mV change in V_{out} at 5 V for the parts exposed to 100% H_2 prior to irradiation. The changes in V_{ref} reflect the changes in V_{out} since $V_{out} \propto V_{ref}$.

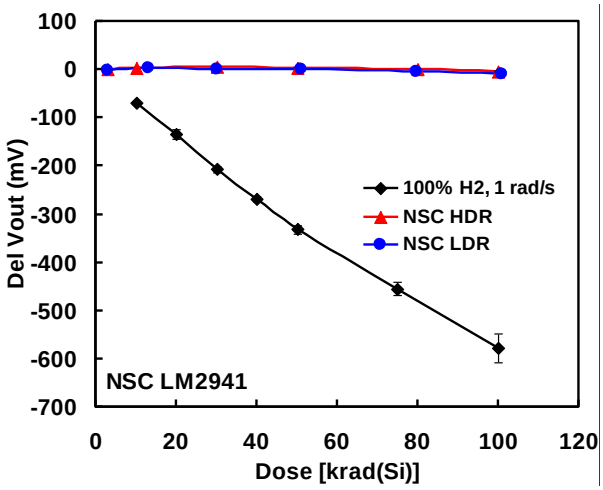


Fig. 73. Average change in output voltage versus dose for the LM2941, comparing NSC data to exposure to 100% H_2.

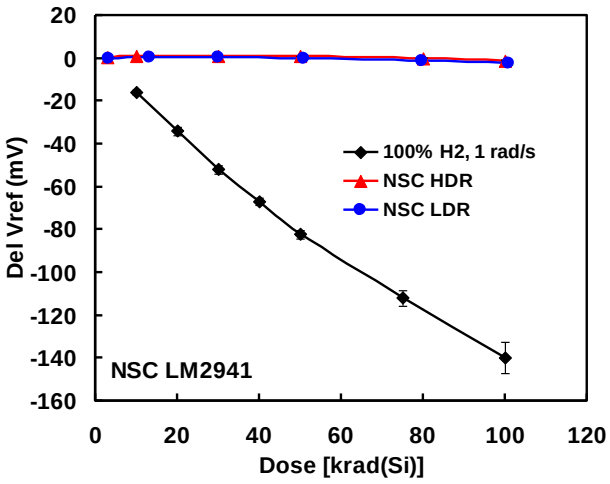


Fig. 74. Average change in reference voltage versus dose for the LM2941, comparing NSC data to exposure to 100% H_2.

6.2.2.5 2N2222A-2N2907 Discrete Bipolar Transistors

Radiation-hardened versions of the widely used discrete bipolar transistors, the *npn* 2N2222A, and the *pnp* 2N2907A were obtained from Semicoa. The HDR and LDR data on the parts not exposed to H_2 were taken by Semicoa. Delidded samples were subjected to 10% H_2 and irradiated, unbiased, at 10 rad/s at JPL. The primary parameters affected by total dose are the forward current gain, Hfe, and the collector-emitter saturation voltage, $V_{ce(sat)}$. The degradation of Hfe is a function of the collector current and degrades much more at low collector current. The normalized percent-gain degradation versus total dose at 1 mA collector current is shown in Figure 75 for the 2N2222A and Figure 76 for the 2N2907A for the baseline Semicoa data at HDR and LDR and the parts soaked in 10% H_2 and irradiated at 10 rad/s.

The amount of degradation with 10% H_2 is much greater than for the baseline parts. Note that in the case of the 2N2907A the baseline degradation at LDR is actually less than at HDR. The average normalized percent increase in $V_{ce(sat)}$ (Ic = 500 mA and Ib = 50 mA) versus dose is shown in Figure 77 for the 2N2222A and Figure 78 for the 2N2907A.

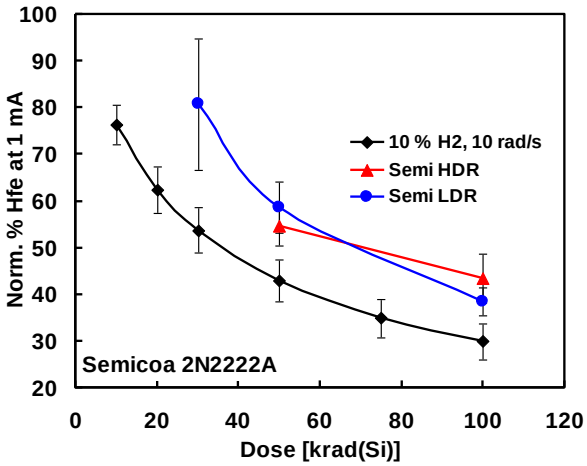


Fig. 75. Average normalized % degradation of H_{fe} at 1 mA for 2N2222A exposed to 10% H_2 compared to baseline Semicoa data.

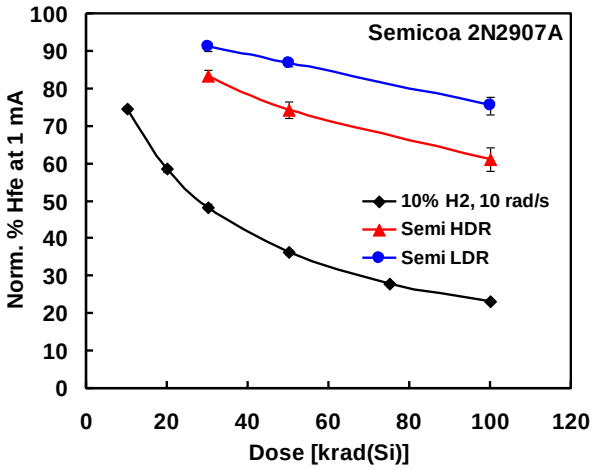


Fig. 76. Average normalized % degradation of H_{fe} at 1 mA for 2N2907A exposed to 10% H_2 compared to baseline Semicoa data.

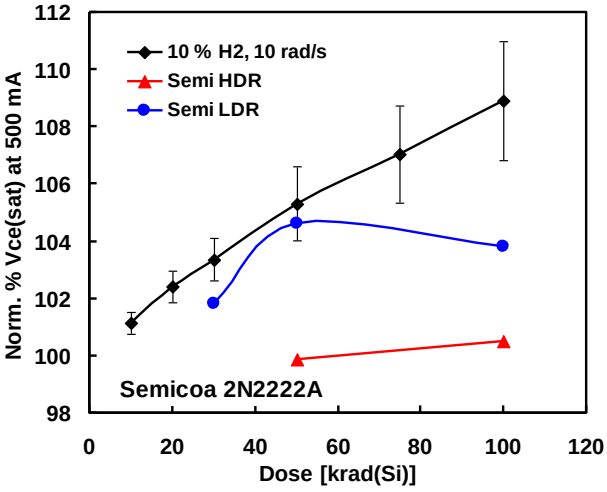


Fig. 77. Average normalized percent increase of $V_{ce(sat)}$ for 2N2222A exposed to 10% H_2 compared to baseline Semicoa data.

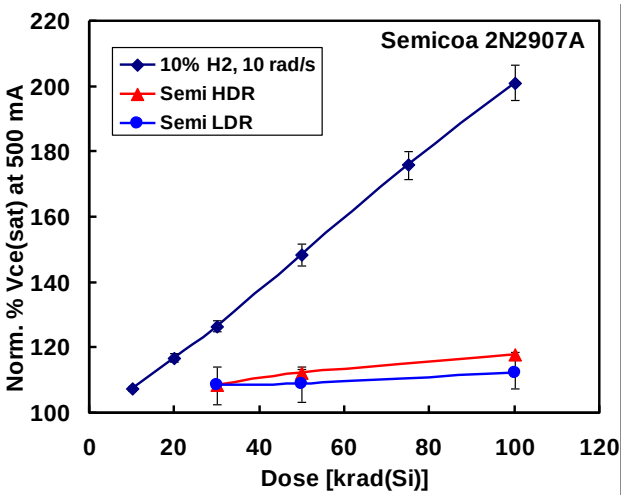


Fig. 78. Average normalized percent increase of $V_{ce(sat)}$ for 2N2907A exposed to 10% H_2 compared to baseline Semicoa data.

Comparing Figures 77 and 78, we see that the increase in $V_{ce(sat)}$ for the H_2 exposed parts is much greater for the 2N2907A than for the 2N2222A, although in both cases the exposure to H_2 does result in enhanced degradation.

6.2.3 Conclusion

The proposed method of accelerated testing using HDR irradiations in environments with elevated concentrations of H_2 have first been tested on six different parts types, all known to exhibit ELDRS, representing a wide variation in manufacturer, process technology, and circuit design (GLPNP, LM193, AD590, LT1019, OP-42, and HSYE-117). For these devices, results are very promising since with the right selection of dose rate and hydrogen concentration, it was possible to bound the LDR response. For ELDRS devices, results indicated that compared to LDR testing, hydrogen-enhanced testing at HDRs can be a very cost-effective approach for part selection during the design phase of space systems. It also may be considered for missions that require higher dose levels for qualification where LDR testing is not practical. For use as a qualification or lot acceptance test method, a characterization test would need to be performed to establish the optimum dose rate and H_2 concentration to bound the LDR response.

Further experiments were performed to consolidate the method, particularly when evaluating ELDRS-free parts; mainly to show that the proposed test would not result in rejection of those parts. The approach was investigated on ELDRS-free parts from three manufacturers, looking at a variety of circuit types (operational amplifier, comparator, voltage references, and low dropout regulator) as well as *npn* and *pnp* discrete bipolar transistors. In all cases, the exposure to H_2 resulted in an increase in the amount of degradation of sensitive electrical parameters. The amount of enhanced degradation over the baseline HDR and LDR degradation varied from a few percent for the 2N2222A to orders of magnitude for some of the ELDRS-free circuits. It is clear that, for characterization, this technique is not applicable as an accelerated technique for the purpose of determining whether a part is ELDRS. In other words, if the only data on a part is at HDR (Condition A of MIL-STD-883, Method 1019) and one wanted to do an accelerated test to determine if the part is ELDRS, this test might produce overly conservative results. However, as an accelerated test to bound the LDR response of a known ELDRS part, it may still be effective, if correlated to the LDR response.

Although the accelerated test using H_2 does not work for ELDRS-free parts, it does illustrate the importance of hydrogen in increasing the amount of degradation in bipolar linear circuits and discrete transistors. In the previous section, we demonstrated that increasing the amount of hydrogen affected the total dose and dose rate response of bipolar linear circuits. With increased hydrogen, the transition dose rate for enhanced LDR degradation moved to higher dose rates and the saturation degradation at LDR increased. Based on this mechanism, we proposed that parts that did not show a significant LDR EF at 10 mrad/s, might begin to show enhanced degradation at dose rates below 10 mrad/s. In a review of ELDRS [111], data on several bipolar circuits were shown where the LDR EF continued to increase at dose rates below 10 mrad/s. Experiments are currently being conducted to investigate the very low dose rate response of several ELDRS-free bipolar circuits to see if the EFs increase down to dose rates of 0.5 mrad/s [95]. To date, the dose levels achieved at 0.5 mrad/s are not sufficiently high to determine whether these ELDRS-free parts will show increased enhanced degradation for dose rates below the current LDR test of MIL-STD-883, Test Method 1019.8, Test Condition D of 10 mrad/s.

This investigation shows that if ELDRS-free parts are purchased in die form for use in a hybrid microcircuit, or to be packaged in a different package from what the manufacturer used for the ELDRS-free qualification, then the parts may not be total-dose-hard or ELDRS-free if subjected to H_2 in the new package form. We have shown in the previous sections that many hermetic package types may have as much as 2–4% H_2 in the package due to outgassing from gold or kovar and that these parts show enhanced degradation compared to parts in packages without H_2.

KEY POINTS ABOUT USING H_2 IRRADIATION AS AN ACCELERATED TESTING TECHNIQUE

* The proposed method of accelerated testing using HDR irradiations in environments with elevated concentrations of H_2 works for ELDRS-sensitive parts with the right combination of dose rate and concentration of H_2.
* For use as a qualification or lot acceptance test method, a characterization test would need to be performed to establish the optimum dose rate and H_2 concentration to bound the LDR response.
* The method does not work for ELDRS-free parts and is overly conservative. Indeed, the amount of enhanced degradation over the baseline HDR and LDR degradation varied from a few percent for the 2N2222A to orders of magnitude for some of the ELDRS-free circuits.
* However, results from experiments conducted on ELDRS-free parts suggest that if parts are purchased in die form for use in a hybrid microcircuit, or to be packaged in a different package from what the manufacturer used for the ELDRS-free qualification, then the parts may not be total-dose-hard or ELDRS-free if they are subjected to H_2 in the new package form.
* 2–4% H_2 in the package can change the degradation of a part by a factor of 10× for some parameters and only if the part does not have nitride passivation. This factor has to be considered for dies that are used in hybrid implementation.

REFERENCES

[1] Schwank, J.R., "Basic Mechanisms of Radiation Effects in the Natural Space Environment," 1994 NSREC Short Course, 1994.

[2] McLean F. B., et al., "Basic Mechanisms of Radiation Effects in Electronic Materials and Devices," Harry Diamond Laboratories Technical Report, No. HDL-TR-2129, September 1987.

[3] Evans R. D., "The Atomic Nucleus," McGraw-Hill, New York, 1955.

[4] Oldham, T.R., "Analysis of Damage in MOS Devices in Several Radiation Environment," IEEE Trans. Nucl. Sci., 1984. **31** (6): p. 1236-1241.

[5] Srour, J.R., "Basic Mechanisms of Radiation Effects on Electronic Materials, Devices, and Integrated Circuits," 1983 IEEE NSREC Short Course, 1983.

[6] Dressendorfer, P.V., "Basic Mechanisms for the New Millenniun," 1998 IEEE NSREC Short Course, 1998.

[7] McLean, F.B. and T.R. Oldham, "Basic Mechanisms of Radiation Effects in Electronic Materials and Devices," Harry Diamond Laboratories Technical Report, 1987. **HDL-TR**: p. 2129.

[8] Shaneyfelt, M.R., et al., "Charge Yield for Cobalt-60 and 10-keV X-Ray Irradiations," IEEE Trans. Nucl. Sci., 1991. **38** (6): p. 1187.

[9] Shaneyfelt, M.R., et al., "Field Dependence of Interface-Trap Buildup in Polysilicon and Metal Gate MOS Devices," IEEE Trans Nucl. Sci., 1990. **37** (6): p. 1632.

[10] Oldham, T. R., "Switching Oxide Traps: Radiation Effects and Soft Errors in Integrated Circuits and Electronic Devices," R.D. Schrimpf and D.M. Fleetwood, Editors. D.M. 2004, World Sci.: New Jersey.

[11] Fleetwood, D.M., "Fast and slow border traps in MOS devices," IEEE Trans Nucl. Sci., 1996. **43** (3): p. 779.

[12] Fleetwood, D.M., "Border traps" in MOS devices," IEEE Trans Nucl. Sci., 1992. **39** (2): p. 269.

[13] Rashkeev, S.N., et al., "Physical model for enhanced interface-trap formation at low dose rates," IEEE Trans Nucl. Sci., 2002. **49** (6): p. 2650.

[14] Tsetseris, L., et al., "Common origin for enhanced low-dose-rate-sensitivity and bias temperature instability under negative bias," IEEE Trans Nucl. Sci., 2005. **52** (6): p. 2265.

[15] Fleetwood, D.M., et al., "Unified model of hole trapping, 1/f noise, and thermally stimulated current in MOS devices," IEEE Trans Nucl. Sci., 2002. **49** (6): p. 2674.

[16] Nicklaw, C.J., et al., "The structure, properties, and dynamics of oxygen vacancies in amorphous SiO_2," IEEE Trans Nucl. Sci., 2002. **49**(6): p. 2667.

[17] Karna, S.P., et al., "Electronic structure theory and mechanisms of the oxide trapped hole annealing process," IEEE Trans Nucl. Sci., 2000. **47**: p. 2316.

[18] Rudra, J.K. and W.B. Fowler, "Oxygen vacancy and the E' center in crystalline SiO_2," Phys. Rev. B, Condens. Matter, 1987. **35**: p. 8223.

[19] Chavez, J.R., et al., "Microscopic structure of the E' center in amorphous SiO_2," IEEE Trans Nucl. Sci., 1997. (6) **44**: p. 1799.

[20] Pantelides, S.T., et al., "Reactions of hydrogen with Si-$SiOs_2$ interfaces," IEEE Trans Nucl. Sci., 2000. (6) **47**: p. 2262.

[21] Lacoe, R.C., et al., "Total dose tolerance of the commercial TSMC 0.35um CMOS process," 2001 IEEE Radiation Effects Data Workshop, 2001. **IEEE cat no. 01TH8588**: p. 72-77.

[22] Fleetwood D. M, et al., "1/f noise and radiation effects in MOS devices," IEEE Trans Elect. Dev., 1994. **41** (11). p. 1953.

[23] Brady F.T., et al., "A Scalable, Radiation Hardened Shallow Trench Isolation," IEEE Trans Nucl. Sci., 1999. **46**(6), p. 1836.

[24] Turowski, M. et al., "Non-uniform Total-Dose-Induced Charge Distribution in Shallow-Trench Isolation Oxides," IEEE Trans Nucl. Sci., 2004. **51**(6).

[25] Wong, J. et al., "A Hardmask STI Etch Process for 0.13um Logic Technology and Beyond," IEEE/SEMI Advanced Semiconductor Manufacturing Conference, 2001. Munich, Germany.

[26] Thompson S.E. and Nishida, T., "Tunneling and Thermal Emission of Elections from a Distribution of Shallow Traps in SiO_2," Appl. Phys. Lett., 1991. **58**: p. 1262.

[27] Lenahan P.M., et al., "Radiation-Induced Interface Traps in MOS Devices: Capture Cross Section and Density of States of P_{b1} Silicon Dangling Bond Centers," IEEE Trans Nucl. Sci., 2002. **49**(6): p. 2708.

[28] Poindexter E.H., et al., "Interface States and Electron Spin Resonance Centers in Thermally Oxidized (1 11) and (100) Silicon Wafers," J. Appl. Phys., 1981. **52**: p. 879.

[29] Saks N.S. and D.B. Brown, "Interface trap formation via the two-stage H+ process," IEEE Trans. Nucl. Sci., 1989. **36**: p. 1848-59

[30] Rashkeev S.N., et al., "Effects of Hydrogen Motion on Interface Trap Formation and Annealing," IEEE Trans Nucl. Sci., 2004. **51** (6): p. 3158.

[31] Paulsen R. E., et al., "Observation of near-interface oxide traps with the charge pumping technique," IEEE Electron Device Letters, **13** (12): 1992.

[32] Fleetwood D. M., et al., "Border Traps: Issues for MOS Radiation Response and Long-Term Reliability," Microelectronics and Reliability **35**, 403-428 (1995).

[33] Oldham T. R., "Ionizing Radiation Effects in MOS Oxides." NJ: World Scientific Publishing Co. Pte. Ltd., 1999.

[34] Muller R. S. and Kamins T.I., "Device Electronics for Integrated Circuits". Second ed. 1986, New York: Wiley.

[35] Cester A., et al., "Switching Oxide Ionizing radiation effects on Ultra-Thin Oxide MOS Structures," R.D. Schrimpf and D.M. Fleetwood, Editors. D.M. 2004, World Sci.: New Jersey.

[36] Lacoe R.C., et al., "Total dose hardness of Three Commercial CMOS Microelectronics Foundries," 4_{th} RADECS, p 265, 1997

[37] Lacoe, R.C., "CMOS Scaling Design Principles and Hardening by Design Methodology," NSREC Short Course, 2003.

[38] Nicollian, E. and Brews, J., Metal Oxide Semiconductor Physics and Technology, NY: John Wiley & Sons, 1982, pp. 319-370 and 550-570.

[39] Fleetwood D. M., et al., "Radiation Effects at Low Electric Fields in Thermal, SIMOX and Bipolar-Base Oxides," IEEE Trans. on Nucl. Sci., 1996. **43** (6): p. 2537.

[40] Cester, A., et al., "Drain Current Decrease in MOSFETs After Heavy Ion Irradiation," IEEE Trans Nucl. Sci., 2004. **51**(6): p. 3150.

[41] Snapp, W., DARPA Rad-Hard-by-Design Program Results, GOMAC Tech 2006, March 20-23, San Diego, CA.

[42] M. R. Shaneyfelt, et al., "Challenges in Hardening Technologies Using Shallow-Trench Isolation," IEEE Trans. Nucl. Sci. 1998. **45** (6): 2583.

[43] I. Sanchez Esqueda, et al., "Two-dimensional Methodology for Modeling Radiation-Induced Off-State Leakage in CMOS Technologies," IEEE Trans. Nucl. Sci. 2004. **45** (6), p. 2583.

[44] P. S. Winokur, et al., "Two-Stage Process for Buildup of Radiation-Induced Interface States," J. Appl. Phys. 1979 **50** (5): p. 3492.

[45] D. M. Fleetwood, et al., "A Reevaluation of the Worst-Case Post irradiation Response for Hardened MOS Transistors," IEEE Trans. Nucl. Sci. 1987 **34** (6), p. 1178.

[46] N. S. Saks, et al., "Effects of Switched Bias on Radiation-Induced Interface Trap Formation," IEEE Trans. Nucl. Sci. vol. 1991. **38** (6), p. 1130.

[47] Fleetwood D. M., et al., "High-Temperature Silicon-On-Insulator Electronics for Space Nuclear Power Systems: Requirements and Feasibility," IEEE Trans. Nucl. Sci. 1988. **35** (5): p. 1099.

[48] Lelis A. J., et al, "Response of Interface Traps During High-Temperature Anneals," IEEE Trans. Nucl. Sci. 1991. **38** (6): p. 1590.

[49] Johnston A. H, "Low Dose Rate Effects in Shallow Trench Isolation Regions," IEEE Trans. Nucl. Sci. 2010, 57 (2010) p. 3279

[50] Cressler J. D., "Using SiGe HBT technology for extreme environment electronics," in Proc. IEEE Bipolar/BiCMOS Circuits and Technology Meeting, 248 – 251, 2005.

[51] Kosier S. L., et al., "Effects of Oxide Charge and Surface Recombination Velocity on the Excess Base Current of BJTs," in Proc. IEEE Bipolar/BiCMOS Circuits and Technology Meeting, 211-214, December 1993.

[51] Nowlin R. N., et al, "Mechanisms of Ionizing-Radiation-Induced Gain Degradation in Modern Bipolar Devices," in Proc. 1991 IEEE Bipolar Circuits and Tech. Mtg., 174-177, December 1991.

[53] Kosier S. L., et al., "Physically Based Comparison of Hot-Carrier-Induced and Ionizing-Radiation-Induced Degradation in BJTs," IEEE Trans. Electron Devices, 1995. **42**: p. 436.

[54] Schmidt D. M., et al., "Modeling Ionizing-Radiation-Induced Gain Degradation of the Lateral PNP Bipolar Junction Transistor," IEEE Trans. Nucl. Sci., 1996. **43** (6), p. 3032.

[55] Barnaby H. J., et al., "Modeling Ionizing Radiation Effects in Lateral PNP BJTs with Non-Uniform Trapped Charge Distributions," in GOMAC Dig., 1998. p. 585.

[56] Barnaby H. J., et al., "The Effects of Emitter-Tied Field Plates on Lateral PNP Ionizing Radiation Response," in IEEE BCTM Proc., 35-38, December 1998.

[57] Barnaby H. J., et al., "Modeling BJT Radiation Response with Non-Uniform Energy Distributions of Interface Traps," IEEE Trans. Nucl. Sci., 2000. **47** (6): p. 514.

[58] Barnaby H. J., et al., "Influence of Interface and Bulk Defect Interaction on Proton Radiation Response of Bipolar Devices," IEEE NSREC, presented at NSREC, 2001.

[59] Rax B. G., et al., "Displacement damage in bipolar linear integrated circuits," IEEE Trans. Nucl. Sci., 1999. **46** (6): p. 1660.

[60] Zoutendyk J. A., et al., "Comparison of the degradation effects of heavy ion, electron, and cobalt-60 irradiation in an advanced bipolar process," IEEE Trans. Nucl. Sci., 35 (6): p. 1428.

[61] Wei A., et al., "Excess Collector Current Due to an Oxide-Trapped-Charge-Induced Emitter in Irradiated NPN BJTs," IEEE Trans. Electron Devices, 1995. **42** (6): p. 923.

[62] Nowlin R. N., et al., "Trends in the Total-Dose Response of Modern Bipolar Transistors," IEEE Trans. Nucl. Sci., 1992. 39 (6): p. 2026.

[63] Pawlikiewicz A. H., "High tolerance to radiation effects and low noise performance of ACUTE-a complementary bipolar SOI IC technology," IEEE Trans. Nucl. Sci., 1196. **43** (6): p. 761.

[64] Beaucour J., et al., "Total Dose Effects on Negative Voltage Regulator," IEEE Trans. Nucl. Sci., 1994. **41** (6): p. 2420.

[65] Johnston A. H., et al., "Enhanced Damage in Linear Bipolar Integrated Circuits at Low Dose Rates," IEEE Trans. Nucl. Sci., 1995. 42 (6): p. 1650.

[66] Schmidt D. M., et al., "Comparison of Ionizing-Radiation-Induced Gain Degradation in Lateral, Substrate, and Vertical PNP BJTs," IEEE Trans. Nucl. Sci., 1995. **42** (6): p. 1541.

[67] Schmidt D. M., et al., "Modeling Ionizing Radiation Induced Gain Degradation of the Lateral Bipolar Junction Transistor," IEEE Trans. Nucl. Sci., 1996. **43**: p. 3032.

[68] Johnston A. H., et al., "Total Dose Effects in Conventional Bipolar Transistors and Linear Integrated Circuits," IEEE Trans. Nuc. Sci. vol. 1994. **41** (6): p. 2427.

[69] Johnston A. H., et al., "Enhanced Damage in Bipolar Devices at Low Dose Rates: Effects at Very Low Dose Rates," IEEE Trans. Nuc. Sci. 1996. **43** (6): p. 3049.

[70] Pease R. L., et al., "Enhanced Low-Dose-Rate Sensitivity of a Low-Dropout Voltage Regulator," IEEE Trans. Nuc. Sci. 1998. **45** (6): p. 2571.

[71] Pease R. L., et al., "Total dose and dose rate response of an AD590 temperature transducer," IEEE Trans. Nucl. Sci., 2007. **54** (4): p. 1049.

[72] Barnaby H., et al., "Analysis of Bipolar Linear Circuit Response Mechanisms for High and Low Dose Rate Total Dose Irradiations," IEEE Trans. Nuc. Sci. 1996. 43 (6): p. 3040.

[73] Barnaby H., et al., "Identification of Degradation Mechanisms in a Bipolar Linear Voltage Comparator Through Correlation of Transistor and Circuit Response," IEEE Trans. Nucl. Sci. 1999. **46** (6): p. 1666.

[74] Flament O., et al., "Enhanced Total Dose Damage in Junction Field Effect Transistors and Related Linear Integrated Circuits," IEEE Trans. Nuc. Sci. vol. 1996. **43** (6): p. 3060-3067.

[75] Johnston A. H., et al., "Total dose effects in op-amps with compensated input stages," IEEE Trans. Nucl. Sci., 2008. **44** (4): p. 1953.

[76] Johnston A. H., et al., "Total Dose Effects in op-amps with compensated input stages," proceeding of Radiation Effects on Components and Systems, pp. 1 – 8, 2007.

[77] Bernard M. F., et al., "Analysis of Total Dose Response of A bipolar Voltage Comparator Combining Radiation Experiments And Design Data," proceeding of Radiation and Its Effects on Components and Systems, pp. 1 – 8, 2005.

[78] Kosier S. L., et al., "Bounding the Total Dose Response of Modern Bipolar Transistors," IEEE Trans. Nuc. Sci. 1994. **41** (6): p. 1864.

[79] Schmidt D. M., et al., "Modeling Ionization Radiation Induced Degradation of the Lateral-PNP Bipolar Junction Transistor," IEEE Trans. Nuc. Sci. 1996. **43** (6): p. 3032.

[80] Enlow E. W., et al., "Response of Advanced Bipolar Processes to Ionizing Radiation," IEEE Trans. Nuc. Sci. 1991. **38** (6): p 1342.

[81] Pease R. L., et al., "A Proposed Hardness Assurance Test Methodology for Bipolar Linear Circuits and Devices in a Space Ionizing Radiation Environment," IEEE Trans. Nucl. Sci. 1997. **44** (6): p. 1981.

[82] Nowlin R. N., et al., "Trends in the Total-Dose Response of Modern Bipolar Transistors," IEEE Trans. Nucl . Sci. 1992. **39**: p. 2026.

[83] Nowlin R. N., et al., "Hardness Assurance and Testing Issues for Bipolar/BiCMOS Devices," IEEE Trans. Nucl. Sci. 1993. **40**: p. 1686.

[84] McClure S., et al., "Dependence of Total Dose Response of Bipolar Linear Microcircuits on Applied Dose Rate," IEEE Trans. Nuc. Sci. 1994. **41** (6): p. 2544.

[85] Johnston A. H., et al., "Total Dose Effects in Conventional Bipolar Transistors and Linear Integrated Circuits," IEEE Trans. Nuc. Sci. vol. 1994. **41** (6): p. 2427.

[86] Beaucour J. T., et al., "Total Dose Effects on Negative Voltage Regulator," IEEE Trans. Nuc. Sci. 1994. **41** (6): p. 2420.

[87] Johnston A. H., et al., "Enhanced Damage in Bipolar Devices at Low Dose Rates: Effects at Very Low Dose Rates," IEEE Trans. Nuc. Sci. 1996. **43** (6): p. 3049.

[88] Pease R. L., et al., "A Compendium of Recent Total Dose Data on Bipolar Linear Microcircuits," IEEE Rad. Effects Data Workshop Record, 28-37, 1996.

[89] Pease R. L., et al., "An Updated Compendium of Enhanced Low Dose Rate Sensitive (ELDRS) Bipolar Linear Circuits," 2001 IEEE Radiation Effects Data Workshop Record, p. 127.

[90] Pease R. L., "A 2008 update to the ELDRS data compendium," Presented at the 2008 RADECS, Jyvaskyla, Finland, September, 2008.

[91] Shaneyfelt M. R., et al., "Thermal-Stress Effects on Enhanced Low Dose Rate Sensitivity in Linear Bipolar ICs," IEEE Trans. Nucl. Sci. NS- 2000. **47** (6): p. 2539.

[92] Fleetwood D. M., et al., "Physical Mechanisms Contributing to Enhanced Bipolar Gain Degradation at Low Dose Rates," IEEE Trans. Nucl. Sci. 1994. **41** (6): p. 1871.

[93] Pease R. L., et al., "Enhanced Low-Dose-Rate Sensitivity of a Low-Dropout Voltage Regulator," IEEE Trans. Nuc. Sci. 1998. **45** (6): p 2571. [94] McClure S. S., et al., "Dose rate and bias dependency of total dose sensitivity of low dropout voltage regulators," IEEE Radiation Effects Data Workshop Record, 100-105, 2000.

[95] Pease R. L., et al., "The effects of hydrogen in hermetically sealed packages on the total dose and dose rate response of bipolar linear circuits," IEEE Trans. Nucl. Sci. 2007. **54** (6): p. 2168.

[96] Chen X. J., et al., "Mechanisms of enhanced radiation-induced degradation due to excess molecular hydrogen in bipolar oxides," IEEE Trans. Nucl. Sci. 2007. **54** (6): p. 1913.

[97] Adell P. C., et al., "Impact of hydrogen contamination on the total dose response of linear bipolar microcircuits," presented at RADECS, September 2007.

[98] Chen X. J., et al., "Annealing behavior of oxide trapped charge in bipolar base oxides after radiation exposure in H_2 environments," IEEE Trans. Nucl. Sci. 2008. **55** (6): p. 3032.

[99] Pease R. L., et al., "Characterization of Enhanced Low Dose Rate Sensitivity (ELDRS) Effects Using Gated Lateral PNP Transistor Structures," IEEE Trans. Nucl. Sci. 2004. **51** (6): p. 3773.

[100] Pease R. L., et al., “The effects of hydrogen on the enhanced low dose rate sensitivity (ELDRS) of bipolar linear circuits,” IEEE Trans. Nucl. Sci. 2008. **55** (6): p. 3169.

[101] Adell P. C., et al., “Irradiation with Molecular Hydrogen as an Accelerated Total Dose Hardness Assurance Test Method for Bipolar Linear Circuit,” IEEE Trans. Nucl. Sci. 2009. **56** (6): p. 3326.

[102] Shaneyfelt M. R., et al., "Impact of Passivation Layers on Enhanced Low-Dose-Rate Sensitivity and Thermal-Stress Effects in Linear Bipolar ICs,” IEEE Trans. Nucl. Sci. 2002. 49 (6): p. 3171.

[103] Barnaby H., et al., "Analysis of Bipolar Linear Circuit Response Mechanisms for High and Low Dose Rate Total Dose Irradiations,” IEEE Trans. Nuc. Sci. 1996. **43** (6): p. 3040.

[104] Batyrev, I. G et al., "Mechanisms of Hydrogen Effects in Bipolar Radiation Response," IEEE Trans. Nuc. Sci. 2008. **55** (6): p. 3039.

[105] Boch, J. et al., "Physical Model for the Low-Dose Rate Effect in Bipolar Devices,” IEEE Trans. Nucl. Sci., 2006 **53** (6): p. 3655.

[106] Stahlbush R. E. et al., "Post-irradiation cracking of H_2 and formation of interface states in irradiated metal-oxide-semiconductor field-effect transistors," J. Appl. Phys., 1993 **73** (6) p. 658.

[107] Mrstik B. J. and Rendell R. W., “Si-SiO_2 interface state generation during X-ray irradiation and during post-irradiation exposure to a hydrogen ambient [MOSFET],” IEEE Trans. Nucl. Sci. 1991 **38** (6) p. 1101.

[108] D. M. Fleetwood et al., “Using laboratory x-ray and cobalt-60 irradiations to predict CMOS device response in strategic and space environments,” IEEE Trans. on Nucl. Sci. 1988 **35** (6): p 487.

[109] Hjalmarson H. P. et al., “Mechanisms for radiation dose-rate sensitivity of bipolar transistors,” IEEE Trans. Nucl. Sci. 2003 **50** (6): p. 1901-1909, 2003.

[110] Hjalmarson H. P. et al., “Calculations of Radiation Dose-Rate Sensitivity of Bipolar Transistors,” IEEE Trans. Nucl. Sci. 2008 **55** (6) p. 3009.

[111] Pease R. L et al., "ELDRS in Bipolar Linear Circuits: A Review,” IEEE Trans. Nucl. 2009 **56** (4): p. 1894.

[112] Rowsey N. L. et al., "A quantitative Model for ELDRS and H2 degradation effects in Irradiated Oxides Based on First Principles Calculations,” IEEE Trans. Nucl. 2011 **58** (6): p. 2937

[113] Witczak S. C. et al., “Dose-rate sensitivity of modern nMOSFETs,” IEEE Trans. on Nucl. Sci. 2005 52 (6): p. 2602.

[114] McLean F. B., "A Framework for Understanding Radiation-Induced Interface States in SiO_2 MOS Structures," IEEE Trans. Nucl. Sci., 1980 27 (6): p. 1651.

[115] Snowden C. M., Introduction to Semiconductor Device Modeling.Hackensack, NJ: World Scientific, 1986.

[116] Esqueda I. S. et al., “Modeling inter-device leakage in 90 nm bulk CMOS devices,” IEEE Trans. Nucl. Sci., 2011 **58** (6): p. 793.

[117] A. G. Revesz, “Chemical and structural aspects of the irradiation behavior of SiO_2 films on silicon,” IEEE Trans. Nucl. Sci., 1977 **24**, p. 2102.

[118] Chen X. J. et al, “Estimation and verification of radiation induced and energy distribution using combined bipolar and MOS characterization methods in gated bipolar devices,” IEEE Trans. Nucl. Sci., 2005 **52** (6) p. 2245.

[119] Mrstik B. J et al., “Comparison of charge trapping in undoped oxides made by low- and high-temperature deposition techniques,” IEEE Trans. Nucl. Sci., 2001 48 p. 2107.

[120] Carriere T., et al., "Evaluation of Accelerated Total Dose Testing of Linear Bipolar Circuits," IEEE Trans. Nuc. Sci. 2000 **47** (6): p. 2350.

[121] Boch J., et al.,”Elevated temperature irradiation at high dose rate of commercial linear bipolar ICs,” IEEE Trans. Nucl. Sci. 2004. **51** (5): p. 2903.

[122] Freitag R. K. and Brown D. B., "Study of Low-Dose-Rate Effects on Commercial Linear Bipolar ICs," IEEE Trans. Nucl. Sci. 1998. **45** (6): p. 2649.

[123] Pershenkov V. S., et al., “Interface trap model for low-dose-rate effect in bipolar devices,” presented at RADECS, Deauville, France, September 10-14, 2007.

[124] Boch J., et al., “Effect of switching from high to low dose rate on linear bipolar technology radiation response,” IEEE Trans. Nucl. Sci. 2004. **51** (5): p. 2896.

[125] Boch J., et al., “Estimation of low-dose-rate degradation on bipolar linear integrated circuits using switching experiments,” IEEE Trans. Nucl. Sci. 2005. **52** (6): p. 2616.

[126] Kruckmeyer K. et al., “Low dose rate test results of National Semiconductor’s ELDRS-free low dropout (LDO) regulator, LM2941, at dose rate of 1 and 10 mrad/s,” in Proc. IEEE Radiation Effects Data Workshop, 2009, pp. 59–64.

[127] Kruckmeyer K. et al., “Low dose rate test results of National Semiconductor’s ELDRS-free LM136-2.5 bipolar reference,” in Proc. IEEE Radiation Effects Data Workshop, pp. 47–50.

[128] Kruckmeyer K. et al., “Low dose rate test results of National Semiconductor’s ELDRS-free bipolar amplifier LM124 and comparators LM139 and LM193,” in Proc. IEEE Radiation Effects Data Workshop Rec., 2008, pp. 110–117.